An Introduction to Analytic Number Theory

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Preface

These notes introduce several foundational topics in analytic number theory. They are intended for readers who have studied elementary number theory and the basics of real and complex analysis. The main subjects are the average orders of arithmetic functions, the prime number theorem, and primes in arithmetic progressions.

The book follows two threads. The first begins with divisibility and Dirichlet convolution and uses Abel summation and the hyperbola method to study mean values of arithmetic functions. The second encodes arithmetic functions as Dirichlet series and proves the prime number theorem by means of Euler products, analytic continuation, and a Tauberian theorem. The final chapter introduces Dirichlet characters and L-functions and proves that every reduced residue class contains infinitely many primes, then establishes their asymptotic distribution for each fixed modulus.

Exercises appear at the end of every chapter; brief hints and complete solutions are collected at the end of the book. When a technical result lies beyond the scope of the course, the text states the precise version needed and labels the ensuing discussion “Proof outline.” Readers already comfortable with elementary number theory may skim Chapter 1, but the convolution notation of Chapter 2 is used throughout the rest of the book.

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Notation and Conventions

Unless otherwise stated, $\mathbb{N} = \{1, 2, 3, \dots\}$, the letter p always denotes a prime, and $d \mid n$ means that the positive integer d divides n. We write $s = \sigma + it \in \mathbb{C}$, where $\sigma, t \in \mathbb{R}$.

Notation	Meaning
$f * g$	Dirichlet convolution of arithmetic functions
$\mathbf{1}$, ε	the constant function 1; the identity for convolution
μ, φ, Λ	the Möbius, Euler, and von Mangoldt functions
$\tau = \mathbf{1} * \mathbf{1}$	the divisor-counting function
$\sigma_k(n)$	the sum of the kth powers of the positive divisors of n
$\nu_p(n)$, $p^\alpha \parallel n$	the exponent of p in n; p^α exactly divides n
$\pi(x)$	the number of primes not exceeding x
$\vartheta(x), \psi(x)$	the Chebyshev functions
$\pi(x; q, a)$, $\vartheta(x; q, a)$, $\psi(x; q, a)$	prime-counting functions in the class $a \pmod q$
σ_c, σ_a	abscissae of ordinary and absolute convergence of a Dirichlet series
$\mathcal{M}f(s)$	the Mellin transform of f
$\mathfrak{g}(\chi)$	the Gauss sum of the Dirichlet character χ
$f(x) = O(g(x))$	there is a constant C such that $\lvert f(x)\rvert \le C\lvert g(x)\rvert$
$f(x) = o(g(x))$	$f(x)/g(x) \to 0$
$f(x) \sim g(x)$	$f(x)/g(x) \to 1$
$A \ll B$	synonymous with $A = O(B)$; subscripts indicate dependence of constants

A sum $\sum_{n \le x}$ is always taken over positive integers n; when its upper endpoint is not an integer, it means $\sum_{n=1}^{\lfloor x \rfloor}$. An empty sum is 0 and an empty product is 1. To avoid ambiguity at endpoints, statements involving Stieltjes integrals or Perron's formula will specify separately whether x is an integer.

Chapter Dependencies

Chapter	Main topics	Main prerequisites
0	Problems, methods, and the big picture	Elementary number theory, calculus
1	Divisibility, congruences, unique factorization	None
2	Arithmetic functions, convolution, Möbius inversion	Chapter 1
3	Abel summation, average orders, hyperbola method	Chapter 2, real analysis
4	Chebyshev functions and elementary prime estimates	Chapters 1–3
5	$\zeta(s)$, zeros, and the prime number theorem	Chapters 3–4, complex analysis
6	Dirichlet series, Mellin transforms, Perron's formula	Chapters 2–5
7	Characters, L-functions, Dirichlet's theorem	Chapters 2, 5, and 6

Chapter 0

The Landscape of Analytic Number Theory

Learning objectives. Recognize the counting and distribution problems typical of analytic number theory, understand the role of Dirichlet series, and compare $\pi(x)$, $\vartheta(x)$, and $\psi(x)$.

Prerequisites. Basic notions concerning primes, infinite series, and complex functions.

Role in the book. This chapter surveys the objects and methods used throughout the book.

0.1 Beginning with prime counting

Define

$$\pi(x) = \#\{p \le x : p \text{ is prime}\}.$$

Euclid's proof shows that $\pi(x) \to \infty$, but it gives no rate of growth. The prime number theorem asserts that

$$\pi(x) \sim \frac{x}{\log x}. \tag{0.1}$$

This formula describes the global distribution of the primes rather than the location of any individual prime. Analytic number theory is concerned with precisely this kind of average law.

The weighted counting functions

$$\vartheta(x) = \sum_{p \le x} \log p, \qquad \psi(x) = \sum_{\substack{p^k \le x \\ k \ge 1}} \log p = \sum_{n \le x} \Lambda(n)$$

interact better with multiplication and series than $\pi(x)$ does. Chapter 4 will prove that $\psi(x) - \vartheta(x) = O(\sqrt{x}\log^2 x)$. Thus the extra contribution of prime powers is negligible on the scale of x. Abel summation then shows that $\vartheta(x) \sim x$ is equivalent to (0.1).

0.2 Three kinds of problems

Counting and averages. Given an arithmetic function $f : \mathbb{N} \to \mathbb{C}$, study

$$S_f(x) = \sum_{n \le x} f(n).$$

For example, when $f = \tau$, the sum $S_f(x)$ counts lattice points beneath the hyperbola $ab \le x$; when $f = \Lambda$, the sum $S_f(x) = \psi(x)$ encodes primes and prime powers.

Distribution in arithmetic progressions. If $(a, q) = 1$, are there infinitely many primes $p \equiv a \pmod{q}$? Dirichlet's theorem gives an affirmative answer. Orthogonality among residue classes will be expressed through Dirichlet characters.

Error terms. A main term describes a macroscopic law, whereas an error term records finer oscillations. The Riemann hypothesis, for example, is closely tied to the error term in the prime number theorem. This course develops that connection but does not aim to prove the Riemann hypothesis.

0.3 Encoding, analysis, and decoding

An arithmetic function $a(n)$ can be encoded by the Dirichlet series

$$F(s) = \sum_{n=1}^{\infty} \frac{a(n)}{n^s}.$$

Taking $a(n) = 1$ gives the Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_{p} (1 - p^{-s})^{-1} \qquad (\sigma > 1).$$

The Euler product on the right translates unique factorization into an analytic identity. Logarithmic differentiation gives in turn

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}, \qquad \sigma > 1.$$

Thus the analytic properties of ζ control the partial sums of Λ. The overall process may be summarized as

arithmetic sequence $\longrightarrow$ generating function $\longrightarrow$ complex analysis $\longrightarrow$ asymptotic formula.

These steps rely, respectively, on convergence, analytic continuation, and Tauberian conditions.

A table of methods

The following table lists several typical correspondences; the relevant hypotheses are supplied in the chapters where the methods are developed.

Arithmetic object	Dirichlet series	Key analytic information	Arithmetic conclusion
$\mathbf{1}(n)$	$\zeta(s)$	A simple pole at $s = 1$ with residue 1	$\sum_{n \le x} 1 = x + O(1)$
$\tau = \mathbf{1} * \mathbf{1}$	$\zeta(s)^2$	A double pole at $s = 1$	The main term of $\sum_{n \le x} \tau(n)$ is $x \log x$
$\Lambda(n)$	$-\zeta'(s)/\zeta(s)$	The pole at $s = 1$ and nonvanishing on the boundary	$\psi(x) \sim x$, the prime number theorem
$\chi(n)$	$L(s, \chi)$	$L(1, \chi) \neq 0$ for nonprincipal characters	Infinitely many primes in each reduced residue class

In the table, the slogan "poles produce main terms" must be used together with Perron's formula or a Tauberian theorem. Boundary estimates for contour integrals, sign conditions on coefficients, and similar hypotheses cannot be omitted. The location of a pole alone does not determine an asymptotic formula for partial sums.

Example 0.1 (Two approaches to the divisor function)**.** *The divisor function satisfies*

$$\tau(n) = \sum_{d|n} 1 = (\mathbf{1} * \mathbf{1})(n).$$

Dirichlet series turn convolution into multiplication, so in $\Re s > 1$*,*

$$\sum_{n\geq 1} \frac{\tau(n)}{n^s} = \zeta(s)^2.$$

Since $\zeta(s) = 1/(s-1) + \gamma + O(s-1)$*, near* $s = 1$ *the right-hand side is*

$$\frac{1}{(s-1)^2} + \frac{2\gamma}{s-1} + O(1).$$

If we use Perron's formula from Chapter 6, these two polar coefficients suggest the main term $x \log x + (2\gamma - 1)x$*. A rigorous error term also requires control of every segment of the contour after the line of integration has been shifted. Chapter 3 will first prove by the entirely elementary hyperbola method that*

$$\sum_{n\leq x} \tau(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x}).$$

The same convolution identity can therefore be used either for lattice-point counting or for multiplication of generating functions.

0.4 The division of labor among three methods

Elementary methods. Here "elementary" usually means avoiding complex analysis, not "simple." The hyperbola method, sieve methods, inclusion-exclusion, and combinatorial identities all belong to this class. They work directly with finite sums, making the sources of error transparent, although they can have difficulty detecting deeper oscillations.

Complex-analytic methods. These methods encode a sequence in an analytic function and then exploit singularities, zeros, analytic continuation, and contour integration. Poles usually determine main terms and zeros control errors, but returning from a function to its coefficients requires convergence and growth conditions to be checked.

Tauberian methods. An Abelian result uses information about coefficients to deduce boundary behavior of a generating function; a Tauberian theorem reverses this implication under additional hypotheses. The Wiener–Ikehara theorem used in Chapter 5 exploits nonnegative coefficients and turns good boundary behavior of $-\zeta'/\zeta$ into $\psi(x) \sim x$. It is neither a purely formal inverse transform nor merely a contour-integral calculation.

Chapters 1–4 rely mainly on elementary methods; beginning with Chapter 5, complex analysis and Tauberian methods take center stage.

Example 0.2. *The harmonic sum already illustrates this division of labor. Comparing the decreasing function $1/t$ with areas of rectangles gives*

$$\sum_{n\le x}\frac{1}{n}=\log x+\gamma+O(1/x).$$

The integral supplies the main term, while the constant records the cumulative discrepancy between the discrete sum and its continuous counterpart. Abel summation in Chapter 3 will extend this idea to sums with arithmetic coefficients.

Chapter Summary

- Analytic number theory studies the counts, averages, and distribution of arithmetic objects rather than evaluating them one by one.
- The functions π, ϑ, and ψ are three differently weighted versions of the same prime-distribution problem.
- Dirichlet series and Euler products bridge additive analysis and multiplicative structure.
- Rigorous convergence arguments and decoding theorems are required to pass from generating functions back to asymptotic formulas.

Common Pitfalls

- Misreading $f(x)\sim g(x)$ as $f(x)-g(x)\to 0$; asymptotic equivalence means that the ratio tends to 1.
- Using the Dirichlet series or Euler product for $\zeta(s)$ directly when $\sigma\le 1$.
- Assuming that an "elementary proof" must be short or weak; the adjective only specifies that complex analysis is not used.
- Reading a main term for partial sums directly from a pole of the generating function at $s=1$ without identifying an inversion formula or Tauberian theorem.

Exercises

Exercise 0.1. Compute $\pi(x)$, $\vartheta(x)$, and $\psi(x)$ for $x=10,30$, and identify the prime powers that account for the differences among the three functions.

Exercise 0.2. Assume that $\vartheta(x)\sim x$. Temporarily allowing the use of Abel summation, derive $\pi(x)\sim x/\log x$ and list the boundary terms that must be controlled.

Exercise 0.3. Show that even if $a(n),b(n)\ge 0$ and their respective Dirichlet series both converge at a real number σ, the series

$$\sum_{n\ge 1}a(n)b(n)n^{-\sigma}$$

need not converge. Which formal manipulations does this phenomenon warn us not to use without additional hypotheses?

Exercise 0.4. Add a row for Euler's totient function φ to the table of methods in this chapter. First use $\mathbf{1}*\varphi=\mathrm{id}$ to predict its Dirichlet series, then use the pole of the resulting expression at $s=2$ to predict the main term of $\sum_{n\le x}\varphi(n)$. No proof of the error term is required at this stage.

Chapter 1

Review of Elementary Number Theory

Learning objectives. Review divisibility, congruences, and unique factorization, and understand the counting interpretation of Euler's totient function.

Prerequisites. Arithmetic with integers and mathematical induction.

Role in the book. Unique factorization is the algebraic basis of Euler products; the Chinese remainder theorem will be used again in Chapter 7.

1.1 Divisibility and greatest common divisors

Definition 1.1 (Divisibility, units, and primes)**.** *Let $a, b \in \mathbb{Z}$. If there is a $c \in \mathbb{Z}$ such that $b = ac$, we say that a* divides *b and write $a \mid b$; otherwise we write $a \nmid b$. The units of the ring $\mathbb{Z}$ are 1 and -1.*

A positive integer $p > 1$ is prime *if its only positive divisors are 1 and p. A positive integer greater than 1 that is not prime is called* composite*.*

Example 1.2. *We have $6 \mid 42$ because $42 = 6 \cdot 7$, whereas $8 \nmid 42$. Also $6 \mid 2 \cdot 3$, but $6 \nmid 2$ and $6 \nmid 3$. Thus the word "prime" in Euclid's lemma cannot be replaced by "composite."*

Definition 1.3 (Greatest common divisor and least common multiple)**.** *For integers a, b that are not both zero, their* greatest common divisor $\gcd(a, b)$ *is the largest positive integer dividing both. For nonzero integers a, b, their* least common multiple $\operatorname{lcm}(a, b)$ *is the least positive integer divisible by both a and b. If $\gcd(a, b) = 1$, we say that a and b are* coprime*.*

For example, $84 = 2^2 \cdot 3 \cdot 7$ and $30 = 2 \cdot 3 \cdot 5$, so $\gcd(84, 30) = 6$ and $\operatorname{lcm}(84, 30) = 420$. Bézout's identity also expresses the greatest common divisor as an integral linear combination of a and b.

Theorem 1.4 (Bézout's identity)**.** *For any $a, b \in \mathbb{Z}$ that are not both zero, there exist $u, v \in \mathbb{Z}$ such that*

$$au + bv = \gcd(a, b).$$

Moreover, $\gcd(a, b)$ is the least positive integer among all integral linear combinations $au + bv$.

Proof. Let $S = \{au + bv > 0 : u, v \in \mathbb{Z}\}$. This set is nonempty, so by the well-ordering principle it has a least element $d = au + bv$. Divide a by d and write $a = qd + r$, where $0 \le r < d$. Since $r = a - q(au + bv)$ is again an integral linear combination of a and b, minimality forces $r = 0$. Thus $d \mid a$, and similarly $d \mid b$. Conversely, every common divisor divides $au + bv = d$, so $d = \gcd(a, b)$. □

Corollary 1.5 (Euclid's lemma)**.** *If a prime p divides ab, then $p \mid a$ or $p \mid b$.*

Proof. If $p \nmid a$, then $\gcd(p, a) = 1$. Choose u, v such that $up + va = 1$ and multiply by b to obtain $upb + vab = b$. Both terms on the left are divisible by p, and hence $p \mid b$. □

1.2 Unique factorization

Once an integer has been factored into prime powers, we need uniform notation for the exponent of each prime.

Definition 1.6 (p-adic valuation and exact divisibility)**.** *Let p be prime and $n \in \mathbb{N}$. Define*

$$\nu_p(n) = \max\{k \geq 0 : p^k \mid n\}.$$

If $\nu_p(n) = \alpha$, we also write $p^\alpha \parallel n$ and say that p^α exactly divides *n. In particular, $p \nmid n$ if and only if $\nu_p(n) = 0$. In a product $\prod_{p^\alpha \parallel n}$, we include only prime divisors of n, so that $\alpha \geq 1$.*

Example 1.7. *Since $360 = 2^3 \cdot 3^2 \cdot 5$,*

$$\nu_2(360) = 3, \qquad \nu_3(360) = 2, \qquad \nu_5(360) = 1,$$

and we may write $2^3 \parallel 360$. Notice that $2^2 \mid 360$ but $2^2 \nmid\!\mid 360$.

Theorem 1.8 (Fundamental theorem of arithmetic)**.** *Every $n \geq 2$ can be written as*

$$n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}, \qquad p_1 < \cdots < p_r, \qquad \alpha_j \geq 1,$$

and this representation is unique apart from the order of the prime factors.

Proof. Existence follows by strong induction. If n is not prime, write $n = ab$ with $1 < a, b < n$ and apply the induction hypothesis to a and b. For uniqueness, if $p_1 \cdots p_k = q_1 \cdots q_\ell$, Euclid's lemma shows that p_1 equals some q_j; cancel this factor and proceed by induction. Grouping repeated primes gives uniqueness in exponential form. □

Unique factorization allows multiplicative questions about integers to be handled one prime at a time. For example, if $m = \prod_p p^{\alpha_p}$ and $n = \prod_p p^{\beta_p}$, with only finitely many nonzero exponents, then

$$\gcd(m, n) = \prod_p p^{\min(\alpha_p, \beta_p)}, \qquad \operatorname{lcm}(m, n) = \prod_p p^{\max(\alpha_p, \beta_p)}.$$

Proposition 1.9 (Basic rules for valuations)**.** *For $m, n \in \mathbb{N}$ and any prime p,*

$$\nu_p(mn) = \nu_p(m) + \nu_p(n),$$

and

$$\nu_p(\gcd(m, n)) = \min\{\nu_p(m), \nu_p(n)\}, \qquad \nu_p(\operatorname{lcm}(m, n)) = \max\{\nu_p(m), \nu_p(n)\}.$$

Consequently $mn = \gcd(m, n) \operatorname{lcm}(m, n)$.

Proof. Write $m = p^a u$ and $n = p^b v$, where $p \nmid uv$. Then $mn = p^{a+b} uv$, proving the first identity. The exponent of p in a common divisor cannot exceed the smaller of a and b, while $p^{\min(a,b)}$ does divide both m and n; the argument for the least common multiple is similar. The final identity holds at the valuation of every prime, so unique factorization shows that the two positive integers are equal. □

1.3 Congruences and the Chinese remainder theorem

Definition 1.10 (Congruence classes and reduced residue systems)**.** *Let $q \geq 1$. If $q \mid a - b$, we say that a and b are* congruent modulo q *and write $a \equiv b \pmod{q}$. The classes defined by this equivalence relation form the ring $\mathbb{Z}/q\mathbb{Z}$.*

A set containing one representative from each congruence class coprime to q is called a reduced residue system modulo q*. The invertible congruence classes form the group of units $(\mathbb{Z}/q\mathbb{Z})^\times$.*

Proposition 1.11 (Criterion for a modular inverse)**.** *A congruence class $[a] \in \mathbb{Z}/q\mathbb{Z}$ is invertible if and only if $\gcd(a, q) = 1$.*

Proof. If $(a, q) = 1$, Bézout's identity gives $au + qv = 1$, and hence $au \equiv 1 \pmod{q}$. Conversely, if $au \equiv 1 \pmod{q}$, then $au + qv = 1$ for some $v \in \mathbb{Z}$, so every common divisor of a and q must divide 1. □

Example 1.12. *The group of units modulo 12 is*

$$(\mathbb{Z}/12\mathbb{Z})^\times = \{[1], [5], [7], [11]\}.$$

Their representatives $1, 5, 7, 11$ form a reduced residue system; for example, $5^2 \equiv 1 \pmod{12}$. The class $[8]$ is not invertible because $(8, 12) = 4$.

Theorem 1.13 (Chinese remainder theorem)**.** *If the positive integers $q_1, \ldots, q_r$ are pairwise coprime, then the map*

$$\mathbb{Z}/(q_1 \cdots q_r)\mathbb{Z} \longrightarrow \prod_{j=1}^{r} \mathbb{Z}/q_j\mathbb{Z}, \qquad x \longmapsto (x \bmod q_1, \ldots, x \bmod q_r)$$

is a ring isomorphism. Thus, for arbitrary a_j, the simultaneous congruences $x \equiv a_j \pmod{q_j}$ have a unique solution modulo $q_1 \cdots q_r$.

Proof. Put $Q = \prod_j q_j$ and $Q_j = Q/q_j$. Since $(Q_j, q_j) = 1$, choose u_j such that $u_j Q_j \equiv 1 \pmod{q_j}$. Then $x = \sum_j a_j u_j Q_j$ satisfies all the congruences. If x and y are both solutions, every q_j divides $x - y$; pairwise coprimality implies $Q \mid x - y$, so the solution is unique. Compatibility with addition and multiplication follows coordinatewise. □

Corollary 1.14 (Chinese remainder decomposition of unit groups)**.** *Under the hypotheses of the Chinese remainder theorem, restricting the ring isomorphism gives a group isomorphism*

$$(\mathbb{Z}/(q_1 \cdots q_r)\mathbb{Z})^\times \cong \prod_{j=1}^{r} (\mathbb{Z}/q_j\mathbb{Z})^\times.$$

In particular, writing $\varphi(q) = \#(\mathbb{Z}/q\mathbb{Z})^\times$, we have $\varphi(q_1 \cdots q_r) = \prod_j \varphi(q_j)$.

Proof. A ring isomorphism preserves multiplicative identities, and an element is invertible if and only if every coordinate is invertible. Counting the elements on both sides proves the final identity. □

Example 1.15. *Solve $x \equiv 2 \pmod{3}$ and $x \equiv 3 \pmod{5}$. Take $Q_1 = 5$, $Q_2 = 3$, and choose $u_1 = u_2 = 2$. Then $x \equiv 2 \cdot 2 \cdot 5 + 3 \cdot 2 \cdot 3 = 38 \equiv 8 \pmod{15}$.*

1.4 Euler's totient function

Definition 1.16 (Euler's totient function)**.** *Euler's totient function $\varphi(n)$ is the number of invertible congruence classes modulo n:*

$$\varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^\times = \#\{1 \le a \le n : (a,n) = 1\}.$$

Here $a = n$ is counted only when $n = 1$; for $n > 1$ the range can equivalently be written $1 \le a < n$. In particular, $\varphi(1) = 1$: the ring modulo 1 has a single element, which is both zero and the multiplicative identity.

Proposition 1.17 (The totient of a prime power)**.** *If p is prime and $k \ge 1$, then*

$$\varphi(p^k) = p^k - p^{k-1} = p^k\left(1 - \frac{1}{p}\right).$$

Proof. Exactly p^{k-1} of the integers $1, \dots, p^k$ are divisible by p; all the others are coprime to p^k. □

Proposition 1.18. *If $n = \prod_{p \mid n} p^{\alpha_p}$, then*

$$\varphi(n) = n \prod_{p \mid n}\left(1 - \frac{1}{p}\right). \tag{1.1}$$

In particular, if $(m,n) = 1$, then $\varphi(mn) = \varphi(m)\varphi(n)$.

Proof. Among $1, \dots, n$, the integers not coprime to n are precisely those divisible by at least one prime $p \mid n$. Inclusion-exclusion shows that the proportion remaining is $\prod_{p \mid n}(1 - 1/p)$, which gives (1.1). When m and n are coprime, the set of prime divisors of mn is the disjoint union of those of m and n, so multiplicativity follows. Alternatively, count unit groups in the Chinese remainder theorem. □

Example 1.19. *Since $36 = 2^2 \cdot 3^2$,*

$$\varphi(36) = 36\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 12.$$

The Chinese remainder theorem pairs the two units modulo 4 with the six units modulo 9 to produce these twelve units. On the other hand, $\varphi(8 \cdot 4) \ne \varphi(8)\varphi(4)$, because multiplicativity applies only to coprime arguments.

Theorem 1.20 (Euler's theorem)**.** *Let $a \in \mathbb{Z}$ and $n \in \mathbb{N}$. If $(a,n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.*

Proof. Multiplication by a permutes the finite group $(\mathbb{Z}/n\mathbb{Z})^\times$. Taking the product over all elements gives $a^{\varphi(n)} \prod_u u = \prod_u u$. The product is a unit and can be cancelled. □

1.5 The language of estimates

Definition 1.21 (Asymptotic notation)**.** *We write $f(x) = O(g(x))$ or $f(x) \ll g(x)$ if, throughout the range under discussion, there is a constant $C > 0$ such that $|f(x)| \le C|g(x)|$. In $O_q(g(x))$, the constant may depend on q. We write $f = o(g)$ if for every $\varepsilon > 0$ and all sufficiently large x, $|f(x)| \le \varepsilon|g(x)|$. If $f, g > 0$ and both $f \ll g$ and $g \ll f$ hold, we write $f \asymp g$; if $f(x)/g(x) \to 1$, we write $f \sim g$, requiring $g(x) \ne 0$ for all sufficiently large x. Unless otherwise stated, limits are taken as $x \to \infty$.*

For example, $x + \sqrt{x} = x + O(\sqrt{x}) \sim x$, whereas $2x \asymp x$ but $2x \not\sim x$. Subscripts matter: $O_q(1)$ may grow rapidly with q and cannot be treated as an absolute constant when uniformity in q is required.

Lemma 1.22. *For $x \geq 1$,*

$$\sum_{n \leq x} \frac{1}{n} = \log x + \gamma + O(1/x),$$

where $\gamma = \lim_{N\to\infty}(\sum_{n\leq N} 1/n - \log N)$.

Proof. Let $a_N = \sum_{n=1}^{N} 1/n - \log N$. Integral comparison gives $a_N \geq 0$ and

$$0 \leq a_n - a_{n+1} = \log\left(1 + \frac{1}{n}\right) - \frac{1}{n+1} \leq \frac{1}{n} - \frac{1}{n+1}.$$

Thus a_N decreases to a limit γ. Summing these differences from $n = N$ to infinity gives $0 \leq a_N - \gamma \leq 1/N$. Now take $N = \lfloor x \rfloor$ and use $0 \leq \log x - \log N \leq 1/N$ and $1/N = O(1/x)$. □

Chapter Summary

- Bézout's identity writes a greatest common divisor as an integral linear combination and gives the criterion for a modular inverse.
- The fundamental theorem of arithmetic decomposes multiplicative structure one prime at a time and is the algebraic source of Euler products.
- The Chinese remainder theorem splits questions with coprime moduli into independent coordinates.
- The valuation ν_p turns multiplication, greatest common divisors, and least common multiples into addition, minima, and maxima of exponents.
- We have $\varphi(n) = n\prod_{p\mid n}(1 - 1/p)$, and φ is multiplicative.

Common Pitfalls

- A multiplicative function is required to multiply only on coprime inputs; this is not the same as complete multiplicativity.
- The implication $a \mid bc \Rightarrow a \mid b$ or $a \mid c$ generally fails when a is composite.
- The constant in O-notation may depend on fixed parameters; uniformity in a parameter must be stated explicitly.

Exercises

Exercise 1.1. Use the Euclidean algorithm to find $\gcd(252, 198)$ and express it as an integral linear combination of 252 and 198.

Exercise 1.2. Prove that $\sum_{d\mid n} \varphi(d) = n$. Hint: classify elements by their orders in the cyclic group $\mathbb{Z}/n\mathbb{Z}$, or classify the integers $1 \leq k \leq n$ by $\gcd(k, n)$.

Exercise 1.3. Prove that if f is multiplicative, then $\sum_{d\mid n} f(d) = \prod_{p^\alpha \| n}(1+f(p)+\cdots+f(p^\alpha))$.

Exercise 1.4. Let n_k be the product of the first k primes. Use (1.1) to study the behavior of $\varphi(n_k)/n_k$ and explain why it is related to the sum of the reciprocals of the primes.

Exercise 1.5. Factor 7560, find every pair (p, α) satisfying $p^\alpha \parallel 7560$ and $\alpha \geq 1$, and compute $\gcd(7560, 5400)$ and $\operatorname{lcm}(7560, 5400)$. Hint: apply Proposition 1.9 one prime at a time.

Exercise 1.6. List a reduced residue system modulo 15, match each unit with an element of $(\mathbb{Z}/3\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times$ using the Chinese remainder theorem, and thereby verify that $\varphi(15) = 8$.

Chapter 2

Arithmetic Functions and Dirichlet Convolution

Learning objectives. Master Dirichlet convolution and Möbius inversion, and compute multiplicative functions from their values at prime powers.

Prerequisites. Unique factorization and Euler's totient function from Chapter 1.

Role in the book. Convolution corresponds to double summation in Chapter 3 and to multiplication of Dirichlet series in Chapter 6.

2.1 Arithmetic functions and standard examples

Definition 2.1. *An* arithmetic function *is a map* $f : \mathbb{N} \to \mathbb{C}$. *It is* multiplicative *if* $f(1) = 1$ *and* $f(mn) = f(m)f(n)$ *whenever* $(m, n) = 1$. *It is* completely multiplicative *if* $f(1) = 1$ *and this equality holds for all* m, n.

Frequently used functions include

$$\mathbf{1}(n) = 1, \qquad \mathrm{id}_\alpha(n) = n^\alpha, \qquad \varepsilon(n) = \begin{cases} 1, & n = 1, \\ 0, & n > 1, \end{cases}$$

Here $\alpha \in \mathbb{C}$, with $n^\alpha = \exp(\alpha \log n)$ defined using the real logarithm. The functions $\mathbf{1}$ and id_α are completely multiplicative, while ε will be the identity for convolution. The divisor functions

$$\tau(n) = \sum_{d \mid n} 1, \qquad \sigma_\alpha(n) = \sum_{d \mid n} d^\alpha$$

are multiplicative but in general not completely multiplicative.

Example 2.2 (Multiplicative versus completely multiplicative)**.** *The function* n^α *is completely multiplicative because* $(mn)^\alpha = m^\alpha n^\alpha$ *for all* m, n. *Euler's totient* φ *is only multiplicative:* $\varphi(2)\varphi(4) = 2$, *whereas* $\varphi(8) = 4$. *The Möbius function is not completely multiplicative either, since* $\mu(p^2) = 0 \neq \mu(p)^2 = 1$. *Prime squares therefore provide a quick test when complete multiplicativity is in doubt.*

Proposition 2.3. *A multiplicative function is uniquely determined by its values at all prime powers* p^k. *Explicitly, if* $n = \prod_p p^{\nu_p(n)}$, *then*

$$f(n) = \prod_{p \mid n} f\big(p^{\nu_p(n)}\big).$$

Proof. Powers of distinct primes are pairwise coprime, so repeated use of multiplicativity gives the formula. Conversely, after specifying $f(1) = 1$ and all values $f(p^k)$, the formula on the right does define a multiplicative function. □

Example 2.4. *If $n = \prod_p p^{\alpha_p}$, choosing the exponent of each prime in a divisor gives*

$$\tau(n) = \prod_{p|n} (\alpha_p + 1), \qquad \sigma_\beta(n) = \prod_{p|n} \left(1 + p^\beta + \cdots + p^{\alpha_p \beta}\right).$$

2.2 Dirichlet convolution

Definition 2.5. *The* Dirichlet convolution *of arithmetic functions f and g is defined by*

$$(f * g)(n) = \sum_{d|n} f(d) g(n/d). \tag{2.1}$$

Proposition 2.6. *The arithmetic functions form a commutative ring under pointwise addition and Dirichlet convolution; the convolution identity is ε. Thus convolution is commutative, associative, and distributive over addition, and $f * \varepsilon = f$.*

Proof. Commutativity follows from the substitution $d \leftrightarrow n/d$. The essential point in associativity is to sum the same finite set in two orders:

$$((f * g) * h)(n) = \sum_{abc=n} f(a) g(b) h(c) = (f * (g * h))(n).$$

Distributivity holds term by term. Finally, $\varepsilon(d)$ is nonzero only for $d = 1$, so $(\varepsilon * f)(n) = f(n)$. □

Example 2.7. *The definitions immediately give*

$$\tau = \mathbf{1} * \mathbf{1}, \qquad \sigma_\alpha = \mathbf{1} * \mathrm{id}_\alpha .$$

For example, the positive divisors of 12 *are* $1, 2, 3, 4, 6, 12$*, and hence* $(\mathbf{1} * \mathrm{id})(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$.

Theorem 2.8 (Closure of multiplicative functions under convolution)**.** *If f and g are multiplicative, then $f * g$ is multiplicative.*

Proof. First, $(f * g)(1) = f(1) g(1) = 1$. Suppose $(m, n) = 1$. Every $d \mid mn$ can be written uniquely as $d = ab$, where $a \mid m$, $b \mid n$, and $(a, b) = 1$; also $(m/a, n/b) = 1$. Therefore

$$\begin{aligned}(f * g)(mn) &= \sum_{a|m} \sum_{b|n} f(ab) g\left(\frac{m}{a} \frac{n}{b}\right) \\ &= \left(\sum_{a|m} f(a) g(m/a)\right) \left(\sum_{b|n} f(b) g(n/b)\right).\end{aligned}$$

□

This theorem reduces global convolution to a local calculation: if f and g are multiplicative, then

$$(f * g)(p^k) = \sum_{j=0}^{k} f(p^j) g(p^{k-j}). \tag{2.2}$$

For standard functions, the local convolutions can be summarized as follows, with $k \geq 0$; μ is defined in the next section:

f and g	$(f*g)(p^k)$	Resulting function
$\mathbf{1},\mathbf{1}$	$k+1$	τ
$\mathbf{1},\mathrm{id}_\alpha$	$1+p^\alpha+\cdots+p^{k\alpha}$	σ_α
$\mu,\mathbf{1}$	1 for $k=0$, 0 for $k\geq 1$	ε
μ,id	1 for $k=0$, p^k-p^{k-1} for $k\geq 1$	φ

Thus a convolution of multiplicative functions generally needs to be computed only at prime powers.

2.3 The Möbius function

Definition 2.9. *The Möbius function is defined by*

$$\mu(n)=\begin{cases}1, & n=1,\\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes},\\ 0, & p^2\mid n \text{ for some prime } p.\end{cases}$$

It is multiplicative because coprime integers have disjoint sets of prime divisors. Its most important identity is the following.

Proposition 2.10.

$$\sum_{d\mid n}\mu(d)=\begin{cases}1, & n=1,\\ 0, & n>1,\end{cases}\qquad \text{that is,}\qquad \mu*\mathbf{1}=\varepsilon.$$

Proof. The sum on the left is multiplicative, so it is enough to check $n=p^k$. If $k\geq 1$, only $d=1,p$ have nonzero Möbius values and their sum is $1-1=0$; for $n=1$ the sum is 1. □

Proposition 2.11 (Existence and recursion for Dirichlet inverses)**.** *An arithmetic function f has an inverse under convolution if and only if $f(1)\neq 0$. If $g=f^{-1}$, then*

$$g(1)=\frac{1}{f(1)},\qquad g(n)=-\frac{1}{f(1)}\sum_{\substack{d\mid n\\ d>1}}f(d)g(n/d)\quad (n>1).$$

Proof. If $f*g=\varepsilon$, evaluation at $n=1$ gives $f(1)g(1)=1$, so $f(1)\neq 0$. Conversely, first define $g(1)$ by the displayed formula. For $n>1$, the equation

$$0=(f*g)(n)=f(1)g(n)+\sum_{\substack{d\mid n\\ d>1}}f(d)g(n/d)$$

gives the recursion. Since $n/d<n$, the right-hand side involves only values already defined. Induction produces a unique g, and the construction ensures that $f*g=\varepsilon$. □

Proposition 2.12 (The inverse of a multiplicative function is multiplicative)**.** *If f is multiplicative, then f^{-1} is multiplicative.*

Proof. Write $g=f^{-1}$. Specify $h(p^k)=g(p^k)$ at prime powers and extend h multiplicatively. Closure under convolution shows that $f*h$ is multiplicative; at every prime power, the local convolution formula gives $(f*h)(p^k)=(f*g)(p^k)=\varepsilon(p^k)$. Hence $f*h=\varepsilon$, and uniqueness in Proposition 2.11 implies $h=g$. □

Proposition 2.10 says precisely that μ is the convolution inverse of the constant function $\mathbf{1}$.

2.4 Möbius inversion

Theorem 2.13 (Möbius inversion)**.** *Let f and g be arithmetic functions. The following two identities are equivalent:*

$$g(n) = \sum_{d\mid n} f(d), \qquad f(n) = \sum_{d\mid n} \mu(d) g(n/d).$$

In convolution notation, $g = \mathbf{1} * f$ if and only if $f = \mu * g$.

Proof. If $g = \mathbf{1} * f$, then associativity and Proposition 2.10 give

$$\mu * g = \mu * (\mathbf{1} * f) = (\mu * \mathbf{1}) * f = \varepsilon * f = f.$$

The reverse direction is identical, or follows by convolving both sides with $\mathbf{1}$. □

Example 2.14 (Recovering $\varphi(n)$ from n)**.** *An exercise in Chapter 1 gives $\sum_{d\mid n} \varphi(d) = n$, or $\mathbf{1} * \varphi = \mathrm{id}$. Inversion yields*

$$\varphi(n) = \sum_{d\mid n} \mu(d)\frac{n}{d} = n \sum_{d\mid n} \frac{\mu(d)}{d}. \tag{2.3}$$

Only squarefree d contribute, so the right-hand side equals $n\prod_{p\mid n}(1 - 1/p)$, recovering the familiar product formula.

Example 2.15 (Counting primitive objects)**.** *Suppose $G(n)$ counts objects whose period divides n, while $F(n)$ counts objects whose least period is exactly n. If every object has a unique least period, then $G(n) = \sum_{d\mid n} F(d)$, so*

$$F(n) = \sum_{d\mid n} \mu(d) G(n/d).$$

There are $G(n) = 2^n$ binary strings of length n. Thus, when a starting position is distinguished, the number of binary cyclic strings with least period exactly n is $\sum_{d\mid n} \mu(d) 2^{n/d}$.

Here "primitive" means that the least period is exactly n. If cyclic shifts are further identified, each orbit of least period n has exactly n elements, so the number of primitive binary necklaces is

$$\frac{1}{n} \sum_{d\mid n} \mu(d) 2^{n/d}.$$

2.5 The von Mangoldt function and the logarithm

Definition 2.16. *The von Mangoldt function is defined by*

$$\Lambda(n) = \begin{cases} \log p, & n = p^k \ (k \geq 1), \\ 0, & \text{otherwise.} \end{cases}$$

Proposition 2.17. *For every $n \geq 1$,*

$$\log n = \sum_{d\mid n} \Lambda(d), \qquad \text{that is,} \qquad \log = \mathbf{1} * \Lambda. \tag{2.4}$$

Consequently $\Lambda = \mu * \log$.

Proof. Write $n = \prod_p p^{\alpha_p}$. The left-hand side is $\sum_p \alpha_p \log p$. On the right, only $d = p^j$ contribute; for a fixed $p \mid n$, the indices $j = 1, \dots, \alpha_p$ contribute a total of $\alpha_p \log p$. The inversion statement follows from Theorem 2.13. □

The function $\log n$ is easy to sum, while $\Lambda(n)$ is concentrated on prime powers. Identity (2.4) will be used in prime estimates and in the expansion of $-\zeta'/\zeta$.

Example 2.18 (Computing five functions)**.** *Take $n = 12 = 2^2 \cdot 3$ and $m = 8 = 2^3$. Prime-by-prime calculation gives*

Input	μ	φ	τ	σ_1	Λ
12	0	4	6	28	0
8	0	4	4	15	$\log 2$

We have $\Lambda(8) = \log 2$ and $\Lambda(12) = 0$ because the former is a prime power. Also $\mu(12) = \mu(8) = 0$ because both integers have a square factor.

Chapter Summary

- A multiplicative function is determined by its local data at prime powers.
- Dirichlet convolution turns "summing over all divisors" into multiplication, and multiplicative functions are closed under convolution.
- Since $\mu * \mathbf{1} = \varepsilon$, Möbius inversion is cancellation under convolution.
- The condition $f(1) \neq 0$ is exactly what guarantees a Dirichlet inverse; the inverse of a multiplicative function is again multiplicative.
- The identities $\mathbf{1} * \varphi = \mathrm{id}$ and $\mathbf{1} * \Lambda = \log$ will be used repeatedly.

Common Pitfalls

- Confusing $(f * g)(n)$ with the pointwise product $f(n)g(n)$.
- Forgetting that the definition of a multiplicative function includes $f(1) = 1$; under our convention, the zero function is not multiplicative.
- Confusing $g(d)$ with $g(n/d)$ in Möbius inversion. One form can be changed into the other, but the argument of μ must be changed at the same time.

Exercises

Exercise 2.1. List all divisors of 12, compute $(\mu * \mathbf{1})(12)$ and $(\mu * \mathrm{id})(12)$ directly, and compare the answers with $\varepsilon(12)$ and $\varphi(12)$.

Exercise 2.2. Prove that $\mu^2(n)$ is the indicator function of the squarefree integers, and prove that $\mu^2(n) = \sum_{d^2 \mid n} \mu(d)$.

Exercise 2.3. Let f be multiplicative with $f(1) = 1$. Prove that its convolution inverse f^{-1} is also multiplicative, and write down $f^{-1}(p)$ and $f^{-1}(p^2)$.

Exercise 2.4. Prove that $\Lambda(n) = -\sum_{d \mid n} \mu(d) \log d$ and explain why this is equivalent to $\Lambda = \mu * \log$.

Exercise 2.5. For $k, n \in \mathbb{N}$, let $J_k(n)$ be the number of k-tuples modulo n that are jointly coprime to n, meaning that $\gcd(a_1, \dots, a_k, n) = 1$. Prove that $J_k(n) = n^k \prod_{p \mid n} (1 - p^{-k})$.

Exercise 2.6. Use Proposition 2.11 to compute the convolution inverse of $f(n) = n$ at $1, p, p^2, pq$, where $p \neq q$ are prime, and compare the answers with $\mu(n)n$.

Exercise 2.7. Let f, g be multiplicative and put $F_p(T) = \sum_{k\geq 0} f(p^k)T^k$ and $G_p(T) = \sum_{k\geq 0} g(p^k)T^k$. Prove that convolution at a prime p corresponds to multiplication of formal power series, and use this perspective to rederive $\mathbf{1}^{-1} = \mu$. Hint: compare the coefficients of T^k.

Chapter 3

Average Orders and Asymptotic Formulas

Learning objectives. Master Abel summation and Dirichlet's hyperbola method, and compute the average orders of several standard arithmetic functions.

Prerequisites. Convolution and Möbius inversion from Chapter 2, and integration by parts from calculus.

Role in the book. This chapter converts convolution identities into asymptotic formulas and establishes the Abel summation formula used later.

3.1 Asymptotic formulas and average orders

Definition 3.1 (Summatory function, average order, and error term)**.** *For an arithmetic function f, its* summatory function *is*

$$F(x) = \sum_{n \le x} f(n).$$

If there is a simpler function g such that

$$\sum_{n \le x} f(n) \sim \sum_{n \le x} g(n),$$

we call g an average order *of f. In an expression $F(x) = M(x) + E(x)$, the function M is the main term and E the error term. The main term gives an asymptotic equivalence $F(x) \sim M(x)$ only when $E(x) = o(M(x))$.*

An average order is not unique. If g is an average order of f and $\sum_{n \le x} h(n) = o(\sum_{n \le x} g(n))$, then $g + h$ is also an average order. We therefore usually give summation formulas with explicit errors instead of merely naming an average order. Also, $O_q(E(x))$ means that the implied constant may depend on the fixed parameter q; if q varies with x, uniformity must be checked again.

Lemma 3.2 (Basic estimates for floors and harmonic sums)**.** *For $x, y \ge 1$,*

$$\lfloor x \rfloor = x + O(1), \qquad \sum_{n \le y} \left\lfloor \frac{x}{n} \right\rfloor = x \sum_{n \le y} \frac{1}{n} + O(y).$$

If $y \ge 2$, the second identity can also be written as

$$x \log y + \gamma x + O(x/y + y).$$

All implied constants are absolute.

Proof. The first identity follows from $0 \le x - \lfloor x \rfloor < 1$. Apply it term by term to x/n and sum to obtain the second identity, then insert the harmonic-sum expansion from Chapter 1. □

Example 3.3. *The function $\tau(n)$ is not pointwise close to $\log n$: at primes, $\tau(p) = 2$, while some integers have many divisors. This chapter will nevertheless prove that*

$$\sum_{n \le x} \tau(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x}),$$

so $\log n + 2\gamma$ is a natural model in the summatory sense.

3.2 Abel summation

Directly estimating a_n is often difficult even when the partial sums $A(x)$ exhibit clear cancellation. Abel summation moves a difference from the sequence to a smooth weight. It is the discrete analogue of integration by parts.

Theorem 3.4 (Abel summation formula)**.** *Let (a_n) be a complex sequence and $A(x) = \sum_{n \le x} a_n$. If $f \in C^1([y, x])$ and $1 \le y < x$, then*

$$\sum_{y < n \le x} a_n f(n) = A(x)f(x) - A(y)f(y) - \int_y^x A(t)f'(t)\,\mathrm{d}t. \tag{3.1}$$

The convention at the endpoints is the same as in the definition of A.

Proof. For $y \le t \le x$, we have $A(t) = A(y) + \sum_{y < n \le x} a_n \mathbf{1}_{n \le t}$. Since this sum is finite, we can integrate term by term:

$$\begin{aligned}
\int_y^x A(t)f'(t)\,\mathrm{d}t &= A(y)\big(f(x) - f(y)\big) + \sum_{y < n \le x} a_n \int_n^x f'(t)\,\mathrm{d}t \\
&= A(y)\big(f(x) - f(y)\big) + \sum_{y < n \le x} a_n \big(f(x) - f(n)\big) \\
&= A(x)f(x) - A(y)f(y) - \sum_{y < n \le x} a_n f(n).
\end{aligned}$$

Rearranging proves the formula. This calculation also handles integer endpoints: the term $n = y$ is already included in $A(y)$, while the integral for $n = x$ is zero. In particular, taking $y = 1$ and adding $a_1 f(1)$ gives

$$\sum_{n \le x} a_n f(n) = A(x)f(x) - \int_1^x A(t)f'(t)\,\mathrm{d}t.$$

□

Proposition 3.5 (Abel summation on an infinite interval)**.** *Let $y \ge 1$, $A(x) = \sum_{n \le x} a_n$, and $f \in C^1([y, \infty))$. If*

$$A(x)f(x) \longrightarrow 0, \qquad \int_y^\infty |A(t)f'(t)|\,\mathrm{d}t < \infty,$$

then $\sum_{n > y} a_n f(n)$ converges and

$$\sum_{n > y} a_n f(n) = -A(y)f(y) - \int_y^\infty A(t)f'(t)\,\mathrm{d}t.$$

Proof. Use x as the upper endpoint in Theorem 3.4. The assumptions make the boundary term tend to zero and the integral converge absolutely. Letting $x \to \infty$ proves the claim. □

Proposition 3.6 (Transferring main terms and errors)**.** *Suppose $A(x) = cx + E(x)$ and $f \in C^1([1, x])$. With the endpoint convention that includes $n = 1$,*

$$\sum_{n \le x} a_n f(n) = c\left(f(1) + \int_1^x f(t)\,\mathrm{d}t\right) + E(x)f(x) - \int_1^x E(t)f'(t)\,\mathrm{d}t.$$

Thus the error consists of a boundary term and a weighted integral, and both must be estimated.

Proof. Start from the finite formula $\sum_{n \le x} a_n f(n) = A(x)f(x) - \int_1^x A(t)f'(t)\,\mathrm{d}t$ and substitute $A(t) = ct + E(t)$. For the main term, use

$$xf(x) - \int_1^x tf'(t)\,\mathrm{d}t = f(1) + \int_1^x f(t)\,\mathrm{d}t.$$

□

Corollary 3.7. *If $A(x) = cx + O(x^\theta)$ with $0 \le \theta < 1$, put $E(t) = A(t) - ct$. Then the constant*

$$C = c + \int_1^\infty \frac{E(t)}{t^2}\,\mathrm{d}t$$

exists, and for $x \ge 1$,

$$\sum_{n \le x} \frac{a_n}{n} = c\log x + C + O(x^{\theta-1}).$$

The implied constant may depend on θ and on the constant in the partial-sum estimate.

Proof. Take $f(t) = 1/t$ in Proposition 3.6 to obtain

$$\sum_{n \le x} \frac{a_n}{n} = c\log x + c + \frac{E(x)}{x} + \int_1^x \frac{E(t)}{t^2}\,\mathrm{d}t.$$

Since $E(t) \ll t^\theta$ and $\theta < 1$, the integral defining C converges absolutely. After subtracting $c\log x + C$, the remainder is $E(x)/x - \int_x^\infty E(t)t^{-2}\,\mathrm{d}t = O(x^{\theta-1})$. □

Example 3.8 (Power sums)**.** *Take $a_n = 1$ and $A(t) = \lfloor t \rfloor = t + O(1)$. For fixed $\alpha > -1$, Abel summation gives*

$$\sum_{n \le x} n^\alpha = \frac{x^{\alpha+1}}{\alpha+1} + O_\alpha(x^\alpha + 1).$$

When $\alpha \ge 0$, the error can be written $O_\alpha(x^\alpha)$; the slightly weaker uniform form also covers the endpoint behavior for $-1 < \alpha < 0$.

3.3 The hyperbola method

If $h = f * g$, then

$$\sum_{n \le x} h(n) = \sum_{ab \le x} f(a)g(b).$$

The expression on the right is a weighted lattice-point sum beneath the hyperbola $ab = x$. Summing directly over a gives

$$\sum_{n\le x}(f*g)(n) = \sum_{a\le x} f(a)G(x/a), \qquad G(u) = \sum_{b\le u} g(b). \tag{3.2}$$

To keep the lengths of both outer sums below roughly $\sqrt{x}$, we split the region symmetrically.

If the estimate on one side is stronger for $a \le u$ and that on the other for $b \le v$, choose u and v, subject to $uv = x$, so that the two errors are of comparable size. In the symmetric case, the natural choice is $u = v = \sqrt{x}$.

Theorem 3.9 (Dirichlet hyperbola identity)**.** *Let $F(u) = \sum_{n\le u} f(n)$ and $G(u) = \sum_{n\le u} g(n)$. If $u, v \ge 1$ and $uv = x$, then*

$$\sum_{n\le x}(f*g)(n) = \sum_{a\le u} f(a)G(x/a) + \sum_{b\le v} g(b)F(x/b) - F(u)G(v). \tag{3.3}$$

Proof. Every point in the region $ab \le x$ satisfies $a \le u$ or $b \le v$; otherwise $ab > uv = x$. The first two sums count these two subregions. Their intersection, the rectangle $a \le u$, $b \le v$, is counted twice, so $F(u)G(v)$ must be subtracted. This is simply inclusion-exclusion on a finite set and involves no interchange of limits. □

3.4 The average order of the divisor function

Because $\tau = \mathbf{1} * \mathbf{1}$,

$$D(x) := \sum_{n\le x}\tau(n) = \#\{(a,b)\in\mathbb{N}^2 : ab \le x\}.$$

Theorem 3.10 (Basic estimate in the Dirichlet divisor problem)**.** *For $x \ge 2$,*

$$D(x) = x\log x + (2\gamma - 1)x + O(\sqrt{x}).$$

Proof. Put $y = \lfloor\sqrt{x}\rfloor$. Since $(y+1)^2 > x$, every lattice point with $ab \le x$ satisfies $a \le y$ or $b \le y$. Since $y^2 \le x$, the intersection is the entire $y \times y$ rectangle. Thus, even when $y^2 \ne x$, the symmetric decomposition gives the exact identity

$$D(x) = 2\sum_{n\le y}\left\lfloor\frac{x}{n}\right\rfloor - y^2.$$

Using $\lfloor x/n\rfloor = x/n + O(1)$ and the harmonic-sum expansion, we obtain

$$D(x) = 2x\left(\log y + \gamma + O(1/y)\right) - y^2 + O(y).$$

Since $y = \sqrt{x} + O(1)$, we have $\log y = \frac{1}{2}\log x + O(x^{-1/2})$ and $y^2 = x + O(\sqrt{x})$. Substitution proves the result. □

The error $O(\sqrt{x})$ comes from estimating the rounding error crudely, one point at a time, along the boundary of the hyperbola. Exploiting cancellation in $\{x/n\}$ improves the error; this is the classical Dirichlet divisor problem. We will not seek the best exponent here.

3.5 Squarefree integers and another average-order example

The square $\mu^2(n)$ of the Möbius function is the indicator of the squarefree integers. The identity from Chapter 2,

$$\mu^2(n) = \sum_{d^2 \mid n} \mu(d),$$

converts the square-divisibility condition into a finite sum that can be rearranged. For a direct verification, only squarefree d can contribute. For each prime with $p^2 \mid n$, we may choose whether or not to include p in d, so the right-hand side is $\prod_{p^2 \mid n}(1+\mu(p))$. This is the empty product 1 when n is squarefree and contains a factor $1-1=0$ otherwise, exactly as required for $\mu^2(n)$.

First we explain the constant used below. Write $\zeta(2) = \sum_{m \geq 1} m^{-2}$ and use the classical evaluation $\zeta(2) = \pi^2/6$; see [1, p. 266]. Since $|\mu(d)| \leq 1$, the following double series converges absolutely, so the convolution identity $\mu * \mathbf{1} = \varepsilon$ gives

$$\left(\sum_{d \geq 1} \frac{\mu(d)}{d^2} \right) \zeta(2) = \sum_{n \geq 1} \frac{1}{n^2} \sum_{d \mid n} \mu(d) = 1.$$

Thus $\sum_{d \geq 1} \mu(d) d^{-2} = 1/\zeta(2) = 6/\pi^2$.

Theorem 3.11 (Density of squarefree integers)**.** *For $x \geq 2$,*

$$\sum_{n \leq x} \mu^2(n) = \frac{6}{\pi^2} x + O(\sqrt{x}).$$

Proof. Rearranging a finite sum gives

$$\begin{aligned} \sum_{n \leq x} \mu^2(n) &= \sum_{d \leq \sqrt{x}} \mu(d) \left\lfloor \frac{x}{d^2} \right\rfloor \\ &= x \sum_{d \leq \sqrt{x}} \frac{\mu(d)}{d^2} + O(\sqrt{x}). \end{aligned}$$

Insert the identity $\sum_{d \geq 1} \mu(d) d^{-2} = 6/\pi^2$ just proved. The omitted tail is $O(\sum_{d > \sqrt{x}} d^{-2}) = O(x^{-1/2})$, and after multiplication by x it is still $O(\sqrt{x})$. □

The sum-of-divisors function $\sigma_1(n) = \sum_{d \mid n} d$ provides another complete example. Unlike τ, the lattice point (d, m) is weighted by d, so the main term has quadratic size.

Theorem 3.12 (Average order of the sum-of-divisors function)**.** *For $x \geq 2$,*

$$\sum_{n \leq x} \sigma_1(n) = \frac{\pi^2}{12} x^2 + O(x \log x).$$

Proof. Since $\sigma_1 = \mathrm{id} * \mathbf{1}$, rearranging by $n = dm$ gives

$$\begin{aligned} \sum_{n \leq x} \sigma_1(n) &= \sum_{m \leq x} \sum_{d \leq x/m} d \\ &= \frac{x^2}{2} \sum_{m \leq x} \frac{1}{m^2} + O\left(x \sum_{m \leq x} \frac{1}{m} \right). \end{aligned}$$

Here we used $\sum_{d \leq y} d = y^2/2 + O(y)$. Since $\sum_{m \leq x} m^{-2} = \zeta(2) + O(1/x) = \pi^2/6 + O(1/x)$, the conclusion follows. □

3.6 The average order of Euler's totient

By (2.3),

$$\sum_{n\le x}\varphi(n)=\sum_{n\le x}n\sum_{d\mid n}\frac{\mu(d)}{d}=\sum_{d\le x}\mu(d)\sum_{m\le x/d}m.$$

The sums are finite, so the rearrangement is unconditional.

Theorem 3.13. *For $x\ge 2$,*

$$\sum_{n\le x}\varphi(n)=\frac{3}{\pi^2}x^2+O(x\log x).$$

Proof. Using $\sum_{m\le y}m=\frac{1}{2}y^2+O(y)$, we obtain

$$\sum_{n\le x}\varphi(n)=\frac{x^2}{2}\sum_{d\le x}\frac{\mu(d)}{d^2}+O\left(x\sum_{d\le x}\frac{1}{d}\right).$$

The identity already proved in this chapter gives

$$\sum_{d=1}^{\infty}\frac{\mu(d)}{d^2}=\frac{1}{\zeta(2)}=\frac{6}{\pi^2}.$$

The absolute value of the truncated tail is at most $\sum_{d>x}d^{-2}=O(1/x)$ and contributes $O(x)$ to the main term; the harmonic sum is $O(\log x)$. Combining these estimates proves the theorem. □

Geometrically, a positive lattice point (a,b) is visible from the origin if and only if $(a,b)=1$. For an integer $N\ge 2$, counting on both sides of the diagonal gives

$$\#\{(a,b):1\le a,b\le N,\ (a,b)=1\}=2\sum_{n\le N}\varphi(n)-1=\frac{6}{\pi^2}N^2+O(N\log N).$$

We subtract 1 because $(1,1)$ is the only coprime point on the diagonal and was counted twice. Thus two integers chosen independently and uniformly from $\{1,\dots,N\}$ are coprime with probability tending to $6/\pi^2$.

3.7 Returning from weighted sums to counting functions

Abel summation can both introduce and remove weights. If $\vartheta(x)=\sum_{p\le x}\log p$, take $a_n=\log n$ for prime n and $a_n=0$ otherwise, and take $f(t)=1/\log t$. For $x\ge 2$ this gives the exact relation

$$\pi(x)=\frac{\vartheta(x)}{\log x}+\int_2^x\frac{\vartheta(t)}{t\log^2 t}\,\mathrm{d}t. \tag{3.4}$$

More explicitly, apply Abel summation first to $2<n\le x$, then add the contribution 1 of the prime 2. This cancels the lower-end term $-\vartheta(2)/\log 2=-1$, leaving no extra constant. If $\vartheta(x)\sim x$, the first term is asymptotic to $x/\log x$. Since $\vartheta(t)\ll t$, splitting the integral at $\sqrt{x}$ gives

$$\int_2^x\frac{\vartheta(t)}{t\log^2 t}\,\mathrm{d}t\ll\int_2^x\frac{\mathrm{d}t}{\log^2 t}\ll\sqrt{x}+\frac{x}{\log^2 x}\ll\frac{x}{\log^2 x}.$$

Hence $\pi(x) \sim x/\log x$. Conversely, if $\pi(x) \sim x/\log x$, the same formula gives

$$\vartheta(x) = \pi(x)\log x - \int_2^x \frac{\pi(t)}{t}\,\mathrm{d}t.$$

Now $\pi(t) \ll t/\log t$. Splitting the integral at $\sqrt{x}$, we obtain

$$\int_2^x \frac{\pi(t)}{t}\,\mathrm{d}t \ll \int_2^x \frac{\mathrm{d}t}{\log t} \ll \sqrt{x} + \frac{x}{\log x} = o(x).$$

Since $\pi(x)\log x \sim x$, it follows that $\vartheta(x) \sim x$. Thus the π- and ϑ-forms of the prime number theorem are equivalent.

Chapter Summary

- Abel summation is discrete integration by parts; it transfers information about partial sums $A(x)$ to weighted sums.
- The hyperbola method exploits the lattice-point region associated with convolution and balances errors between two variables.
- The average order of τ is $\log n + 2\gamma$, and its basic error comes from the boundary of the hyperbola.
- We have $\sum_{n\le x}\varphi(n) \sim 3x^2/\pi^2$, with the constant arising from $1/\zeta(2)$.
- Abel summation converts $\vartheta(x) \sim x$ and $\pi(x) \sim x/\log x$ into one another.

Common Pitfalls

- Rearranging a doubly infinite series as though it were finite; all rearrangements in this chapter are first made in the finite region $ab \le x$.
- Summing y separate $O(1)$ errors and still writing $O(1)$ instead of the usual $O(y)$.
- Forgetting that an outside coefficient may magnify the error in $\lfloor x\rfloor = x + O(1)$.
- Using only the boundary term when deriving the prime number theorem from $\vartheta(x) \sim x$, without estimating the Abel integral.

Exercises

Exercise 3.1. Use Abel summation to prove that, for fixed $\alpha > -1$, $\sum_{n\le x} n^\alpha \sim x^{\alpha+1}/(\alpha+1)$.

Exercise 3.2. Deduce $D(x) = \sum_{n\le x}\lfloor x/n\rfloor$ from (3.2), and explain its relation to counting divisor pairs (a,b).

Exercise 3.3. Check the proof of Theorem 3.11 line by line, and explain why merely knowing $\sum_{d\le y}\mu(d) = O(y)$ is still insufficient to improve the $O(\sqrt{x})$ error.

Exercise 3.4. Suppose $a_n \ge 0$ and $A(x) = \sum_{n\le x} a_n = O(x)$. Prove that $\sum_{n\le x} a_n/n = O(\log x)$ for $x \ge 2$. Where is nonnegativity used?

Exercise 3.5. In Theorem 3.9, take $u = x^\alpha$ and $v = x^{1-\alpha}$ with $0 \le \alpha \le 1$. Do the bounds $F(t) \ll t^\rho$ and $G(t) \ll t^\eta$ alone determine an optimal α? If two error terms additionally satisfy $E_1 \ll x^A u^B$ and $E_2 \ll x^C u^{-D}$ with $B, D > 0$, find the α that minimizes the larger exponent in these two bounds, and explain how to check the contribution of $F(u)G(v)$.

Exercise 3.6. For fixed $\alpha > 0$, use $\sigma_\alpha = \mathrm{id}_\alpha * \mathbf{1}$ to predict and prove the main term of $\sum_{n\le x}\sigma_\alpha(n)$. Hint: treat separately the accumulation of the power-sum error when $\alpha < 1$, $\alpha = 1$, and $\alpha > 1$.

Chapter 4

Elementary Results on the Distribution of Primes

Learning objectives. Compare π, ϑ, and ψ, prove Chebyshev's estimates using binomial coefficients, and determine the main term of the sum of reciprocal primes.

Prerequisites. Unique factorization, the von Mangoldt function, Abel summation, and basic asymptotic notation.

Role in the book. This chapter proves $\pi(x) \asymp x/\log x$; the asymptotic equivalence will be established in Chapter 5.

4.1 How many primes are there? Two first steps

Theorem 4.1 (Euclid)**.** *There are infinitely many primes.*

Proof. If the only primes were $p_1, \dots, p_r$, then $N = p_1 \cdots p_r + 1 > 1$ would have a prime divisor q. But division of N by any p_j leaves remainder 1, so q is not on the list, a contradiction. □

The contradiction above does not yet quantify the growth of the number of primes. We next prove the stronger fact that the sum of their reciprocals diverges.

Theorem 4.2 (Euler)**.**

$$\sum_{p} \frac{1}{p} = \infty.$$

Proof. Suppose instead that $\sum_p 1/p$ converges. For sufficiently large p, $-\log(1 - 1/p) \le 1/p + 1/p^2$, so the infinite product $\prod_p (1 - 1/p)^{-1}$ converges to a finite value. On the other hand, unique factorization and expansion of a finite product give, for every N,

$$\prod_{p \le N} \left(1 - \frac{1}{p}\right)^{-1} = \sum_{\substack{n \ge 1 \\ p \mid n \Rightarrow p \le N}} \frac{1}{n} \ge \sum_{n \le N} \frac{1}{n}.$$

The right-hand side diverges with N, a contradiction. Nonnegative terms justify the expansion of the finite product; no conditional rearrangement is involved. □

4.2 Three counting functions

Definition 4.3 (Prime-counting and Chebyshev functions)**.** *For $x \ge 1$, define*

$$\pi(x) = \sum_{p \le x} 1, \qquad \vartheta(x) = \sum_{p \le x} \log p, \qquad \psi(x) = \sum_{n \le x} \Lambda(n) = \sum_{p^k \le x} \log p.$$

Here p always ranges over primes and $k \geq 1$. The function ϑ weights only primes, whereas ψ also weights prime powers; all three are right-continuous step functions.

Example 4.4. *For $x = 10$,*

$$\pi(10) = 4, \qquad \vartheta(10) = \log 2 + \log 3 + \log 5 + \log 7,$$

while $\psi(10)$ also counts $2^2, 2^3, 3^2$, and therefore $\psi(10) = 3\log 2 + 2\log 3 + \log 5 + \log 7$. The contribution of prime powers is not small over short ranges, but it is negligible on an asymptotic scale relative to x.

Proposition 4.5 (Exact relation between the Chebyshev functions)**.** *For $x \geq 1$,*

$$\psi(x) = \sum_{k \geq 1} \vartheta(x^{1/k}), \tag{4.1}$$

where all terms with $k > \log x / \log 2$ vanish.

Proof. On the right, a prime p contributes $\log p$ once for every k such that $p^k \leq x$. Its total contribution is therefore exactly the part of $\Lambda(p) + \Lambda(p^2) + \cdots$ lying below x. Since the smallest prime is 2, there are no nonzero terms once $2^k > x$. □

Proposition 4.6. *For $x \geq 2$,*

$$0 \leq \psi(x) - \vartheta(x) \ll \sqrt{x} \log^2 x.$$

Proof. The crude estimate $\vartheta(y) \leq \pi(y) \log y \leq y \log y$ suffices. In (4.1), we have $x^{1/k} \leq \sqrt{x}$ for $k \geq 2$, and at most $\log x / \log 2$ terms are nonzero. Their total is therefore $O(\sqrt{x} \log^2 x)$. □

This error is already $o(x)$. Thus $\psi(x) \sim x$ if and only if $\vartheta(x) \sim x$, and by (3.4) both are equivalent to the prime number theorem.

Proposition 4.7 (Least-common-multiple interpretation)**.** *For $x \geq 1$,*

$$\psi(x) = \log \operatorname{lcm}(1, 2, \ldots, \lfloor x \rfloor). \tag{4.2}$$

Proof. The exponent of p in the least common multiple is the largest k for which $p^k \leq x$. After taking logarithms, the contribution of a fixed p is $k \log p$, or one copy of $\log p$ from each of $p, p^2, \ldots, p^k$. □

Proposition 4.8 (Equivalence of three forms of the prime number theorem)**.** *The following statements are equivalent:*

$$\psi(x) \sim x, \qquad \vartheta(x) \sim x, \qquad \pi(x) \sim \frac{x}{\log x}.$$

Proof. Proposition 4.6 gives $\psi(x) - \vartheta(x) = o(x)$, so the first two statements are equivalent. The Abel relation (3.4) shows that $\vartheta(x) \sim x$ implies $\pi(x) \sim x / \log x$. Conversely, if $\pi(x) \sim x / \log x$, then

$$\vartheta(x) = \pi(x) \log x - \int_2^x \frac{\pi(t)}{t} \, \mathrm{d}t.$$

The first term is asymptotic to x. The assumption also gives $\pi(t) \ll t / \log t$ for $t \geq 2$. For $x \geq 4$, estimating separately on $[2, \sqrt{x}]$ and $[\sqrt{x}, x]$ gives

$$\int_2^x \frac{\mathrm{d}t}{\log t} \leq \frac{\sqrt{x}}{\log 2} + \frac{2x}{\log x} = O(x / \log x).$$

Thus the original integral is $O(x / \log x) = o(x)$, and $\vartheta(x) \sim x$ follows. □

4.3 The binomial-coefficient method

The exponent of a prime in a binomial coefficient can be found from its exponent in factorials. We first prove the needed formula of Legendre.

Lemma 4.9 (Legendre's formula)**.** *If p is prime and $n \geq 1$ is an integer, then*

$$\nu_p(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor .$$

The sum on the right is in fact finite.

Proof. Among $1, \ldots, n$, there are $\lfloor n/p \rfloor$ integers contributing at least one factor of p, $\lfloor n/p^2 \rfloor$ contributing one additional factor, and so on. Layer-by-layer counting accumulates exactly $\nu_p(m)$ from each integer m. Every term vanishes once $p^j > n$. □

Example 4.10.

$$\nu_2(10!) = \left\lfloor \frac{10}{2} \right\rfloor + \left\lfloor \frac{10}{4} \right\rfloor + \left\lfloor \frac{10}{8} \right\rfloor = 5 + 2 + 1 = 8.$$

Directly, $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$*, in agreement with the formula.*

The central binomial coefficient has both a combinatorial estimate and a prime factorization:

$$\frac{4^n}{2n+1} \leq \binom{2n}{n} \leq 4^n. \tag{4.3}$$

The upper bound comes from the binomial expansion. For the lower bound, the largest among the $2n+1$ terms in $\sum_{k=0}^{2n} \binom{2n}{k} = 4^n$ is the central term.

Lemma 4.11. *For every integer $n \geq 1$,*

$$\vartheta(2n) - \vartheta(n) \leq 2n \log 2.$$

Proof. Every prime $n < p \leq 2n$ occurs once in $(2n)!$ and in neither copy of $n!$, so $p \mid \binom{2n}{n}$. These primes are distinct, and hence

$$\exp\bigl(\vartheta(2n) - \vartheta(n)\bigr) \leq \binom{2n}{n} \leq 4^n.$$

Taking logarithms proves the assertion. □

Proposition 4.12 (Chebyshev upper bound)**.** *For $x \geq 2$, we have $\vartheta(x) \leq 4(\log 2)x$, and consequently $\psi(x) \ll x$.*

Proof. First let $x = 2^m$ and sum Lemma 4.11 from 2^{j-1} to 2^j:

$$\vartheta(2^m) \leq \sum_{j=1}^{m} 2^j \log 2 < 2^{m+1} \log 2.$$

For general x, choose the least $2^m \geq x$. Then $2^m < 2x$, and monotonicity gives the stated upper bound.

Now (4.1) and $\vartheta(y) \ll y$ yield

$$\psi(x) \ll x + \sum_{2 \leq k \leq \log_2 x} x^{1/k} \ll x + \sqrt{x} \log x \ll x.$$

The bounded range of small x can be absorbed into the constant. □

Proposition 4.13 (Chebyshev lower bound)**.** *There is an absolute constant $c > 0$ such that $\psi(x) \geq cx$ for all sufficiently large x.*

Proof. For each prime p, Lemma 4.9 gives

$$\nu_p\binom{2n}{n} = \sum_{j\geq 1}\left(\left\lfloor\frac{2n}{p^j}\right\rfloor - 2\left\lfloor\frac{n}{p^j}\right\rfloor\right).$$

Each summand is 0 or 1, and can be nonzero only when $p^j \leq 2n$. Therefore

$$\log\binom{2n}{n} \leq \sum_{p^j\leq 2n}\log p = \psi(2n).$$

Together with (4.3), this gives

$$\psi(2n) \geq 2n\log 2 - \log(2n+1).$$

For arbitrary large x, take $n = \lfloor x/2\rfloor$ and use monotonicity of ψ. After discarding finitely many x, any fixed $c < \log 2$ will do. □

Theorem 4.14 (Chebyshev's estimates)**.** *There are absolute constants $0 < c < C$ such that, for all sufficiently large x,*

$$cx \leq \vartheta(x) \leq Cx, \qquad c\frac{x}{\log x} \leq \pi(x) \leq C\frac{x}{\log x}.$$

In brief, $\vartheta(x) \asymp x$ and $\pi(x) \asymp x/\log x$.

Proof. Propositions 4.13 and 4.6 give $\vartheta(x) \gg x$, while Proposition 4.12 gives the upper bound.

Since $\vartheta(x) \leq \pi(x)\log x$, the lower bound for π follows immediately. For the upper bound, split the primes at $\sqrt{x}$:

$$\pi(x) \leq \sqrt{x} + \frac{\vartheta(x) - \vartheta(\sqrt{x})}{\log\sqrt{x}} \leq \sqrt{x} + \frac{2\vartheta(x)}{\log x} \ll \frac{x}{\log x}.$$

□

4.4 Two estimates of Mertens

The Chebyshev upper bound and the convolution identity $\log = \mathbf{1} * \Lambda$ already determine the main term of the sum of reciprocal primes. Only the elementary tools developed above are needed.

Theorem 4.15 (Mertens estimates)**.** *For real $x \geq 2$,*

$$\sum_{n\leq x}\frac{\Lambda(n)}{n} = \log x + O(1), \qquad \sum_{p\leq x}\frac{\log p}{p} = \log x + O(1).$$

There is a real constant B_1 such that

$$\sum_{p\leq x}\frac{1}{p} = \log\log x + B_1 + O\left(\frac{1}{\log x}\right).$$

Proof. First let $N \geq 2$ be an integer. Sum $\log n = \sum_{d \mid n} \Lambda(d)$ and interchange the finite sums to obtain

$$\log(N!) = \sum_{d \leq N} \Lambda(d) \left\lfloor \frac{N}{d} \right\rfloor = N \sum_{d \leq N} \frac{\Lambda(d)}{d} + O(\psi(N)).$$

Proposition 4.12 gives $\psi(N) = O(N)$. Since $\log t$ is increasing,

$$\int_1^N \log t \, \mathrm{d}t \leq \log(N!) \leq \int_1^N \log t \, \mathrm{d}t + \log N.$$

Thus $\log(N!) = N \log N - N + O(\log N)$. Division by N proves the first estimate; taking $N = \lfloor x \rfloor$ extends it to real x. Higher prime powers contribute a bounded amount:

$$\sum_p \sum_{k \geq 2} \frac{\log p}{p^k} = \sum_p \frac{\log p}{p(p-1)} \leq 2 \sum_{n \geq 2} \frac{\log n}{n^2} < \infty.$$

Removing them proves the second estimate.

Write $A(x) = \sum_{p \leq x} (\log p)/p = \log x + E(x)$, where $E(x) = O(1)$. Abel summation with weight $1/\log t$ gives

$$\begin{aligned} \sum_{p \leq x} \frac{1}{p} &= \frac{A(x)}{\log x} + \int_2^x \frac{A(t)}{t \log^2 t} \, \mathrm{d}t \\ &= \log \log x + 1 - \log \log 2 + \frac{E(x)}{\log x} + \int_2^x \frac{E(t)}{t \log^2 t} \, \mathrm{d}t. \end{aligned}$$

Because E is bounded, the following integral converges absolutely and defines

$$B_1 = 1 - \log \log 2 + \int_2^\infty \frac{E(t)}{t \log^2 t} \, \mathrm{d}t.$$

Its tail is $O(1/\log x)$, which proves the last formula. □

Remark 4.16. The number B_1 is the Mertens constant for the sum of reciprocal primes. This formula sharpens the divergence in Theorem 4.2 to a vanishing error, without using the prime number theorem. Determining the main term of a weighted average does not by itself justify removing the weight to obtain an asymptotic formula for $\pi(x)$.

4.5 How far are we from the prime number theorem?

Corollary 4.17. *If* $\lim_{x \to \infty} \pi(x) \log x / x$ *exists, it must equal* 1.

Proof. Call the limit ℓ; Chebyshev's estimates give $0 < \ell < \infty$. Abel summation yields

$$\sum_{p \leq x} \frac{1}{p} = \frac{\pi(x)}{x} + \int_2^x \frac{\pi(t)}{t^2} \, \mathrm{d}t.$$

Write $\pi(t) = (\ell + \varepsilon(t))t/\log t$, where $\varepsilon(t) \to 0$. Splitting off a fixed finite interval and then bounding $|\varepsilon(t)|$ on the tail gives $\int_2^x \varepsilon(t)/(t \log t) \, \mathrm{d}t = o(\log \log x)$. Since $\pi(x)/x = O(1/\log x)$, the reciprocal-prime sum is $\ell \log \log x + o(\log \log x)$. Comparison with Theorem 4.15 forces $\ell = 1$. □

Chebyshev's estimates identify the order of magnitude, and Mertens' estimates determine the only possible limiting value. It remains to prove convergence. In terms of ψ, the elementary argument gives only

$$0 < c \leq \frac{\psi(x)}{x} \leq C,$$

whereas the prime number theorem asks us to prove that this ratio tends to 1. Chapter 5 encodes Λ in $-\zeta'/\zeta$; the absence of zeros on $\Re s = 1$ will rule out oscillations on the main scale that could prevent the limit from existing.

The structure of the proof explains why the binomial-coefficient method stops at $\asymp$. The upper bound crudely compresses the contribution of many primes into $\binom{2n}{n} \leq 4^n$, while the lower bound uses only $\binom{2n}{n} \geq 4^n/(2n+1)$. These inequalities provide positive linear constants, but no mechanism forcing the upper and lower limits of $\vartheta(x)/x$ to coincide. Determining the constant 1 requires finer cancellation across scales or boundary information from a generating function.

Chapter Summary

- Divergence of the sum of reciprocal primes is quantitatively stronger than the infinitude of primes.
- The function ψ is both $\sum_{p^k \leq x} \log p$ and $\log \operatorname{lcm}(1, \ldots, \lfloor x \rfloor)$.
- Legendre's formula counts prime exponents in factorials one layer at a time and is the arithmetic basis of the binomial-coefficient estimate.
- The central binomial coefficient gives $\vartheta(x) \asymp x$, and hence $\pi(x) \asymp x/\log x$.
- Mertens' estimates give $\sum_{p \leq x} 1/p = \log\log x + B_1 + O(1/\log x)$.
- If $\pi(x)\log x/x$ has a limit, it is 1; the prime number theorem must still prove that the limit exists.

Common Pitfalls

- Inferring $\pi(x) \sim x/\log x$ directly from $\sum_p 1/p = \infty$; divergence supplies only very weak aggregate information.
- Treating ψ as a sum over primes only; it includes all prime powers.
- Using $\pi(x) \leq \vartheta(x)/\log 2$ as the upper bound. This gives only $O(x)$; small primes must be separated to obtain $O(x/\log x)$.

Exercises

Exercise 4.1. Verify (4.2), and compute both sides when $x = 10$.

Exercise 4.2. Prove that $\vartheta(x) \le \pi(x)\log x$ and that $\vartheta(x) \ge (\pi(x) - \pi(\sqrt{x}))\tfrac{1}{2}\log x$.

Exercise 4.3. Improve the crude argument in Proposition 4.6 to prove $\psi(x) - \vartheta(x) = O(\sqrt{x}\log x)$. Hint: first treat the square terms separately, then estimate the number and size of the terms with $k \ge 3$.

Exercise 4.4. Prove that if $\pi(x) \sim x/\log x$, then the nth prime satisfies $p_n \sim n\log n$.

Exercise 4.5. Extract explicit constants from the proof of Theorem 4.14. Which steps cause the largest losses?

Exercise 4.6. Use Legendre's formula to compute $\nu_2(100!)$ and $\nu_5(100!)$, and find the number of trailing zeros of $100!$.

Exercise 4.7. Prove that $\int_2^x \mathrm{d}t/\log t = O(x/\log x)$. Hint: split at $\sqrt{x}$; bound the integrand by $1/\log 2$ on the first part and by $2/\log x$ on the second. Use the estimate to complete the reverse implication in Proposition 4.8.

Chapter 5

The Riemann Zeta Function and the Prime Number Theorem

Learning objectives. Establish the Euler product and analytic continuation of ζ, prove nonvanishing on the boundary line, and deduce the prime number theorem from the Wiener–Ikehara theorem.

Prerequisites. The Chebyshev functions from Chapter 4, together with uniform convergence, zeros of analytic functions, and meromorphic functions from complex analysis.

Role in the book. This chapter gives an analytic proof of the prime number theorem; the next chapter organizes the Dirichlet-series tools used here.

5.1 The half-plane of absolute convergence

We first record the normal-convergence result needed for termwise differentiation.

Lemma 5.1 (Normal convergence and termwise differentiation)**.** *Suppose each f_n is analytic in a region Ω and the series $\sum_n f_n$ converges normally: for each compact $K \subset \Omega$, $\sum_n \sup_{s\in K} |f_n(s)| < \infty$. Then $\sum_n f_n$ is analytic in Ω, the derivative series also converges normally, and*

$$\left(\sum_n f_n\right)' = \sum_n f_n'.$$

Proof. The first assertion is Weierstrass's theorem. Given a compact set $K \subset \Omega$, choose $r > 0$ so that its closed r-neighborhood K_r lies in Ω. Cauchy's estimate gives $\sup_K |f_n'| \le r^{-1} \sup_{K_r} |f_n|$, proving normal convergence of the derivative series. On the circle of radius r centered at each $s \in K$, interchange summation and integration in Cauchy's integral formula to obtain the stated derivative identity. □

Definition 5.2. *For $s = \sigma + it$ with $\sigma > 1$, define*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad n^{-s} = \exp(-s \log n).$$

Since $\sum n^{-\sigma}$ converges, the series converges normally on every closed half-plane $\sigma \ge 1 + \delta$. It therefore defines an analytic function and can be differentiated term by term.

Theorem 5.3 (Euler product)**.** *For $\sigma > 1$,*

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \tag{5.1}$$

where the product converges uniformly on compact subsets and $\zeta(s) \neq 0$.

Proof. For a finite set of primes P, multiplication of geometric series and unique factorization give

$$\prod_{p\in P}(1-p^{-s})^{-1} = \sum_{\substack{n\geq 1\\ p\mid n\Rightarrow p\in P}} n^{-s}.$$

As P exhausts the primes, the right-hand side tends to $\sum n^{-s}$ because the absolutely convergent series $\sum n^{-\sigma}$ controls the tail uniformly. More quantitatively, if P contains every prime $p \leq y$, then every omitted integer is greater than y, so on $\sigma \geq 1+\delta$ the error is at most $\sum_{n>y} n^{-1-\delta} = O_\delta(y^{-\delta})$. Moreover, $\sum_p |\log(1-p^{-s})|$ converges uniformly on $\sigma \geq 1+\delta$, where $\log(1-z) = -\sum_{k\geq 1} z^k/k$ denotes the analytic branch for $|z|<1$; its general term is $O_\delta(p^{-1-\delta})$. The product therefore converges to a nonzero analytic function. □

In the same region, absolute convergence justifies taking logarithms of the product and then differentiating.

Theorem 5.4 (Logarithmic derivative)**.** *For $\sigma > 1$,*

$$-\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{k\geq 1} \frac{\log p}{p^{ks}} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}. \tag{5.2}$$

Proof. By Theorem 5.3, on any closed half-plane $\sigma \geq 1+\delta$ we may choose the logarithm determined by the condition that it tends to 0 as $s \to +\infty$, and write termwise

$$\log\zeta(s) = \sum_p \sum_{k\geq 1} \frac{1}{kp^{ks}}.$$

The double series and its termwise derivative converge uniformly and absolutely on $\sigma \geq 1+\delta$; the latter is dominated by $\sum_{n\geq 2} (\log n) n^{-1-\delta} < \infty$. Differentiation gives

$$\frac{\zeta'}{\zeta}(s) = -\sum_p \sum_{k\geq 1} \frac{\log p}{p^{ks}}.$$

Collecting coefficients according to $n = p^k$ gives (5.2); absolute convergence justifies the rearrangement. □

Identity (5.2) corresponds to $\mathbf{1} * \Lambda = \log$ from Chapter 2: convolution becomes multiplication, and the coefficients of the logarithmic derivative are Λ.

5.2 Analytic continuation into the right half-plane

For $\sigma > 1$, first regard the series as a Stieltjes integral against the step function $\lfloor x \rfloor$:

$$\zeta(s) = \int_{1^-}^{\infty} x^{-s}\, \mathrm{d}\lfloor x \rfloor.$$

Integrate by parts on $[1, N]$ and let $N \to \infty$. Since $\lfloor N \rfloor N^{-s} = O(N^{1-\sigma}) \to 0$, we obtain

$$\begin{aligned} \zeta(s) &= s\int_1^{\infty} \lfloor x \rfloor x^{-s-1}\, \mathrm{d}x \\ &= \frac{s}{s-1} - s\int_1^{\infty} \{x\} x^{-s-1}\, \mathrm{d}x. \end{aligned} \tag{5.3}$$

The second integral converges absolutely and locally uniformly for $\sigma > 0$ because $0 \le \{x\} < 1$. More explicitly, given a compact $K \subset \{\Re s > 0\}$, choose $\delta > 0$ with $\Re s \ge \delta$ on K. The integrand and each fixed-order derivative with respect to s are bounded by $x^{-1-\delta}$ and $(\log x)^j x^{-1-\delta}$, respectively, all integrable on $[1, \infty)$. Differentiation under the integral sign therefore proves analyticity of the integral, so the right-hand side gives a meromorphic continuation of ζ to $\sigma > 0$.

It agrees with the original series for $\sigma > 1$ and therefore continues the same function. In $0 < \sigma \le 1$, the original Dirichlet series generally does not converge.

Proposition 5.5. *The function $\zeta(s)$ is analytic in $\Re s > 0$ except at $s = 1$, where it has a simple pole with residue 1. More precisely,*

$$\zeta(s) = \frac{1}{s-1} + h(s),$$

where h is analytic in $\Re s > 0$ and $h(1) = \gamma$.

Proof. Analyticity and the pole follow directly from (5.3), which also gives $h(1) = 1 - \int_1^\infty \{x\} x^{-2}\,\mathrm{d}x$. For an integer $N \ge 1$, finite Abel summation yields

$$\sum_{n \le N} \frac{1}{n} - \log N = 1 - \int_1^N \frac{\{x\}}{x^2}\,\mathrm{d}x.$$

As $N \to \infty$, the left side tends to Euler's constant γ and the right side tends to $h(1)$. Hence $h(1) = \gamma$. □

Remark 5.6. The full functional equation continues ζ to the entire complex plane and introduces the critical strip and the Riemann hypothesis. Its proof requires the Gamma function and either Poisson summation or theta functions. Our proof of the prime number theorem needs only continuation to $\Re s > 0$, so the functional equation is not a logical prerequisite.

5.3 Zeros, poles, and logarithmic derivatives

The nonvanishing proof compares the principal parts of several logarithmic derivatives. The next local statement fixes the signs and orders once and for all.

Definition 5.7 (Orders of zeros and poles)**.** *Let m be a positive integer and let g be analytic near s_0, with $g(s_0) \neq 0$. If $f(s) = (s - s_0)^m g(s)$, then s_0 is a zero of order m of f. If $f(s) = (s - s_0)^{-m} g(s)$, then s_0 is a pole of order m of f; the latter identity holds in a punctured neighborhood of s_0.*

Example 5.8. *The function $f(s) = (s-1)^2(s+2)$ has a zero of order two at $s = 1$, so the principal term of f'/f near 1 is $2/(s-1)$. The function $1/(s-1)^3$ has a pole of order three, and the principal term of its logarithmic derivative is $-3/(s-1)$. Zeros and poles have opposite signs.*

Proposition 5.9 (Local principal part of a logarithmic derivative)**.** *If a meromorphic function f has a zero of order m at s_0, then*

$$\frac{f'}{f}(s) = \frac{m}{s - s_0} + H(s);$$

if it has a pole of order m, the principal term is $-m/(s - s_0)$. In each case H is analytic near s_0.

Proof. Write $f(s) = (s - s_0)^{\pm m} g(s)$, where $g(s_0) \neq 0$. Logarithmic differentiation gives

$$\frac{f'}{f}(s) = \frac{\pm m}{s - s_0} + \frac{g'}{g}(s).$$

After shrinking the neighborhood, g has no zeros there, so g'/g is analytic. □

5.4 Nonvanishing on the boundary line

The coefficients $3, 4, 1$ make a linear combination of the Dirichlet series for $-\zeta'/\zeta(\sigma + kit)$ termwise nonnegative. At the same time, a hypothetical boundary zero would contribute a simple-pole principal part with coefficient at most -4, while the pole at $s = 1$ contributes a simple-pole principal part with coefficient 3. The identity $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2$ achieves both purposes.

Theorem 5.10 (Hadamard–de la Vallée Poussin)**.** *There are no zeros of ζ on $\Re s = 1$. More precisely, $\zeta(1 + it) \neq 0$ for $t \neq 0$, while $s = 1$ is a simple pole.*

Proof. For $\sigma > 1$, (5.2) and

$$3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2 \geq 0$$

give, for every real t,

$$-3\frac{\zeta'}{\zeta}(\sigma) - 4\Re\frac{\zeta'}{\zeta}(\sigma + it) - \Re\frac{\zeta'}{\zeta}(\sigma + 2it) \geq 0. \tag{5.4}$$

Indeed, after expansion, the contribution from each n is $\Lambda(n)n^{-\sigma}[3 + 4\cos(t\log n) + \cos(2t\log n)] \geq 0$.

Suppose, to the contrary, that $t_0 \neq 0$ and $\zeta(1 + it_0) = 0$, with order $m \geq 1$. Let $\sigma \downarrow 1$. Proposition 5.5 gives

$$-\frac{\zeta'}{\zeta}(\sigma) = \frac{1}{\sigma - 1} + O(1),$$

while Proposition 5.9 gives

$$\Re\frac{\zeta'}{\zeta}(\sigma + it_0) = \frac{m}{\sigma - 1} + O(1).$$

Since $t_0 \neq 0$, the point $1 + 2it_0$ is not a pole of ζ. Let $m_2 \geq 0$ be its order as a zero, with $m_2 = 0$ if it is not a zero. The same local expansion gives $\Re(\zeta'/\zeta)(\sigma + 2it_0) = m_2/(\sigma - 1) + O(1)$. Thus the left-hand side of (5.4) equals

$$\frac{3 - 4m - m_2}{\sigma - 1} + O(1),$$

which is negative when σ is sufficiently close to 1, a contradiction. □

5.5 The Tauberian bridge

Returning from boundary behavior to asymptotics of coefficients requires a Tauberian theorem. We use the following Dirichlet-series version.

Theorem 5.11 (Wiener–Ikehara theorem: Dirichlet-series form)**.** *Let $a_n \geq 0$, and suppose*

$$F(s) = \sum_{n=1}^{\infty} a_n n^{-s}$$

converges in $\Re s > 1$. If there is an $A \geq 0$ such that $F(s) - A/(s-1)$ extends continuously to the closed half-plane $\Re s \geq 1$ and is analytic in the open half-plane $\Re s > 1$, then

$$\sum_{n \leq x} a_n = Ax + o(x) \qquad (x \to \infty).$$

For $A > 0$ this is equivalent to $\sum_{n \leq x} a_n \sim Ax$; for $A = 0$ the conclusion is $\sum_{n \leq x} a_n = o(x)$.

Proof outline. Put $B(u) = \sum_{n \leq \mathrm{e}^u} a_n$ for $u \geq 0$ and $B(u) = 0$ for $u < 0$. Given $\sigma = \Re s > 1$, choose $1 < \sigma_0 < \sigma$. Nonnegativity and convergence give $B(u) \leq \mathrm{e}^{\sigma_0 u} F(\sigma_0)$, so the boundary term $\mathrm{e}^{-su} B(u)$ in integration by parts tends to zero at infinity and the resulting ordinary integral converges absolutely. Hence

$$F(s) = \int_{0^-}^{\infty} \mathrm{e}^{-su} \, \mathrm{d}B(u) = s \int_0^{\infty} B(u) \mathrm{e}^{-su} \, \mathrm{d}u \qquad (\Re s > 1).$$

Thus F is the Laplace–Stieltjes transform of B, while its ordinary Laplace transform is $F(s)/s$. Majorants and minorants with compactly supported Fourier transforms control weighted partial sums. After subtraction of the polar term, boundary continuity and the Riemann–Lebesgue lemma make the smoothed remainder tend to zero. Monotonicity of B then yields $\mathrm{e}^{-u} B(u) \to A$. For the complete proof and this Dirichlet-series formulation, see [6, §8.3, Corollary 8.8]; we quote the theorem without proving the Fourier approximation result. Boundary continuity is a local condition at each finite boundary point; it does not require a uniform bound on the remainder along the entire line.

Nonnegativity makes $S(x) = \sum_{n \leq x} a_n$ monotone. Fourier analysis first controls smooth averages of S; if the coefficients may change sign arbitrarily, sharp local oscillations can disappear under smoothing, and the average no longer determines pointwise asymptotics.

5.6 The prime number theorem

The equivalence of the three common formulations was proved in Chapter 4. The analytic argument need only handle ψ, which corresponds directly to a Dirichlet series.

Proposition 5.12 (Equivalent forms of the prime number theorem)**.** *As $x \to \infty$, the following are equivalent:*

$$\psi(x) \sim x, \qquad \vartheta(x) \sim x, \qquad \pi(x) \sim \frac{x}{\log x}.$$

Proof. This is Proposition 4.8. The key ingredients are $\psi(x) - \vartheta(x) = o(x)$ and the Abel summation formula (3.4); neither step uses the zeta function. □

Theorem 5.13 (Prime number theorem)**.** *As $x \to \infty$,*

$$\psi(x) \sim x, \qquad \vartheta(x) \sim x, \qquad \pi(x) \sim \frac{x}{\log x}.$$

Proof. For $\sigma > 1$, Theorem 5.4 gives

$$F(s) := -\frac{\zeta'}{\zeta}(s) = \sum_{n \geq 1} \Lambda(n) n^{-s},$$

with coefficients $\Lambda(n) \geq 0$. Since $\zeta(s) = 1/(s-1) + h(s)$, write $\zeta(s) = (s-1)^{-1} g(s)$, where $g(s) = 1 + (s-1)h(s)$ is analytic near 1 and $g(1) = 1$. In a smaller neighborhood g has no zeros, so

$$-\frac{\zeta'}{\zeta}(s) = \frac{1}{s-1} - \frac{g'}{g}(s),$$

where the remainder is analytic; the coefficient 1 comes from the order of the pole of ζ. Theorem 5.10 ensures that ζ does not vanish at any other boundary point $1 + it$. Since ζ is analytic there, each such point has a small zero-free neighborhood. Hence $F(s) - 1/(s-1)$ is analytic locally and extends continuously to the entire line $\Re s = 1$. These pointwise neighborhoods suffice; a zero-free strip of fixed width to the left of the whole line is not required.

The hypotheses of Wiener–Ikehara are verified as follows:

1. $a_n = \Lambda(n) \geq 0$;
2. the series converges absolutely in $\Re s > 1$ and equals $F(s)$;
3. F is analytic in $\Re s > 1$ because the Euler product has no zeros there;
4. the singularity of $F(s) - 1/(s-1)$ at $s = 1$ has been removed, and continuity at all other boundary points follows from the nonvanishing theorem;
5. the polar coefficient is $A = 1$.

Theorem 5.11, with $a_n = \Lambda(n)$ and $A = 1$, therefore gives $\psi(x) = \sum_{n \leq x} \Lambda(n) \sim x$.

Finally, Proposition 5.12 gives the other two formulations. □

The Euler product and logarithmic derivative were used only in $\sigma > 1$; boundary information came from (5.3) and the nonvanishing theorem. The proof of the Wiener–Ikehara theorem is the only external input here.

Mean cancellation of the Möbius function

The same nonvanishing result also controls cancellation between the signs of $\mu(n)$. We first make the coefficients nonnegative so that the stated Tauberian theorem applies.

Corollary 5.14. *If $M(x) = \sum_{n \leq x} \mu(n)$, then $M(x) = o(x)$.*

Proof. Unique factorization and absolute convergence give, for $\Re s > 1$,

$$\frac{1}{\zeta(s)} = \prod_p (1 - p^{-s}) = \sum_{n \geq 1} \frac{\mu(n)}{n^s}.$$

The nonnegative coefficients $b_n = 1 + \mu(n)$ therefore have Dirichlet series

$$G(s) = \sum_{n \geq 1} \frac{b_n}{n^s} = \zeta(s) + \frac{1}{\zeta(s)}.$$

Near $s = 1$, write $\zeta(s) = 1/(s-1) + h(s)$. Then $1/\zeta(s) = (s-1)/(1 + (s-1)h(s))$ is analytic and vanishes at 1. At every other boundary point, Theorem 5.10 makes $1/\zeta$ analytic. Thus $G(s) - 1/(s-1)$ extends continuously to $\Re s \geq 1$. By Theorem 5.11,

$$\lfloor x \rfloor + M(x) = \sum_{n \leq x} b_n = x + o(x).$$

Subtracting $\lfloor x \rfloor = x + O(1)$ proves the result. □

Together with Theorem 3.11, this has a concrete counting interpretation:

$$\#\{n \leq x : \mu(n) = 1\} = \#\{n \leq x : \mu(n) = -1\} + o(x) = \frac{3}{\pi^2} x + o(x).$$

Indeed, the two counts are respectively $\frac{1}{2} \sum_{n \leq x} (\mu(n)^2 + \mu(n))$ and $\frac{1}{2} \sum_{n \leq x} (\mu(n)^2 - \mu(n))$. The signs are equally frequent on the scale of the main term; this gives no square-root bound for their cancellation.

5.7 Error terms and zeros: a further direction

The prime number theorem uses only nonvanishing on $\Re s = 1$. A wider zero-free region and growth estimates for ζ'/ζ allow the line of integration in Perron's formula to be shifted and produce explicit errors for $\psi(x) - x$. The Riemann hypothesis asserts that every nontrivial zero has real part $1/2$ and is closely related to an estimate near $\psi(x) = x + O(x^{1/2}\log^2 x)$. We mention only the direction and do not use this deep result in the proof above.

Chapter Summary

- The Euler product and the Dirichlet-series expansion of $-\zeta'/\zeta$ hold directly only in $\Re s > 1$.
- The floor-function integral continues ζ meromorphically to $\Re s > 0$ and exhibits the residue 1 at $s = 1$.
- The nonnegative trigonometric polynomial $3 + 4\cos\theta + \cos 2\theta$ rules out boundary zeros.
- The Wiener–Ikehara theorem converts boundary analyticity and nonnegative coefficients into $\psi(x) \sim x$.
- Removing prime powers and applying Abel summation then gives $\pi(x) \sim x/\log x$.
- Applying the same Tauberian theorem to $1 + \mu(n) \geq 0$ gives $\sum_{n\leq x} \mu(n) = o(x)$.

Common Pitfalls

- Using the Euler product directly on $\Re s = 1$, where it does not converge absolutely.
- Assuming that the analytically continued values are still given term by term by the original Dirichlet series.
- Jumping from "nonvanishing on the boundary" to the prime number theorem without specifying a Tauberian theorem or inversion device.
- Ignoring the hypothesis $a_n \geq 0$ in Wiener–Ikehara; oscillating coefficients generally require stronger assumptions.

Exercises

Exercise 5.1. Prove that $\zeta(\sigma) > 1$ and $-\zeta'(\sigma) = \sum_{n\geq 2}(\log n)n^{-\sigma} > 0$ for $\sigma > 1$.

Exercise 5.2. Use (5.3) to prove that $\lim_{s\to 1}(s-1)\zeta(s) = 1$.

Exercise 5.3. Expand (5.4) in detail, verify that the coefficient of every n is nonnegative, and explain why the coefficients in $3 + 4\cos\theta + \cos 2\theta$ are exactly strong enough to overwhelm one boundary zero.

Exercise 5.4. Assume $\psi(x) \sim x$. Without citing the conclusion in the text, give a complete proof that $\pi(x) \sim x/\log x$.

Exercise 5.5. Compare the Wiener–Ikehara route with the classical route using Perron's formula and a truncated integral. Which boundary properties of ζ must be controlled in each method?

Exercise 5.6. Let $m \geq 1$ be an integer, let g be analytic near s_0 with $g(s_0) \neq 0$, and put $f(s) = (s - s_0)^m g(s)$. Compute $f'/f - m/(s - s_0)$ directly and explain why it is analytic at s_0.

Exercise 5.7. In the proof of Theorem 5.13, justify each of the five checks, citing numbered theorems, propositions, or equations where possible. Explain the logical dependencies among these checks, and determine whether $A = 1$ depends on the order or on the residue of the pole of ζ at 1.

Chapter 6

Dirichlet Series, Mellin Transforms, and Perron's Formula

Learning objectives. Master convergence and multiplication of Dirichlet series, understand the Mellin transform, and learn to use Perron's formula.

Prerequisites. Abel summation, Dirichlet convolution, complex integration, and the residue theorem.

Role in the book. This chapter organizes the generating-function techniques used in the preceding chapter; Chapter 7 applies them to L-functions.

6.1 Half-planes of convergence

Definition 6.1 (Dirichlet series and their abscissae of convergence)**.** *A series of the form*

$$F(s) = \sum_{n=1}^{\infty} a_n n^{-s}$$

is called a Dirichlet series*. If it converges at s, we speak of ordinary convergence; if $\sum |a_n| n^{-\Re s} < \infty$, we speak of absolute convergence. Its* abscissa of convergence σ_{c} *and* abscissa of absolute convergence σ_{a} *are the infima of the boundaries of the right half-planes in which the series converges ordinarily and absolutely, respectively.*

The abscissae may be $\pm\infty$. For $\zeta(s)$, both are 1. The natural variable of a Dirichlet series is $n^{-s} = \mathrm{e}^{-s\log n}$. Convergence on a boundary line must be considered separately.

Theorem 6.2 (Partial-sum criterion)**.** *Let $A(x) = \sum_{n \le x} a_n$. If $A(x) = O(x^{\alpha})$, then $F(s)$ converges and is analytic in $\Re s > \alpha$, and*

$$F(s) = s \int_1^{\infty} A(x) x^{-s-1}\,\mathrm{d}x. \tag{6.1}$$

For every $\delta > 0$, the integral converges locally uniformly on the closed half-plane $\Re s \ge \alpha + \delta$.

Proof. Abel summation with a finite upper limit N gives

$$\sum_{n \le N} a_n n^{-s} = A(N) N^{-s} + s \int_1^{N} A(x) x^{-s-1}\,\mathrm{d}x.$$

If $\sigma > \alpha$, the boundary term tends to zero and the integrand is bounded by a constant times $x^{\alpha-\sigma-1}$. Uniform control on compact sets also gives analyticity. □

This also proves the half-plane property for ordinary convergence. If the series converges at s_0, put $b_n = a_n n^{-s_0}$; the partial sums $\sum_{n\le x} b_n$ are bounded. Applying the theorem to $\sum b_n n^{-(s-s_0)}$ shows that the original series converges and is analytic in $\Re s > \Re s_0$. Thus every point to the right of the abscissa is a point of convergence; the abscissa alone does not determine convergence on the boundary.

Theorem 6.3 (Termwise differentiation in the region of absolute convergence)**.** *Suppose $F(s) = \sum a_n n^{-s}$ has abscissa of absolute convergence $\sigma_{\mathrm{a}} < +\infty$. For every real $b > \sigma_{\mathrm{a}}$, the original series and all the derivative series below converge uniformly and absolutely on $\Re s \ge b$. The function F is analytic in $\Re s > \sigma_{\mathrm{a}}$, and for every integer $k \ge 0$,*

$$F^{(k)}(s) = \sum_{n=1}^{\infty} a_n (-\log n)^k n^{-s} \qquad (\Re s > \sigma_{\mathrm{a}}).$$

If $\sigma_{\mathrm{a}} = -\infty$, the conclusion holds for every real b, and F is entire.

Proof. Fix a real $b > \sigma_{\mathrm{a}}$, choose a real σ_0 with $\sigma_{\mathrm{a}} < \sigma_0 < b$, and put $\delta = b - \sigma_0 > 0$. Then $\sum |a_n| n^{-\sigma_0}$ converges. For $\sigma \ge b$,

$$|a_n| n^{-\sigma} (\log n)^k \le |a_n| n^{-\sigma_0} \frac{(\log n)^k}{n^{\delta}}.$$

The second factor on the right is bounded. The Weierstrass test gives uniform absolute convergence, and Lemma 5.1 permits termwise differentiation. □

Example 6.4. *For $\zeta(s)$ the theorem gives*

$$\zeta^{(k)}(s) = \sum_{n\ge 1} \frac{(-\log n)^k}{n^s} \qquad (\Re s > 1).$$

This series alone cannot enlarge the region $\Re s > 1$. The analytic continuation in Chapter 5 is a different representation of the function, not a consequence of termwise differentiation.

Corollary 6.5. *We always have*

$$0 \le \sigma_{\mathrm{a}} - \sigma_{\mathrm{c}} \le 1,$$

when the abscissae are finite.

Proof. The lower bound is immediate. If the series converges at s_0, then $a_n n^{-s_0} \to 0$, so $|a_n| \ll n^{\Re s_0}$. It therefore converges absolutely whenever $\sigma > \Re s_0 + 1$. Let s_0 approach σ_{c} from the right. □

Example 6.6. *The sums over a period of a nonprincipal Dirichlet character χ vanish, so its partial sums are bounded. Theorem 6.2 therefore makes $\sum \chi(n) n^{-s}$ converge in $\Re s > 0$. Its absolute-value series is $\sum_{(n,q)=1} n^{-\sigma}$ and converges exactly when $\sigma > 1$: at $\sigma = 1$, the terms with $n \equiv 1 \pmod q$ already form a divergent subseries.*

Proposition 6.7 (Uniqueness of coefficients)**.** *Suppose $F(s) = \sum_{n\ge 1} a_n n^{-s}$ converges absolutely in some right half-plane. For every integer $m \ge 1$,*

$$a_m = \lim_{\sigma \to +\infty} m^{\sigma} \left(F(\sigma) - \sum_{n<m} \frac{a_n}{n^{\sigma}} \right).$$

Consequently, two such Dirichlet series that represent the same function in a right half-plane have identical coefficients.

Proof. Fix a real b at which the series converges absolutely. For $\sigma \geq b$, multiplying the expression in parentheses by m^σ gives

$$a_m + \sum_{n>m} a_n \left(\frac{m}{n}\right)^\sigma.$$

Each term in the tail tends to zero as $\sigma \to \infty$, and

$$\sum_{n>m} |a_n| \left(\frac{m}{n}\right)^\sigma \leq m^b \sum_{n>m} |a_n| n^{-b} < \infty.$$

This supplies a summable bound independent of σ, so dominated convergence makes the whole tail tend to zero. Recovering the coefficients successively from $m = 1$ proves uniqueness. □

A Dirichlet series therefore retains all the information in its arithmetic coefficients. This justifies comparing coefficients of convergent series; recovering partial sums at finite height requires the Perron formula developed below.

6.2 Multiplication and Euler products

Doubly infinite series cannot be rearranged freely under conditional convergence, so the following multiplication theorem assumes absolute convergence.

Theorem 6.8 (Multiplication of Dirichlet series)**.** *If $F(s) = \sum f(n)n^{-s}$ and $G(s) = \sum g(n)n^{-s}$ both converge absolutely at a point s, then*

$$F(s)G(s) = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^s}.$$

Proof. Absolute convergence gives

$$\sum_{a,b\geq 1} \left|\frac{f(a)g(b)}{(ab)^s}\right| = \left(\sum_a \frac{|f(a)|}{a^\sigma}\right)\left(\sum_b \frac{|g(b)|}{b^\sigma}\right) < \infty.$$

We may therefore rearrange according to $n = ab$. The coefficient of n is exactly $\sum_{d|n} f(d)g(n/d)$. □

Theorem 6.9 (General Euler product)**.** *If f is multiplicative and $\sum |f(n)|n^{-\sigma} < \infty$, then*

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_p \left(\sum_{k=0}^{\infty} \frac{f(p^k)}{p^{ks}}\right).$$

If f is completely multiplicative, the local factor is $(1 - f(p)p^{-s})^{-1}$.

Proof. First multiply finitely many local factors. Unique factorization gives the sum over integers with no prime factors outside this finite set. Absolute convergence makes the tail tend to zero as the set of primes expands and justifies every rearrangement. If f is completely multiplicative, absolute convergence also implies $\sum_{k\geq 0} |f(p)p^{-s}|^k < \infty$, so $|f(p)p^{-s}| < 1$. This justifies summing each local factor as a geometric series. □

Example 6.10. *In the corresponding regions of absolute convergence, the convolution identities give*

$$\sum_{n\geq 1} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)} \quad (\sigma > 1), \qquad \sum_{n\geq 1} \frac{\tau(n)}{n^s} = \zeta(s)^2 \quad (\sigma > 1),$$

and

$$\sum_{n\geq 1}\frac{\varphi(n)}{n^s}=\frac{\zeta(s-1)}{\zeta(s)}\quad(\sigma>2).$$

The last identity comes from $\mathbf{1}*\varphi=\mathrm{id}$. *We write* $\sigma>2$ *because absolute convergence of* $\sum\varphi(n)n^{-s}$ *follows from* $\varphi(n)\leq n$ *only in that region.*

6.3 The Gamma and Beta functions

Definition 6.11. *For* $\Re s>0$, *Euler's Gamma function is defined by*

$$\Gamma(s)=\int_0^\infty \mathrm{e}^{-x}x^{s-1}\,\mathrm{d}x.$$

For $\Re u,\Re v>0$, *the Beta function is defined by*

$$B(u,v)=\int_0^1 t^{u-1}(1-t)^{v-1}\,\mathrm{d}t.$$

Convergence at the endpoints follows respectively from integrability of $x^{\sigma-1}$ near 0, rapid exponential decay at infinity, and integrability of $t^{\Re u-1}$ and $(1-t)^{\Re v-1}$.

More explicitly, split the Gamma integral into $(0,1)$ and $(1,\infty)$. The first part is controlled by $\int_0^1 x^{\sigma-1}\,\mathrm{d}x=1/\sigma$, while on the second part exponential decay dominates every power. This also shows that the integral definition does not apply when $\Re s\leq 0$; values there require analytic continuation.

Proposition 6.12. *For* $\Re s>0$, $\Gamma(s+1)=s\Gamma(s)$, *and hence* $\Gamma(n+1)=n!$ *for* $n\geq 0$; *also* $\Gamma(s)\neq 0$. *For* $\Re u,\Re v>0$,

$$B(u,v)=\frac{\Gamma(u)\Gamma(v)}{\Gamma(u+v)}. \tag{6.2}$$

Proof. On compact subsets of $\Re s>0$, the Gamma integral is dominated by a common integrable function, so it defines an analytic function. Integration by parts gives the recurrence, with vanishing boundary terms in this region, and $\Gamma(1)=1$. To prove nonvanishing for complex parameters, first integrate the Beta integral by parts to obtain $B(s,N+1)=(N/s)B(s+1,N)$. The boundary terms vanish because $\Re s>0$ and $N\geq 1$. Iterate this identity, ending with $B(s+N,1)=1/(s+N)$, to obtain

$$B(s,N+1)=\frac{N!}{s(s+1)\cdots(s+N)}\qquad(N\geq 1).$$

Substitute $t=x/N$. Since $(1-x/N)^N\leq \mathrm{e}^{-x}$ for $0\leq x\leq N$, dominated convergence gives

$$\frac{N!N^s}{s(s+1)\cdots(s+N)}=\int_0^N x^{s-1}(1-x/N)^N\,\mathrm{d}x\longrightarrow\Gamma(s).$$

On the other hand, the expression on the left equals

$$\frac{1}{s}\exp\left(s(\log N-H_N)-\sum_{k=1}^{N}\left[\log(1+s/k)-s/k\right]\right),\qquad H_N=\sum_{k=1}^{N}\frac{1}{k}.$$

Here the principal logarithm is used; all the numbers $1+s/k$ lie in the right half-plane because $\Re s>0$. Moreover, $H_N-\log N\to\gamma$, and each bracketed term is $O_s(k^{-2})$, so the exponent has a finite limit. The resulting limit is nonzero, proving $\Gamma(s)\neq 0$.

For the Beta identity, begin with the absolutely convergent double integral

$$\Gamma(u)\Gamma(v) = \int_0^\infty \int_0^\infty \mathrm{e}^{-(x+y)} x^{u-1} y^{v-1}\,\mathrm{d}x\,\mathrm{d}y.$$

Set $r = x + y$ and $t = x/(x+y)$. Then $x = rt$, $y = r(1-t)$, the first quadrant becomes $r > 0$, $0 < t < 1$, and the absolute value of the Jacobian is r. The integral separates as $\Gamma(u+v)B(u,v)$. Since $\Re(u+v) > 0$, we may divide by the nonzero value $\Gamma(u+v)$ to obtain (6.2). □

The recurrence also gives a meromorphic continuation of the Gamma function. Choose $m \geq 1$ such that $\Re(s+m) > 0$ and define

$$\Gamma(s) = \frac{\Gamma(s+m)}{s(s+1)\cdots(s+m-1)}.$$

The recurrence makes the expressions for different m compatible. At $s = -n$, where $n = 0, 1, \ldots$, the continuation has a simple pole with residue $(-1)^n/n!$; it is analytic and nonzero everywhere else. Indeed, near $s = -n$, take $m = n+1$. The numerator tends to $\Gamma(1) = 1$, while the product of all denominator factors other than $s+n$ tends to $(-1)^n n!$. This proves both the simplicity of the pole and the stated residue.

Proposition 6.13 (Special values of the Gamma function)**.**

$$\Gamma(1) = 1, \qquad \Gamma(n+1) = n!, \qquad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Proof. The first two identities follow from the definition and recurrence. Let $I = \int_0^\infty \mathrm{e}^{-x^2}\,\mathrm{d}x$. Tonelli's theorem and polar coordinates give

$$I^2 = \int_0^\infty \int_0^\infty \mathrm{e}^{-(x^2+y^2)}\,\mathrm{d}x\,\mathrm{d}y = \int_0^{\pi/2} \int_0^\infty \mathrm{e}^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta = \frac{\pi}{4}.$$

Thus $I = \sqrt{\pi}/2$. Substituting $x = u^2$ in the Gamma integral gives $\Gamma(1/2) = 2I = \sqrt{\pi}$. □

Example 6.14. *The Beta–Gamma identity gives*

$$B\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{\Gamma(1/2)^2}{\Gamma(1)} = \pi.$$

The direct substitution $t = \sin^2\theta$ also reduces the integral to $2\int_0^{\pi/2}\,\mathrm{d}\theta = \pi$, providing an independent check of the change of variables.

6.4 The Mellin transform

Definition 6.15 (Mellin transform and fundamental strip)**.** *Let $f : (0,\infty) \to \mathbb{C}$ be measurable and locally integrable. Its Mellin transform is*

$$\mathcal{M}f(s) = \int_0^\infty f(x) x^{s-1}\,\mathrm{d}x,$$

whenever the integral converges. If the region of absolute convergence has nonempty interior, its largest open vertical strip $\alpha < \Re s < \beta$ is called the fundamental strip*; its endpoints are determined by the growth of f near 0 and ∞, respectively. Convergence on the boundary is considered separately. The integral converges uniformly on compact subsets of the fundamental strip, so $\mathcal{M}f$ is analytic there.*

Uniform convergence in the fundamental strip can be checked directly. If the real parts on a compact set lie in $[a, b]$, with $\alpha < a \le b < \beta$, dominate the integrand on $(0, 1)$ by $|f(x)|x^{a-1}$ and on $(1, \infty)$ by $|f(x)|x^{b-1}$. Both functions are integrable, so the analytic parameter-integral theorem applies. The two bounds cannot be interchanged: the ordering of powers reverses between $x < 1$ and $x > 1$.

This is a Fourier/Laplace type transform on the multiplicative group $(0, \infty)$. Convergence must be checked separately at both endpoints. Even if f decays rapidly at infinity, the behavior $f(x) \sim x^{-2}$ as $x \downarrow 0$ still forces $\Re s > 2$.

Proposition 6.16 (Mellin scaling formula)**.** *If $a > 0$ and $\mathcal{M}f(s)$ converges absolutely at a point s, then*

$$\mathcal{M}[f(a\,\cdot)](s) = a^{-s}\mathcal{M}f(s).$$

Proof. Set $u = ax$ in the integral. Then $x^{s-1}\,\mathrm{d}x = a^{-s}u^{s-1}\,\mathrm{d}u$. Absolute convergence justifies the substitution. □

Example 6.17 (Computing fundamental strips)**.** *Let $f(x) = \mathrm{e}^{-x}$. Near 0, $f(x) \asymp 1$, while at infinity it decays exponentially. Thus the fundamental region is $\Re s > 0$, and $\mathcal{M}f(s) = \Gamma(s)$. If $f(x) = 1/(1+x)$, the behavior near 0 again requires $\Re s > 0$, but at infinity the integrand is comparable to x^{s-2} and also requires $\Re s < 1$. The fundamental strip is therefore $0 < \Re s < 1$.*

Example 6.18 (The Gamma factor and ζ)**.** *The substitution $u = nx$ gives*

$$\int_0^\infty \mathrm{e}^{-nx}x^{s-1}\,\mathrm{d}x = \Gamma(s)n^{-s}.$$

When $\sigma > 1$, absolute convergence permits summation over n, and $\sum_{n\ge1}\mathrm{e}^{-nx} = 1/(\mathrm{e}^x - 1)$. Therefore

$$\Gamma(s)\zeta(s) = \int_0^\infty \frac{x^{s-1}}{\mathrm{e}^x - 1}\,\mathrm{d}x, \qquad \sigma > 1. \tag{6.3}$$

Near 0, the denominator is comparable to x, so the integral requires exactly $\sigma > 1$; exponential decay controls infinity.

Proposition 6.19 (A sufficient condition for Mellin inversion)**.** *Let c be a real number in the fundamental strip, and suppose that $u \mapsto \mathcal{M}f(c+iu)$ belongs to $L^1(\mathbb{R})$. Then, at every continuity point $x > 0$ of f,*

$$f(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \mathcal{M}f(s)x^{-s}\,\mathrm{d}s.$$

The integral on the right converges absolutely.

Proof. Put $h(v) = \mathrm{e}^{cv}f(\mathrm{e}^v)$. The substitution $x = \mathrm{e}^v$ gives $\int_{\mathbb{R}}|h(v)|\,\mathrm{d}v = \int_0^\infty |f(x)|x^{c-1}\,\mathrm{d}x < \infty$ and

$$\mathcal{M}f(c+iu) = \int_{\mathbb{R}} h(v)\mathrm{e}^{iuv}\,\mathrm{d}v.$$

The L^1 Fourier inversion theorem gives $h(v) = (2\pi)^{-1}\int_{\mathbb{R}}\mathcal{M}f(c+iu)\mathrm{e}^{-iuv}\,\mathrm{d}u$ at every continuity point of h. Multiply by e^{-cv} and set $v = \log x$ to obtain the claimed formula. For the transform convention and standard form of Mellin inversion, see [7, §1.14(iv)]. □

6.5 Perron's formula

Perron's formula is Mellin inversion for the sharp cutoff $\mathbf{1}_{n\le x}$.

Definition 6.20 (Symmetric limit of a Perron integral)**.** *If G is continuous on the line $\Re s = c$, the notation*

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} G(s)\,\mathrm{d}s$$

means the symmetric limit

$$\lim_{T\to\infty}\frac{1}{2\pi i}\int_{c-iT}^{c+iT} G(s)\,\mathrm{d}s,$$

provided that the limit exists. It does not automatically denote an absolutely convergent integral; the two endpoints must tend to infinity at the same rate.

Lemma 6.21 (Perron kernel)**.** *If $c>0$, $y>0$, and $y\neq 1$, then*

$$\lim_{T\to\infty}\frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{y^s}{s}\,\mathrm{d}s = \begin{cases}1, & y>1,\\ 0, & 0<y<1.\end{cases}$$

At $y=1$, the symmetric limit is $1/2$.

Proof. Denote the integral at finite height by $I_T(y)$ and write $u=\log y$. Fix $T>0$. If $u>0$, close to the left with the rectangle whose vertices are $c\pm iT$ and $-R\pm iT$. The function e^{us}/s has residue 1 at $s=0$. As $R\to\infty$, the left vertical integral tends to zero, while the sum of the absolute values of the two horizontal integrals is at most

$$\frac{2}{T}\int_{-\infty}^{c}\mathrm{e}^{u\sigma}\,\mathrm{d}\sigma = \frac{2y^c}{T|u|}.$$

If $u<0$, close to the right; no pole is enclosed, and the same estimate uses $\int_c^{\infty}\mathrm{e}^{u\sigma}\,\mathrm{d}\sigma = y^c/|u|$. Thus, for $y\neq 1$, $I_T(y)-\mathbf{1}_{y>1}=O(y^c/(T|u|))$, which also proves the stated limit.

For $u=0$, direct calculation gives

$$\frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{\mathrm{d}s}{s} = \frac{1}{\pi}\arctan(T/c)\longrightarrow\frac{1}{2}.$$

To obtain a bound that remains useful near $y=1$, conjugate symmetry gives

$$I_T(y)=\frac{y^c}{\pi}\int_0^T\frac{c\cos(tu)+t\sin(tu)}{c^2+t^2}\,\mathrm{d}t.$$

When $T|u|\le 1$, the absolute value of the first integral is at most $\pi/2$; using $|\sin(tu)|\le t|u|$ bounds the second by $T|u|\le 1$. Also $1\le y^c$ when $y>1$. Combining this with the earlier estimate away from the jump yields

$$I_T(y)=\mathbf{1}_{y>1}+O\left(y^c\min\left\{1,\frac{1}{T|\log y|}\right\}\right)\quad(y\neq 1),$$

with an absolute implied constant. □

Theorem 6.22 (Truncated Perron formula)**.** *Let $c>0$, and suppose $F(s)=\sum a_n n^{-s}$ converges absolutely on $\Re s=c$. Let $x>0$ be nonintegral and $T\ge 1$. Then*

$$\begin{aligned}\sum_{n\le x}a_n &= \frac{1}{2\pi i}\int_{c-iT}^{c+iT}F(s)\frac{x^s}{s}\,\mathrm{d}s\\ &\quad+O\left(\sum_{n\ge 1}|a_n|\left(\frac{x}{n}\right)^c\min\left\{1,\frac{1}{T|\log(x/n)|}\right\}\right).\end{aligned}\tag{6.4}$$

Proof. Absolute convergence permits the interchange of sum and integral on the finite segment:

$$\frac{1}{2\pi i}\int_{c-iT}^{c+iT} F(s)\frac{x^s}{s}\,\mathrm{d}s = \sum_{n\geq 1} a_n I_T(x/n).$$

Apply the truncated estimate from the Perron-kernel proof term by term, then sum using the triangle inequality. Since x is not an integer, $\log(x/n)$ never vanishes; the total error is controlled by $\sum |a_n|(x/n)^c < \infty$. □

Theorem 6.23 (Perron's formula)**.** *Let $c > 0$, and suppose $F(s) = \sum a_n n^{-s}$ converges absolutely on $\Re s = c$. If $x > 0$ is not an integer, then*

$$\sum_{n\leq x} a_n = \lim_{T\to\infty} \frac{1}{2\pi i}\int_{c-iT}^{c+iT} F(s)\frac{x^s}{s}\,\mathrm{d}s. \tag{6.5}$$

If x is an integer, the right-hand side is $\sum_{n<x} a_n + a_x/2$.

Proof. At finite height, absolute convergence permits termwise integration, producing a weighted sum of the Perron kernel at $y = x/n$. For every $n \neq x$, the error bound from Lemma 6.21 gives

$$|a_n|\,|I_T(x/n) - \mathbf{1}_{n<x}| \ll |a_n|(x/n)^c.$$

The bound is summable in n and independent of T, while each error tends to zero. Dominated convergence for series therefore permits the interchange of limit and sum. If x is an integer, first separate the term $n = x$; since $I_T(1) = \pi^{-1}\arctan(T/c) \to 1/2$, it contributes half its weight. □

In actual estimates, one first shifts a contour at finite height T and controls the truncation error. Residues from crossed poles give main terms; the new vertical integral and the horizontal segments contribute to the error.

Reading a main term from a pole: a smoothed example

The sharp Perron kernel decays only like $1/|s|$, so moving its contour directly requires delicate estimates on the horizontal segments. First consider the smoothed version with an extra factor $(s+1)^{-1}$. For $c > 0$ and $y > 0$,

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{y^s}{s(s+1)}\,\mathrm{d}s = \begin{cases} 1 - y^{-1}, & y > 1, \\ 0, & 0 < y \leq 1. \end{cases}$$

Again use a finite-height rectangle, keeping track of its boundary terms. Put $u = \log y \neq 0$, fix $T > 0$, and close to the left or right according to the sign of u. Send the far vertical side to infinity; its integral vanishes by exponential decay. On both horizontal sides, $|s(s+1)| \geq T^2$, so the sum of their absolute integrals is at most

$$\frac{2y^c}{T^2|\log y|}.$$

Now let $T \to \infty$. Closing to the left gives residues 1 and $-y^{-1}$; closing to the right encloses no poles. The integrand on the vertical line decays like $|t|^{-2}$, giving absolute convergence. An integrable bound uniform in a fixed neighborhood of $y = 1$ permits passage to the limit there as well.

Take $F(s) = \zeta(s)$, $c > 1$, and any $x > 0$. Since

$$x^c \sum_{n\geq 1} n^{-c} \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{|(c+it)(c+1+it)|} < \infty,$$

we may integrate term by term on the entire vertical line, obtaining

$$\sum_{n\leq x} \left(1 - \frac{n}{x}\right) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \zeta(s) \frac{x^s}{s(s+1)}\,\mathrm{d}s. \tag{6.6}$$

Moving the line of integration to $0 < c' < 1$ crosses the simple pole of ζ at $s = 1$, whose residue is

$$\operatorname*{Res}_{s=1} \zeta(s) \frac{x^s}{s(s+1)} = \frac{x}{2}.$$

We can justify this contour shift explicitly. Splitting off the terms up to the integer N in (5.3) gives

$$\zeta(s) = \sum_{n\leq N} n^{-s} + \frac{N^{1-s}}{s-1} - s\int_N^{\infty} \{u\} u^{-s-1}\,\mathrm{d}u \qquad (\Re s > 0,\ s \neq 1).$$

On the fixed strip $c' \leq \sigma \leq c$, take $N = \lceil |t| \rceil$ for $|t| \geq 2$. Bounding the three terms separately gives $\zeta(\sigma + it) = O_{c',c}(|t|^{1-c'})$. Thus the integrand on the new vertical line is $O(x^{c'}|t|^{-1-c'})$ and is absolutely integrable. For fixed x, each horizontal integral is $O_{x,c',c}(T^{-1-c'}) \to 0$. The new line therefore contributes $O_{c'}(x^{c'}) = o(x)$. Direct evaluation of the left-hand side gives the sharper estimate

$$\lfloor x \rfloor - \frac{\lfloor x \rfloor (\lfloor x \rfloor + 1)}{2x} = \frac{x}{2} + O(1).$$

Thus the pole at $s = 1$ supplies the main term of the smoothed sum. Unsmooth partial sums with general coefficients require either the error control in Theorem 6.22 or a Tauberian hypothesis.

Chapter Summary

- Growth of the partial sums $A(x)$ determines a half-plane of convergence for a Dirichlet series.
- In a half-plane of absolute convergence, a Dirichlet series uniquely determines its coefficients.
- Absolute convergence makes Dirichlet convolution correspond to multiplication of series and makes multiplicativity produce Euler products.
- Perron's formula recovers partial sums from a vertical integral, with half weight at integer endpoints.
- The truncated Perron formula expresses the finite-height error in terms of the logarithmic distance from n to x.
- Before shifting a contour, one must retain the truncation height and verify the boundary integrals; a formal residue calculation is not enough.

Common Pitfalls

- Confusing ordinary with absolute convergence and rearranging a double series in a region of conditional convergence.
- Assuming that an analytic continuation to the left is still equal to the original Dirichlet series.
- Ignoring the distinct convergence restrictions imposed by the endpoints 0 and ∞ in a Mellin transform.
- Forgetting the half weight at integer points in Perron's formula, or using a finite-height integral without its truncation error.

Exercises

Exercise 6.1. Use Theorem 6.8 and $\mathbf{1} * \Lambda = \log$ to derive $-\zeta'(s) = \zeta(s) \sum \Lambda(n) n^{-s}$ for $\sigma > 1$.

Exercise 6.2. Without citing Proposition 6.13, prove again that $\Gamma(1/2) = \sqrt{\pi}$, explaining step by step why Tonelli's theorem and the change to polar coordinates apply.

Exercise 6.3. Let $a_n = (-1)^{n-1}$. Find the abscissae of convergence and absolute convergence of the corresponding Dirichlet series, and use Theorem 6.2 to describe its region of analyticity.

Exercise 6.4. Split the integral in (6.3) into $(0, 1)$ and $(1, \infty)$, and explain how subtracting $1/x$ continues the right-hand side to a larger half-plane.

Exercise 6.5. For $F(s) = \zeta(s)$, formally move the Perron integral from $c > 1$ to $0 < c' < 1$. Identify the crossed pole and its main term, and list the three types of boundary integrals that must be controlled to turn the formal argument into a rigorous estimate.

Exercise 6.6. Find the Mellin transforms and fundamental strips of $f(x) = \mathbf{1}_{(0,1)}(x)$ and $g(x) = \mathbf{1}_{(1,\infty)}(x) x^{-2}$. Why are the two fundamental strips determined by different endpoints?

Exercise 6.7. Prove the smoothed-kernel formula used in (6.6) and compute the residues at $s = 0$ and $s = -1$. Hint: treat separately the contour-closing directions for $y > 1$ and $0 < y < 1$.

Chapter 7

Dirichlet Characters, L-Functions, and Primes in Arithmetic Progressions

Learning objectives. Use character orthogonality to isolate residue classes, establish the basic properties and boundary nonvanishing of $L(s,\chi)$, and prove the prime number theorem in progressions for fixed moduli.

Prerequisites. The Chinese remainder theorem, finite abelian groups, Dirichlet series, and the pole of ζ.

Role in the book. This chapter applies generating-function methods to reduced residue classes modulo q, proving Dirichlet's theorem and their asymptotic distribution.

7.1 Characters on finite groups

Throughout this chapter, q is a positive integer and a, n are integers. In asymptotic statements the modulus q is fixed unless stated otherwise.

Definition 7.1 (Dirichlet, principal, and conjugate characters)**.** *A Dirichlet character modulo q is the extension to all integers of a homomorphism $\chi : (\mathbb{Z}/q\mathbb{Z})^\times \to \mathbb{C}^\times$: extend it periodically with period q and set $\chi(n) = 0$ when $(n,q) > 1$. The extension of the trivial homomorphism, which is identically 1, is the* principal character χ_0*, so $\chi_0(n) = 1$ if and only if $(n,q) = 1$. The function $\overline{\chi}(n) = \overline{\chi(n)}$ is again a character modulo q and is called the* conjugate character*. If $\chi = \overline{\chi}$, the character is* real*; otherwise it is* nonreal*.*

Every element of a finite group has finite order, so $\chi(n)$ is a root of unity when $(n,q) = 1$; in particular, $|\chi(n)| = 1$. The extended function χ is completely multiplicative.

Example 7.2 (Modulo 4)**.** *There are two characters modulo 4: the principal character χ_0 and*

$$\chi_4(n) = \begin{cases} 0, & 2 \mid n, \\ 1, & n \equiv 1 \pmod 4, \\ -1, & n \equiv 3 \pmod 4. \end{cases}$$

The latter distinguishes the two odd residue classes.

Proposition 7.3 (Dual of a finite abelian group)**.** *The characters of a finite abelian group G form its dual group $\widehat{G}$, and $|\widehat{G}| = |G|$. Consequently, there are exactly $\varphi(q)$ Dirichlet characters modulo q.*

Proof. By the structure theorem for finite abelian groups, $G \cong C_{m_1} \times \cdots \times C_{m_r}$. A character of the cyclic group $C_m = \langle g \rangle$ is determined by $\chi(g)$, which may be any mth root of unity, so there are exactly m choices. Characters on a direct product multiply coordinatewise. Thus $|\widehat{G}| = m_1 \cdots m_r = |G|$. Take $G = (\mathbb{Z}/q\mathbb{Z})^\times$ for the last statement. □

Example 7.4 (Character table modulo 5)**.** *The element* 2 *generates* $(\mathbb{Z}/5\mathbb{Z})^\times$*. Define* $\chi_1(2) = i$*. The four characters are* $\chi_0, \chi_1, \chi_1^2, \chi_1^3$*:*

	1	2	3	4
χ_0	1	1	1	1
χ_1	1	i	$-i$	-1
χ_1^2	1	-1	-1	1
χ_1^3	1	$-i$	i	-1

The four values cannot be assigned arbitrarily; they must respect multiplication, for instance $\chi(4) = \chi(2)^2$*.*

Theorem 7.5 (Character orthogonality)**.** *For characters modulo* q*,*

$$\sum_{a \bmod q} \chi(a)\overline{\psi(a)} = \begin{cases} \varphi(q), & \chi = \psi, \\ 0, & \chi \neq \psi, \end{cases}$$

and, when $(a, q) = 1$*,*

$$\frac{1}{\varphi(q)} \sum_{\chi \bmod q} \overline{\chi(a)}\chi(n) = \begin{cases} 1, & n \equiv a \pmod q, \\ 0, & \text{otherwise.} \end{cases} \tag{7.1}$$

Proof. If ρ is a nontrivial character, choose $b \in G$ with $\rho(b) \neq 1$. The substitution $a \mapsto ba$ gives $\sum_G \rho(a) = \rho(b) \sum_G \rho(a)$, so the sum is zero; for the trivial character it is $|G|$. Apply this to $\rho = \chi\overline{\psi}$ to obtain the first identity. For the second identity, suppose $(n, q) = 1$ and put $g = na^{-1} \in G$. If $g \neq 1$, at least one coordinate of g in the cyclic decomposition is nontrivial. Choose a character on that cyclic factor that is nontrivial at this coordinate, and the trivial character on every other factor. This gives a character ρ with $\rho(g) \neq 1$. Thus $\chi \mapsto \chi(g)$ is a nontrivial character on $\widehat{G}$, and its sum is zero. If $g = 1$, each summand is 1, giving $\varphi(q)$. If $(n, q) > 1$, every $\chi(n)$ is zero. □

7.2 Primitive characters, conductors, and Gauss sums

The same arithmetic oscillation can be represented by characters with different moduli. Its conductor records the smallest modulus on which it genuinely depends.

Definition 7.6 (Induction, primitive characters, and conductors)**.** *Let* $q^* \mid q$*. A character* χ *modulo* q *is* induced *by a character* χ^* *modulo* q^* *if*

$$\chi(n) = \chi^*(n)\chi_0^{(q)}(n) \qquad (n \in \mathbb{Z}).$$

The least positive modulus q^* *that can induce* χ *is its* conductor*. If the conductor equals* q*, then* χ *is* primitive modulo q*; otherwise it is imprimitive.*

The minimum is taken among positive divisors of q. It exists because q itself always induces χ. Induction is transitive, so a character at the least inducing modulus must be primitive, or the modulus could be reduced further. For each $d \mid q$, the reduction map $(\mathbb{Z}/q\mathbb{Z})^\times \to (\mathbb{Z}/d\mathbb{Z})^\times$

is surjective: given $(b,d)=1$, the Chinese remainder theorem supplies $n \equiv b \pmod d$ and $n \equiv 1 \pmod p$ for all primes $p \mid q$ with $p \nmid d$, and then $(n,q)=1$. Thus the inducing character at a fixed modulus is uniquely determined by χ; in particular, a principal character can only be induced by a principal character.

Example 7.7. *The nonprincipal character χ_4 modulo 4 induces a character modulo 8: on odd integers it retains the periodic values $1,-1,1,-1$, and on even integers it is zero. This character has conductor 4 and is therefore imprimitive when regarded modulo 8. The principal character modulo q has conductor 1.*

Definition 7.8 (Gauss sum)**.** *The Gauss sum of a character χ modulo q is*

$$\mathfrak{g}(\chi) = \sum_{a \bmod q} \chi(a) \mathrm{e}^{2\pi i a/q}.$$

More generally, its finite Fourier transform at frequency n is

$$\mathfrak{g}_n(\chi) = \sum_{a \bmod q} \chi(a) \mathrm{e}^{2\pi i a n/q}.$$

Proposition 7.9 (Basic transformation of Gauss sums)**.** *If $(n,q)=1$, then*

$$\mathfrak{g}_n(\chi) = \overline{\chi(n)}\, \mathfrak{g}(\chi).$$

Proof. The substitution $b \equiv an \pmod q$ permutes the residue classes modulo q, and $\chi(a) = \chi(bn^{-1}) = \chi(b)\overline{\chi(n)}$. Substitute this identity in the sum. □

For primitive characters one can also prove $|\mathfrak{g}(\chi)| = \sqrt{q}$ and vanishing properties of the corresponding Fourier coefficients when $(n,q)>1$. These facts provide another classical route to functional equations and nonvanishing for real characters. Dirichlet's theorem here requires only the definitions and induction relation above.

7.3 Dirichlet L-functions

Definition 7.10. *For a character χ modulo q, define*

$$L(s,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \qquad \Re s > 1.$$

Proposition 7.11 (Partial sums of nonprincipal characters)**.** *If $\chi \neq \chi_0$, then for every $x \geq 1$,*

$$\left| \sum_{n \leq x} \chi(n) \right| \leq q.$$

Consequently, the original series for $L(s,\chi)$ converges and is analytic in $\Re s > 0$.

Proof. By orthogonality, the sum over a complete period is $\sum_{a=1}^{q} \chi(a) = 0$. Divide $\{1,\dots,\lfloor x \rfloor\}$ into complete blocks of length q and a final incomplete block. Every complete period cancels, and the remainder has at most q terms, each of absolute value at most 1. Analyticity follows from Theorem 6.2. □

Proposition 7.12. *For $\sigma > 1$,*

$$L(s,\chi) = \prod_p (1 - \chi(p)p^{-s})^{-1}. \tag{7.2}$$

This product has no zeros in $\sigma > 1$. If $\chi \neq \chi_0$, the original series analytically continues $L(s,\chi)$ to $\sigma > 0$; whereas

$$L(s,\chi_0) = \zeta(s) \prod_{p|q} (1 - p^{-s})$$

is meromorphic in $\sigma > 0$, with its only pole a simple pole at $s = 1$ of residue $\varphi(q)/q$.

Proof. Complete multiplicativity and absolute convergence give the Euler product. More precisely,

$$\ell_\chi(s) = \sum_p \sum_{k\geq 1} \frac{\chi(p)^k}{kp^{ks}}$$

converges absolutely and uniformly on compact subsets of $\sigma > 1$, and $L(s,\chi) = \exp(\ell_\chi(s))$. This proves nonvanishing and specifies the analytic branch of $\log L(s,\chi)$ used below; on the real axis it tends to 0 as $s \to +\infty$. For nonprincipal χ, Proposition 7.11 gives analytic continuation. The formula for the principal character deletes the Euler factors for $p \mid q$; its residue is $\prod_{p|q}(1 - 1/p) = \varphi(q)/q$. □

Proposition 7.13 (Finite Euler factors for induced characters)**.** *Suppose a character χ modulo q is induced by a primitive character χ^* of conductor q^*. Then, for $\Re s > 1$ and by analytic continuation wherever both sides are defined,*

$$L(s,\chi) = L(s,\chi^*) \prod_{\substack{p|q \\ p\nmid q^*}} \left(1 - \frac{\chi^*(p)}{p^s}\right).$$

In particular, all these finite factors are nonzero at $s = 1$. If χ is nonprincipal, then χ^* is also nonprincipal, and $L(1,\chi) = 0$ if and only if $L(1,\chi^*) = 0$. If χ is principal, both sides have a simple pole at $s = 1$, so $L(1,\chi)$ cannot be treated as a finite function value.

Proof. For $p \nmid q$, the two characters agree at p and its powers. For $p \mid q$, the local Euler factor of the induced character is 1. If $p \mid q^*$, then $\chi^*(p) = 0$ and no change is needed. If $p \mid q$ but $p \nmid q^*$, the local factor $(1 - \chi^*(p)p^{-s})^{-1}$ must be deleted from $L(s,\chi^*)$, producing the finite product. At $s = 1$, $|\chi^*(p)/p| \leq 1/p < 1$, so every factor is nonzero. □

Example 7.14. *For the nonprincipal character modulo 4,*

$$L(s,\chi_4) = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \cdots.$$

At $s = 1$, the Leibniz series gives $L(1,\chi_4) = \pi/4 \neq 0$. The general result for arbitrary moduli depends on the nonvanishing theorem below.

7.4 Nonvanishing at the boundary point 1

The case of real characters requires an analytic lemma that depends only on nonnegativity of the coefficients.

Lemma 7.15 (Landau's lemma)**.** *Let $F(s) = \sum_{n\geq 1} a_n n^{-s}$ have coefficients $a_n \geq 0$ and finite abscissa of convergence σ_c. Then the real point $s = \sigma_c$ is a singularity of F.*

Proof. Suppose instead that F is analytic in the disk $|s-\sigma_{\mathrm{c}}| < r$. Put $s_0 = \sigma_{\mathrm{c}} + r/4$. The disk centered at s_0 with radius $r/2$ lies inside this disk and extends to the left of σ_{c}. Because the coefficients are nonnegative, convergence at a real point is absolute convergence. Termwise differentiation is therefore valid at s_0, and

$$(-1)^k F^{(k)}(s_0) = \sum_{n\ge 1} a_n (\log n)^k n^{-s_0} \ge 0.$$

Indeed, choose $0 < \eta < (s_0 - \sigma_{\mathrm{c}})/2$. For fixed k and $|s - s_0| \le \eta$,

$$a_n (\log n)^k |n^{-s}| \ll_{k,\eta} a_n n^{-(s_0 - 2\eta)}.$$

The right side is summable because $s_0 - 2\eta > \sigma_{\mathrm{c}}$, giving uniform convergence of the derivative series in this neighborhood. For $h > 0$ within the disk, expand in a Taylor series and use Tonelli's theorem to interchange the two nonnegative sums:

$$\begin{aligned} F(s_0 - h) &= \sum_{k\ge 0} \frac{(-h)^k}{k!} F^{(k)}(s_0) \\ &= \sum_{n\ge 1} a_n n^{-s_0} \sum_{k\ge 0} \frac{(h \log n)^k}{k!} = \sum_{n\ge 1} a_n n^{-(s_0 - h)}. \end{aligned}$$

The left side is a finite analytic value, so the series on the right converges. Choosing $r/4 < h < r/2$ makes the original series converge to the left of its abscissa, a contradiction. □

Theorem 7.16 (Dirichlet nonvanishing theorem)**.** *If χ is a nonprincipal Dirichlet character, then*

$$L(1, \chi) \ne 0.$$

Proof. The Euler product is not absolutely convergent at $s = 1$, so it cannot decide $L(1, \chi)$ directly. First suppose χ is nonreal. Taking logarithms of Euler products for $\sigma > 1$ gives

$$\begin{aligned} &3 \log \zeta(\sigma) + 4 \Re \log L(\sigma, \chi) + \Re \log L(\sigma, \chi^2) \\ &\quad = \sum_p \sum_{k\ge 1} \frac{3 + 4\Re(\chi(p)^k) + \Re(\chi(p)^{2k})}{k p^{k\sigma}}. \end{aligned}$$

If $p \nmid q$, write $\chi(p)^k = \mathrm{e}^{i\theta}$; the numerator is $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2 \ge 0$. If $p \mid q$, it is 3. The right-hand side is therefore nonnegative, and exponentiation gives

$$\zeta(\sigma)^3 |L(\sigma, \chi)|^4 |L(\sigma, \chi^2)| \ge 1. \tag{7.3}$$

If $L(1, \chi) = 0$, the orders of the zeros and poles near 1 make the left-hand side tend to zero: ζ^3 supplies a pole of order at most three, while $|L(\chi)|^4$ supplies a zero of order at least four. Since a nonreal character cannot satisfy $\chi^2 = \chi_0$, the last factor has no pole. This contradicts (7.3).

Now suppose χ is a real nonprincipal character. Multiplication of Dirichlet series gives, for $\sigma > 1$,

$$\zeta(s) L(s, \chi) = \sum_{n\ge 1} \frac{a(n)}{n^s}, \qquad a(n) = \sum_{d \mid n} \chi(d).$$

The function a is multiplicative. At prime powers,

$$a(p^k) = 1 + \chi(p) + \cdots + \chi(p)^k = \begin{cases} k + 1, & \chi(p) = 1, \\ 1, & \chi(p) = 0, \\ 1 \ (k \text{ even}),\ 0 \ (k \text{ odd}), & \chi(p) = -1. \end{cases}$$

Thus $a(n) \geq 0$, and $a(m^2) \geq 1$. The series consequently diverges at $s = 1/2$, so its abscissa of convergence satisfies $1/2 \leq \sigma_{\mathrm{c}} \leq 1$; in particular, it is finite.

If $L(1, \chi) = 0$, this zero cancels the simple pole of ζ at 1. Proposition 7.12 also says that $L(s, \chi)$ is analytic in $\Re s > 0$, so $\zeta(s)L(s, \chi)$ would be analytic throughout $\Re s > 0$. This contradicts Landau's lemma, which says that the real point $\sigma_{\mathrm{c}} \geq 1/2$ must be a singularity. To justify the continuation in this last step, the product and the nonnegative-coefficient series agree initially in $\Re s > 1$, and hence throughout $\Re s > \sigma_{\mathrm{c}}$ by the identity theorem. The product therefore supplies precisely the continuation across the abscissa that Landau's lemma forbids. Both the real and nonreal cases are now complete. □

7.5 Dirichlet's theorem

Definition 7.17 (Counting functions in arithmetic progressions)**.** *Let* $(a, q) = 1$. *Define*

$$\pi(x; q, a) = \sum_{\substack{p \leq x \\ p \equiv a \,(q)}} 1, \qquad \vartheta(x; q, a) = \sum_{\substack{p \leq x \\ p \equiv a \,(q)}} \log p,$$

and

$$\psi(x; q, a) = \sum_{\substack{n \leq x \\ n \equiv a \,(q)}} \Lambda(n).$$

Dirichlet's theorem asserts that $\pi(x; q, a) \to \infty$*; the prime number theorem for arithmetic progressions also specifies the asymptotic rate.*

Example 7.18. *Modulo* 4, $\pi(20; 4, 1) = 3$, *corresponding to* $5, 13, 17$, *whereas* $\pi(20; 4, 3) = 4$, *corresponding to* $3, 7, 11, 19$. *It is easy to count the residue representative* 1 *by mistake, but* 1 *is not prime; the counting function always ranges over primes only.*

Theorem 7.19 (Primes in arithmetic progressions)**.** *If* $q \geq 1$ *and* $(a, q) = 1$, *then the congruence class*

$$p \equiv a \pmod{q}$$

contains infinitely many primes.

Complete deduction from the nonvanishing theorem. For $\sigma > 1$, take logarithms of the Euler products, choosing the continuous branch that tends to 0 as $\sigma \to \infty$:

$$\log L(\sigma, \chi) = \sum_{p} \sum_{k \geq 1} \frac{\chi(p)^k}{k p^{k\sigma}}.$$

Weight by $\overline{\chi(a)}$ and sum over all characters modulo q. Orthogonality (7.1) makes the $k = 1$ term isolate exactly the primes $p \equiv a \pmod{q}$:

$$\sum_{\chi \bmod q} \overline{\chi(a)} \log L(\sigma, \chi) = \varphi(q) \sum_{p \equiv a \,(q)} \frac{1}{p^{\sigma}} + O_q(1). \tag{7.4}$$

The error comes from $k \geq 2$ and is bounded in absolute value by $\varphi(q) \sum_p \sum_{k \geq 2} (k p^{k\sigma})^{-1}$. More explicitly, for $\sigma \geq 1$,

$$\sum_{p} \sum_{k \geq 2} \frac{1}{k p^{k\sigma}} \leq \sum_{p} \frac{p^{-2}}{1 - p^{-1}} \leq 2 \sum_{n \geq 2} \frac{1}{n^2} < \infty,$$

so the $O_q(1)$ error is uniformly bounded as $\sigma \downarrow 1$.

As $\sigma \downarrow 1$, the principal-character term satisfies

$$\log L(\sigma, \chi_0) = \log \frac{1}{\sigma - 1} + O_q(1).$$

For every nonprincipal character, Proposition 7.12 and Theorem 7.16 give a small disk about 1 in which $L(s, \chi)$ is analytic and nonzero, so it has an analytic logarithm g_χ there. On the intersection with $\Re s > 1$, the chosen ℓ_χ differs from g_χ by a constant integer multiple of $2\pi i$. Thus $\ell_\chi(\sigma)$ remains bounded as $\sigma \downarrow 1$. Taking real parts in (7.4) now gives

$$\sum_{p \equiv a\,(q)} \frac{1}{p^\sigma} = \frac{1}{\varphi(q)} \log \frac{1}{\sigma - 1} + O_q(1). \tag{7.5}$$

The right-hand side tends to infinity, so the residue class cannot contain only finitely many primes. □

Remark 7.20. For a set of primes $\mathcal{P}$, its Dirichlet density among the primes is the limit

$$\delta_D(\mathcal{P}) = \lim_{\sigma \downarrow 1} \frac{\sum_{p \in \mathcal{P}} p^{-\sigma}}{\log(1/(\sigma - 1))},$$

when this limit exists. Taking logarithms of the Euler product and bounding higher prime powers gives $\sum_p p^{-\sigma} = \log(1/(\sigma-1)) + O(1)$, so the denominator can equivalently be replaced by $\sum_p p^{-\sigma}$. Formula (7.5) gives each reduced residue class Dirichlet density $1/\varphi(q)$. We next prove nonvanishing on the whole boundary and obtain the same value for the relative natural density $\lim_{x \to \infty} \pi(x; q, a)/\pi(x)$.

7.6 Boundary nonvanishing and primes in progressions

At $s = 1$, a real character has χ^2 principal, which introduces another pole into the trigonometric inequality and requires the Landau argument. Off the real axis that pole is no longer at the point under consideration, so $3 + 4\cos\theta + \cos 2\theta \geq 0$ treats all characters.

Theorem 7.21 (Boundary nonvanishing of Dirichlet L-functions)**.** *For every Dirichlet character χ modulo q and real $t \neq 0$, we have $L(1 + it, \chi) \neq 0$. Together with Theorem 7.16, this shows that nonprincipal L-functions have no zeros on $\Re s = 1$; the principal L-function has a simple pole at $s = 1$.*

Proof. Fix $t \neq 0$. Expanding logarithms of Euler products in the region of absolute convergence $\sigma > 1$ gives

$$\begin{aligned} &3 \log \zeta(\sigma) + 4 \log |L(\sigma + it, \chi)| + \log |L(\sigma + 2it, \chi^2)| \\ &\quad = \sum_p \sum_{k \geq 1} \frac{3 + 4\Re(\chi(p)^k p^{-ikt}) + \Re(\chi(p)^{2k} p^{-2ikt})}{k p^{k\sigma}} \geq 0. \end{aligned}$$

If $p \nmid q$, put $z = \chi(p)^k p^{-ikt}$. Then $|z| = 1$, and the numerator is $3 + 4\Re z + \Re z^2 = 2(1 + \Re z)^2$. If $p \mid q$, the numerator is 3. Exponentiating yields

$$\zeta(\sigma)^3 |L(\sigma + it, \chi)|^4 |L(\sigma + 2it, \chi^2)| \geq 1.$$

If $L(1 + it, \chi) = 0$, analyticity from Proposition 7.12 gives $L(\sigma + it, \chi) = O(\sigma - 1)$. Since $1 + 2it \neq 1$, we also have $L(\sigma + 2it, \chi^2) = O(1)$, whether or not χ^2 is principal. Together with $\zeta(\sigma) = O((\sigma - 1)^{-1})$, this makes the left side $O(\sigma - 1)$ as $\sigma \downarrow 1$, contradicting its lower bound of 1. □

Theorem 7.22 (Prime number theorem in arithmetic progressions)**.** *Fix integers $q \geq 1$ and a with $(a, q) = 1$. As $x \to \infty$,*

$$\psi(x; q, a) \sim \frac{x}{\varphi(q)}, \qquad \vartheta(x; q, a) \sim \frac{x}{\varphi(q)}, \qquad \pi(x; q, a) \sim \frac{x}{\varphi(q) \log x}.$$

Proof. Take logarithmic derivatives of the absolutely convergent Euler products in $\Re s > 1$ and use orthogonality to obtain

$$F_a(s) := \sum_{\substack{n \geq 1 \\ n \equiv a \,(q)}} \frac{\Lambda(n)}{n^s} = \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \overline{\chi(a)} \left(-\frac{L'}{L}(s, \chi) \right).$$

The coefficients $\Lambda(n)\mathbf{1}_{n \equiv a \,(q)}$ on the left are nonnegative. The simple pole of the principal character at 1 gives $-L'/L$ the principal part $1/(s-1)$ there: write $L(s, \chi_0) = H(s)/(s-1)$, where H is analytic near 1 and $H(1) = \varphi(q)/q \neq 0$. Then

$$-\frac{L'}{L}(s, \chi_0) = \frac{1}{s-1} - \frac{H'(s)}{H(s)}.$$

The nonprincipal logarithmic derivatives are analytic near 1 by Theorem 7.16. Near every $1 + it$ with $t \neq 0$, all the logarithmic derivatives are analytic by Theorem 7.21. Consequently, $F_a(s) - 1/(\varphi(q)(s-1))$ extends continuously to $\Re s \geq 1$, and Theorem 5.11 yields $\psi(x; q, a) \sim x/\varphi(q)$. Only continuation near each boundary point is needed; no zero-free strip of uniform width is required. The Tauberian theorem is applied to the full series F_a, whose coefficients are nonnegative. The coefficients $\chi(n)\Lambda(n)$ of an individual twisted series need not be nonnegative, so this version of the theorem cannot be applied to them directly.

By Proposition 4.6, already proved in the main text,

$$0 \leq \psi(x; q, a) - \vartheta(x; q, a) \leq \psi(x) - \vartheta(x) = O(\sqrt{x} \log^2 x) = o(x),$$

so $\vartheta(x; q, a) \sim x/\varphi(q)$. Abel summation gives

$$\pi(x; q, a) = \frac{\vartheta(x; q, a)}{\log x} + \int_2^x \frac{\vartheta(t; q, a)}{t \log^2 t} \, \mathrm{d}t.$$

Since $\vartheta(t; q, a) \leq \vartheta(t) \ll t$, the integral is $O(x/\log^2 x)$, proving the claimed asymptotic formula for π. □

Remark 7.23. Theorem 7.22 and the prime number theorem give $\pi(x; q, a)/\pi(x) \to 1/\varphi(q)$: the reduced residue classes share the primes equally in the main term. The modulus q is fixed throughout; this proof supplies no error uniform in a growing modulus $q = q(x)$. As subsets of the positive integers, these sets of primes still have natural density 0. See also [8, Chs. 3–4] for boundary nonvanishing and the prime number theorem in progressions.

7.7 The proof route

For fixed $(a, q) = 1$, the logical chain is

$$\left.\begin{array}{c} \text{orthogonality and logarithms of Euler products} \\ \text{the principal pole and nonprincipal nonvanishing at } 1 \end{array}\right\} \Longrightarrow \sum_{p \equiv a(q)} p^{-\sigma} \to \infty.$$

Orthogonality isolates residue classes, while the nonvanishing theorem independently resolves the analytic question at $s = 1$. Together these yield the displayed divergence, which rules out the possibility of only finitely many primes in the progression.

Theorem 7.21 then extends nonvanishing to the whole boundary. Applying orthogonality to $-L'/L$ produces the nonnegative series for a residue class; Wiener–Ikehara yields the asymptotic equidistribution in Theorem 7.22.

Chapter Summary

- Dirichlet characters form a Fourier basis on $(\mathbb{Z}/q\mathbb{Z})^\times$, and orthogonality isolates residue classes.
- The function $L(s,\chi)$ has an Euler product for $\sigma > 1$; the series for a nonprincipal character continues to $\sigma > 0$.
- The statement $L(1,\chi) \neq 0$ for nonprincipal characters is the key arithmetic input and cannot be obtained from a formal product outside the region of absolute convergence.
- Applying orthogonality to $\log L$, the pole of the principal character forces the reciprocal-prime sum in every reduced residue class to diverge.
- Nonvanishing on the whole boundary lets the Tauberian theorem act on the nonnegative series for a residue class, giving the prime number theorem in progressions for each fixed modulus.

Common Pitfalls

- Forgetting that a character is defined to be zero when $(n,q) > 1$, and consequently misapplying orthogonality.
- Confusing "nonprincipal" with "nonreal"; real quadratic characters are precisely the delicate case in the nonvanishing proof.
- Expanding $\prod_p (1-\chi(p)/p)^{-1}$ directly at $s=1$ and declaring it nonzero.
- Claiming from Dirichlet's theorem alone that primes in each reduced residue class have relative natural density $1/\varphi(q)$ within the set of all primes. As subsets of the positive integers, these sets of primes all have natural density 0.

Exercises

Exercise 7.1. List all characters modulo 3, verify both orthogonality relations, and write the first six nonzero terms of the L-series for the nonprincipal character.

Exercise 7.2. Check Proposition 7.11 line by line, and use Abel summation on an infinite interval to write an integral representation of $L(1,\chi)$.

Exercise 7.3. Prove (7.3) in detail. Hint: for every p^k, use $3 + 4\Re(\chi(p)^k) + \Re(\chi(p)^{2k}) \geq 0$.

Exercise 7.4. For $q = 4$ and $a = 3$, write out (7.4) and explain how $L(1,\chi_4) = \pi/4$ forces infinitely many primes $p \equiv 3 \pmod 4$.

Exercise 7.5. Use Proposition 7.13 to relate the L-function of the character modulo 8 induced by χ_4 to $L(s,\chi_4)$, and verify that the finite product is empty, hence nonzero at $s = 1$.

Exercise 7.6. Find the conjugate of each character in the character table modulo 5, and verify that $\overline{\chi(n)} = \chi(n)^{-1}$ when $(n,5) = 1$. Which characters are real?

Exercise 7.7. Give a complete justification for interchanging the Taylor series and the double nonnegative sum in Landau's lemma, and explain why $a(m^2) \geq 1$ forces the abscissa of convergence of $\zeta(s)L(s,\chi)$ to be at least $1/2$.

Hints to the Exercises

Each hint gives only a point of entry. When an infinite process is involved, the convergence hypotheses of the cited theorem must still be checked.

Chapter 0

Exercise 0.1. List the primes and prime powers not exceeding the given bound. In ψ, the term $\log p$ appears $\lfloor \log x / \log p \rfloor$ times.

Exercise 0.2. Begin with (3.4). After substituting $\vartheta(t) = t + o(t)$, split the integral at a fixed T; on the large interval use $|\vartheta(t) - t| \le \varepsilon t$, and finally let $\varepsilon \downarrow 0$.

Exercise 0.3. Fix $\sigma > 0$ and make the coefficients nonzero only at $n = 2^k$, $k \ge 1$. Choose $a(2^k) = b(2^k) = 2^{k\sigma/2}/\sqrt{k}$ and compare the two original series with the series of pointwise products.

Exercise 0.4. From $\mathbf{1} * \varphi = \mathrm{id}$, predict $\sum \varphi(n) n^{-s} = \zeta(s-1)/\zeta(s)$. The residue at $s = 2$ is $1/\zeta(2)$, while the Perron kernel contributes another factor $1/2$.

Chapter 1

Exercise 1.1. Perform division with remainder until the remainder is zero, then substitute backward from the last nonzero remainder to find the Bézout coefficients.

Exercise 1.2. Classify by $d = \gcd(k, n)$. For fixed $d \mid n$, write $k = dm$; the conditions become $1 \le m \le n/d$ and $(m, n/d) = 1$, giving exactly $\varphi(n/d)$ elements.

Exercise 1.3. List the divisors when $n = p^\alpha$, then multiply the choices at distinct primes. In general, $f(p^j)$ cannot be replaced by $f(p)^j$.

Exercise 1.4. We have $\varphi(n_k)/n_k = \prod_{j \le k}(1 - 1/p_j)$. Take logarithms and use $\log(1-u) = -u + O(u^2)$. Since the sum of reciprocal prime squares converges, the behavior of the product is governed by $\sum 1/p$.

Exercise 1.5. Compute the exponents of $2, 3, 5, 7$ separately. Take the smaller exponent prime by prime for the greatest common divisor and the larger for the least common multiple; check numerically with $mn = (m, n)[m, n]$.

Exercise 1.6. There are two units modulo 3 and four modulo 5. Use the Chinese remainder theorem to combine the eight coordinate pairs.

Chapter 2

Exercise 2.1. List the six divisors of 12. If d has a square factor then $\mu(d) = 0$; the value $(\mu * \mathrm{id})(12)$ should agree with $\varphi(12)$ from Chapter 1.

Exercise 2.2. Both sides are multiplicative, so check only $n = p^k$. On the right, only $d = 1, p$ can be nonzero; treat $k = 0, 1$ and $k \geq 2$ separately.

Exercise 2.3. Use the convolution-inverse recursion at p and p^2. To prove multiplicativity, construct a candidate prime by prime and use uniqueness of the inverse.

Exercise 2.4. Expand $\log(n/d) = \log n - \log d$ in $\Lambda = \mu * \log$. For $n > 1$, use $\sum_{d|n} \mu(d) = 0$ to remove the term containing $\log n$; check $n = 1$ directly.

Exercise 2.5. Use the Möbius indicator identity $\mathbf{1}_{(a_1,\ldots,a_k,n)=1} = \sum_{d|(a_1,\ldots,a_k,n)} \mu(d)$. After interchanging finite sums, there are $(n/d)^k$ tuples all divisible by d.

Exercise 2.6. Guess $g(n) = \mu(n)n$. Complete multiplicativity of id verifies $(\mathrm{id} * (\mu\,\mathrm{id}))(n) = n \sum_{d|n} \mu(d)$.

Exercise 2.7. The local formula (2.2) is exactly the Cauchy product of two formal power series. The local series of $\mathbf{1}$ is $(1 - T)^{-1}$, whose inverse is $1 - T$.

Chapter 3

Exercise 3.1. In Abel summation, take $A(t) = \lfloor t \rfloor$ and $f(t) = t^\alpha$, and substitute $\lfloor t \rfloor = t + O(1)$ into both the boundary term and the integral.

Exercise 3.2. The value $\tau(n)$ is the number of ordered pairs of positive integers satisfying $ab = n$; summing over $n \leq x$ changes the condition to $ab \leq x$.

Exercise 3.3. The two main error sources are both $O(\sqrt{x})$: the sum of individual floor errors and the tail $\sum_{d>\sqrt{x}} d^{-2}$ multiplied by x.

Exercise 3.4. Once $A(x) = O(x)$ has been assumed, nonnegativity is no longer needed; it is usually only an easy way to establish that partial-sum bound. Apply Abel summation directly with $f(t) = 1/t$.

Exercise 3.5. The partial-sum bounds directly control only $F(u)G(v)$; they do not specify the errors in the two truncated sums. For the additional error bounds, compare the intersection of $A + B\alpha$ and $C - D\alpha$ with the endpoints of $[0, 1]$.

Exercise 3.6. Start from $\sum_{m \leq x} \sum_{d \leq x/m} d^\alpha$. The power-sum error is $O((x/m)^\alpha + 1)$; when summing over m, the behavior of $\sum m^{-\alpha}$ changes at $\alpha = 1$.

Chapter 4

Exercise 4.1. The exponent of p in the least common multiple is $\lfloor \log x / \log p \rfloor$. Taking logarithms turns the product into a sum over $p^k \leq x$.

Exercise 4.2. For the upper bound use $\log p \leq \log x$ term by term. For the lower bound retain only $\sqrt{x} < p \leq x$, where every weight is at least $\frac{1}{2} \log x$.

Exercise 4.3. Estimate the square term separately as $\vartheta(\sqrt{x}) = O(\sqrt{x} \log x)$. For $k \geq 3$, we have $x^{1/k} \leq x^{1/3}$ and there are $O(\log x)$ terms.

Exercise 4.4. From $n = \pi(p_n)$ obtain $n \sim p_n/\log p_n$. First compare $\log p_n$ with $\log n$, then substitute back.

Exercise 4.5. Retain the constants $2\log 2$, the rounding errors, and the constants for bounded ranges at every step. The largest losses come from compressing the prime weights of an entire dyadic interval into the central binomial coefficient and discarding small primes when returning from ϑ to π.

Exercise 4.6. The number of trailing zeros is $\min\{\nu_2(100!), \nu_5(100!)\}$. Since factors of 2 are more abundant than factors of 5, it is enough to compute $\lfloor 100/5\rfloor + \lfloor 100/25\rfloor$ carefully.

Exercise 4.7. For $x \geq 4$, split the integral at $\sqrt{x}$. The two parts are at most $\sqrt{x}/\log 2$ and $2x/\log x$, and $\sqrt{x} = O(x/\log x)$.

Chapter 5

Exercise 5.1. On $\sigma \geq 1+\delta$, dominate the original series by $n^{-1-\delta}$ and the derivative series by $(\log n)n^{-1-\delta}$; the latter also converges.

Exercise 5.2. Multiply (5.3) by $s-1$. The integral containing the fractional part is bounded near $s = 1$, so its product with $s-1$ tends to zero.

Exercise 5.3. Expand all three logarithmic derivatives using (5.2), take real parts, and fix one n. A zero of order m gives a negative principal coefficient of at least $4m$, while the pole contributes only 3.

Exercise 5.4. From $\psi - \vartheta = o(x)$ obtain $\vartheta(x) \sim x$, and hence $\vartheta(t) \ll t$. Split the integral in Abel summation at $\sqrt{x}$ to obtain $O(x/\log^2 x)$.

Exercise 5.5. Wiener–Ikehara needs only boundary continuity and nonnegative coefficients. The classical Perron route must also control the finite height, the vertical growth of ζ'/ζ, and the horizontal segments after the contour is shifted.

Exercise 5.6. Write $f'/f = m/(s-s_0) + g'/g$. Since $g(s_0) \neq 0$, continuity makes g nonzero in a smaller neighborhood.

Exercise 5.7. Use $\Lambda \geq 0$, the logarithmic-derivative series, the nonvanishing Euler product, boundary nonvanishing, and the simple pole of ζ at 1. Compare absolute convergence with analyticity, and compute the logarithmic derivative of $C/(s-1)$ for $C \neq 0$.

Chapter 6

Exercise 6.1. Apply multiplication of Dirichlet series to $\mathbf{1} * \Lambda = \log$. On the other hand, termwise differentiation gives $-\zeta'(s) = \sum(\log n)n^{-s}$; compare the two formulas in $\Re s > 1$.

Exercise 6.2. After squaring the Gaussian integral, Tonelli's theorem applies because the integrand is nonnegative. The Jacobian in polar coordinates is r, and the angular range for the first quadrant is $[0, \pi/2]$.

Exercise 6.3. The partial sums are always 0 or 1, so the ordinary abscissa of convergence is at most 0; at the real point $s = 0$, the terms do not tend to zero. The absolute-value series has the same order as $\zeta(\sigma)$.

Exercise 6.4. Near 0, write $(\mathrm{e}^x - 1)^{-1} = x^{-1} + O(1)$. Subtracting x^{-1} from the integrand improves it by one power and permits continuation across the original boundary.

Exercise 6.5. The residue crossed at $s = 1$ is x. The finite rectangle also has a new vertical side and two horizontal sides, all of which must be estimated together with the Perron truncation error.

Exercise 6.6. The first integral is $\int_0^1 x^{s-1}\,\mathrm{d}x$ and the second is $\int_1^\infty x^{s-3}\,\mathrm{d}x$. Read the fundamental strips from the respective integrability conditions at 0 and ∞.

Exercise 6.7. For $y > 1$, close to the left, enclosing 0 and -1; the sum of the residues is $1 - y^{-1}$. For $0 < y < 1$, close to the right, enclosing no poles.

Chapter 7

Exercise 7.1. The group $(\mathbb{Z}/3\mathbb{Z})^\times$ is cyclic of order two, so there are only the principal character and one real character taking the value -1. Add two complete periods to verify orthogonality quickly.

Exercise 7.2. Complete periods sum to zero, leaving a tail of fewer than q terms. To evaluate $L(1, \chi)$, take $f(t) = 1/t$ in Proposition 3.5.

Exercise 7.3. Start from the logarithmic expansion of the Euler product and fix each p^k. If $p \nmid q$, put $\chi(p)^k = \mathrm{e}^{i\theta}$; if $p \mid q$, the numerator is simply 3.

Exercise 7.4. There are only two characters modulo 4. The principal L-function has a pole at 1, while the nonprincipal value is $\pi/4$; substitute the two terms separately in (7.4).

Exercise 7.5. At the prime 2, both the induced character modulo 8 and χ_4 vanish, so the finite Euler product is empty. New factors arise only from primes $p \mid q$ with $p \nmid q^*$.

Exercise 7.6. Take complex conjugates row by row. A root of unity satisfies $\overline{z} = z^{-1}$; modulo 5, χ_0 and χ_1^2 are real, while χ_1 and χ_1^3 are conjugates.

Exercise 7.7. Differentiate term by term at a real point $s_0 > \sigma_{\mathrm{c}}$. After multiplication by $(-h)^k$, every Taylor term is nonnegative, so Tonelli's theorem applies; the square-indexed subseries gives a lower bound by $\zeta(2s)$.

Solutions to the Exercises

Each problem is restated under its exercise number, followed immediately by a complete solution. Whenever series, integrals, or rearrangements occur, the relevant convergence conditions are also recorded.

Chapter 0

Exercise 0.1. Compute $\pi(x)$, $\vartheta(x)$, and $\psi(x)$ for $x = 10, 30$, and identify the prime powers that account for the differences among the three functions.

Solution. The primes not exceeding 10 are $2, 3, 5, 7$, so

$$\pi(10) = 4, \qquad \vartheta(10) = \log(2 \cdot 3 \cdot 5 \cdot 7) = \log 210.$$

Besides these primes, the prime powers not exceeding 10 are $4, 8, 9$. Hence

$$\psi(10) = 3\log 2 + 2\log 3 + \log 5 + \log 7 = \log 2520.$$

The primes not exceeding 30 are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29$, and therefore

$$\pi(30) = 10, \qquad \vartheta(30) = \log 6469693230.$$

The additional prime powers are $4, 8, 16, 9, 27, 25$, so

$$\psi(30) = \vartheta(30) + 3\log 2 + 2\log 3 + \log 5 = \log 2329089562800.$$

The function π gives each prime weight 1, while ϑ gives it weight $\log p$; ψ also counts $p^2, p^3, \dots$, with weight $\log p$ each time.

Exercise 0.2. Assume that $\vartheta(x) \sim x$. Temporarily allowing the use of Abel summation, derive $\pi(x) \sim x/\log x$ and list the boundary terms that must be controlled.

Solution. The Abel summation formula (3.4) gives

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t \log^2 t}\,\mathrm{d}t.$$

Write $\vartheta(t) = t + E(t)$, where $E(t) = o(t)$. The first term is

$$\frac{x}{\log x} + o\left(\frac{x}{\log x}\right).$$

The main integral satisfies

$$\int_2^x \frac{\mathrm{d}t}{\log^2 t} = O\left(\frac{x}{\log^2 x}\right) = o\left(\frac{x}{\log x}\right);$$

for example, estimate $[2, \sqrt{x}]$ and $[\sqrt{x}, x]$ separately. For the error integral, given $\varepsilon > 0$, choose a fixed T such that $|E(t)| \le \varepsilon t$ for $t \ge T$. Then

$$\int_2^x \frac{E(t)}{t \log^2 t}\,\mathrm{d}t = O_T(1) + O\left(\varepsilon \frac{x}{\log^2 x}\right) = o\left(\frac{x}{\log x}\right).$$

The upper boundary term contributes $x/\log x + o(x/\log x)$; both integrals and any constant from the lower endpoint are $o(x/\log x)$. Hence $\pi(x) \sim x/\log x$.

Exercise 0.3. Show that even if $a(n), b(n) \ge 0$ and their respective Dirichlet series both converge at a real number σ, the series

$$\sum_{n\ge 1} a(n)b(n)n^{-\sigma}$$

need not converge. Which formal manipulations does this phenomenon warn us not to use without additional hypotheses?

Solution. Fix $\sigma > 0$ and make the coefficients nonzero only at $n = 2^k$, $k \ge 1$, by setting

$$a(2^k) = b(2^k) = \frac{2^{k\sigma/2}}{\sqrt{k}}.$$

Then

$$\sum_{n\ge 1} a(n)n^{-\sigma} = \sum_{k\ge 1} \frac{2^{-k\sigma/2}}{\sqrt{k}}.$$

For $\sigma > 0$, this series converges absolutely, and the same is true for b. Their pointwise product, however, gives

$$\sum_{n\ge 1} a(n)b(n)n^{-\sigma} = \sum_{k\ge 1} \frac{2^{k\sigma}}{k} 2^{-k\sigma} = \sum_{k\ge 1} \frac{1}{k} = \infty.$$

The choice $\sigma > 0$ is essential. If $\sigma \le 0$, convergence of the original series and $n^{-\sigma} \ge 1$ imply that $a(n)$ is bounded. Hence

$$\sum_{n} a(n)b(n)n^{-\sigma} \le \big(\sup_n a(n)\big) \sum_n b(n)n^{-\sigma} < \infty.$$

Convergence of two nonnegative Dirichlet series does not imply convergence of the series formed from pointwise products of their coefficients. Rearrangement, termwise integration, and termwise differentiation require appropriate convergence hypotheses; multiplication of Dirichlet series produces Dirichlet convolution instead.

Exercise 0.4. Add a row for Euler's totient function φ to the table of methods in this chapter. First use $\mathbf{1} * \varphi = \mathrm{id}$ to predict its Dirichlet series, then use the pole of the resulting expression at $s = 2$ to predict the main term of $\sum_{n\le x} \varphi(n)$. No proof of the error term is required at this stage.

Solution. In the region of absolute convergence, the convolution identity $\mathbf{1} * \varphi = \mathrm{id}$ becomes

$$\zeta(s) \sum_{n\ge 1} \frac{\varphi(n)}{n^s} = \zeta(s-1),$$

and hence

$$\sum_{n\ge 1} \frac{\varphi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)} \qquad (\Re s > 2).$$

The function $\zeta(s-1)$ has residue 1 at $s=2$, and $\zeta(2)\neq 0$, so the residue of the right-hand side is $1/\zeta(2)=6/\pi^2$. The Perron kernel x^s/s contributes a further factor $x^2/2$ at $s=2$, suggesting

$$\sum_{n\le x}\varphi(n)\sim\frac{1}{\zeta(2)}\frac{x^2}{2}=\frac{3}{\pi^2}x^2.$$

Theorem 3.13 in Chapter 3 proves this main term by elementary rearrangement.

Chapter 1

Exercise 1.1. Use the Euclidean algorithm to find $\gcd(252,198)$ and express it as an integral linear combination of 252 and 198.

Solution. The Euclidean algorithm gives

$$252=198+54,\quad 198=3\cdot 54+36,\quad 54=36+18,$$

so $\gcd(252,198)=18$. Back-substitution gives

$$18=54-36=54-(198-3\cdot 54)=4\cdot 54-198=4\cdot 252-5\cdot 198.$$

Thus the required Bézout representation is $18=4\cdot 252-5\cdot 198$.

Exercise 1.2. Prove that $\sum_{d\mid n}\varphi(d)=n$. Hint: classify elements by their orders in the cyclic group $\mathbb{Z}/n\mathbb{Z}$, or classify the integers $1\le k\le n$ by $\gcd(k,n)$.

Solution. For $1\le k\le n$, classify according to $d=n/\gcd(k,n)$. For fixed $d\mid n$, the condition $n/\gcd(k,n)=d$ is equivalent to

$$k=\frac{n}{d}m,\qquad 1\le m\le d,\qquad (m,d)=1.$$

There are exactly $\varphi(d)$ such m. These disjoint classes cover all k, so

$$n=\sum_{d\mid n}\varphi(d).$$

Equivalently, d is the order of k in the cyclic group $\mathbb{Z}/n\mathbb{Z}$, and there are $\varphi(d)$ elements of order d.

Exercise 1.3. Prove that if f is multiplicative, then $\sum_{d\mid n}f(d)=\prod_{p^\alpha\|n}(1+f(p)+\cdots+f(p^\alpha))$.

Solution. Write $n=\prod_{p^\alpha\|n}p^\alpha$. Every divisor $d\mid n$ has a unique expression $d=\prod p^{j_p}$ with $0\le j_p\le\alpha$. Since powers of distinct primes are coprime and f is multiplicative,

$$f(d)=\prod_{p^\alpha\|n}f(p^{j_p}).$$

Summing independently over all exponents j_p gives

$$\sum_{d\mid n}f(d)=\prod_{p^\alpha\|n}\sum_{j=0}^{\alpha}f(p^j)=\prod_{p^\alpha\|n}\bigl(1+f(p)+\cdots+f(p^\alpha)\bigr).$$

Only multiplicativity was used; $f(p^j)$ cannot in general be replaced by $f(p)^j$.

Exercise 1.4. Let n_k be the product of the first k primes. Use (1.1) to study the behavior of $\varphi(n_k)/n_k$ and explain why it is related to the sum of the reciprocals of the primes.

Solution. If $n_k = p_1 \cdots p_k$, then (1.1) gives

$$\frac{\varphi(n_k)}{n_k} = \prod_{j \le k} \left(1 - \frac{1}{p_j}\right).$$

Each new prime multiplies the product by a positive number less than 1, so the ratio is strictly decreasing. Taking logarithms and using $\log(1-u) = -u + O(u^2)$ gives

$$\log \frac{\varphi(n_k)}{n_k} = -\sum_{j \le k} \frac{1}{p_j} + O\left(\sum_{j \le k} \frac{1}{p_j^2}\right) = -\sum_{j \le k} \frac{1}{p_j} + O(1),$$

because $\sum_p p^{-2} \le \sum_{n \ge 2} n^{-2} < \infty$. Thus the product tends to zero exactly when the sum of reciprocal primes diverges. Euler's argument in Chapter 4 proves $\sum_p 1/p = \infty$, so $\varphi(n_k)/n_k \to 0$.

Exercise 1.5. Factor 7560, find every pair (p, α) satisfying $p^\alpha \parallel 7560$ and $\alpha \ge 1$, and compute $\gcd(7560, 5400)$ and $\operatorname{lcm}(7560, 5400)$. Hint: apply Proposition 1.9 one prime at a time.

Solution. Successive factorization gives

$$7560 = 2^3 \cdot 3^3 \cdot 5 \cdot 7,$$

so the required pairs are $(2,3), (3,3), (5,1), (7,1)$. Also

$$5400 = 2^3 \cdot 3^3 \cdot 5^2.$$

Take the smaller exponent prime by prime for the greatest common divisor and the larger for the least common multiple. Thus

$$\gcd(7560, 5400) = 2^3 \cdot 3^3 \cdot 5 = 1080,$$

and

$$\operatorname{lcm}(7560, 5400) = 2^3 \cdot 3^3 \cdot 5^2 \cdot 7 = 37800.$$

As a check, $1080 \cdot 37800 = 7560 \cdot 5400$.

Exercise 1.6. List a reduced residue system modulo 15, match each unit with an element of $(\mathbb{Z}/3\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times$ using the Chinese remainder theorem, and thereby verify that $\varphi(15) = 8$.

Solution. A reduced residue system modulo 15 is

$$1, 2, 4, 7, 8, 11, 13, 14.$$

Under the Chinese remainder isomorphism, these correspond to

x	1	2	4	7	8	11	13	14
$(x \bmod 3, x \bmod 5)$	$(1,1)$	$(2,2)$	$(1,4)$	$(1,2)$	$(2,3)$	$(2,1)$	$(1,3)$	$(2,4)$

The right-hand side runs through the $2 \cdot 4 = 8$ elements of $(\mathbb{Z}/3\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times$. Hence $\varphi(15) = \varphi(3)\varphi(5) = 2 \cdot 4 = 8$.

Chapter 2

Exercise 2.1. List all divisors of 12, compute $(\mu * \mathbf{1})(12)$ and $(\mu * \mathrm{id})(12)$ directly, and compare the answers with $\varepsilon(12)$ and $\varphi(12)$.

Solution. The divisors of 12 are $1, 2, 3, 4, 6, 12$, with Möbius values $1, -1, -1, 0, 1, 0$. Hence

$$(\mu * \mathbf{1})(12) = 1 - 1 - 1 + 0 + 1 + 0 = 0 = \varepsilon(12).$$

On the other hand,

$$\begin{aligned}(\mu * \mathrm{id})(12) &= \sum_{d \mid 12} \mu(d) \frac{12}{d} \\ &= 12 - 6 - 4 + 0 + 2 + 0 = 4 = \varphi(12).\end{aligned}$$

The values agree with $\varepsilon(12)$ and $\varphi(12)$, respectively.

Exercise 2.2. Prove that $\mu^2(n)$ is the indicator function of the squarefree integers, and prove that $\mu^2(n) = \sum_{d^2 \mid n} \mu(d)$.

Solution. By definition, $\mu(n)^2 = 1$ if and only if every prime exponent of n is 0 or 1, and is 0 otherwise. It is therefore exactly the indicator of the squarefree integers. The function $n \mapsto \sum_{d^2 \mid n} \mu(d)$ is also multiplicative, so it is enough to check $n = p^k$:

$$\sum_{d^2 \mid p^k} \mu(d) = \sum_{0 \le j \le k/2} \mu(p^j) = \begin{cases} 1, & k = 0, 1, \\ 1 - 1 = 0, & k \ge 2. \end{cases}$$

This agrees with $\mu^2(p^k)$ at every prime power. Multiplicativity proves the identity for all n.

Exercise 2.3. Let f be multiplicative with $f(1) = 1$. Prove that its convolution inverse f^{-1} is also multiplicative, and write down $f^{-1}(p)$ and $f^{-1}(p^2)$.

Solution. The recursion for the convolution inverse gives

$$f^{-1}(p) = -f(p),$$

and

$$f^{-1}(p^2) = -\big(f(p) f^{-1}(p) + f(p^2) f^{-1}(1)\big) = f(p)^2 - f(p^2).$$

To prove multiplicativity, use the recursion at each prime p to define uniquely the sequence $h(p^0), h(p^1), \ldots$, and set

$$h\left(\prod_p p^{\alpha_p}\right) = \prod_p h(p^{\alpha_p}).$$

This h is multiplicative. Its convolution with f is multiplicative as well, and the recursion gives $(f * h)(p^k) = 0$ for every $k \ge 1$, while $(f * h)(1) = 1$. Hence $f * h = \varepsilon$. Uniqueness of the convolution inverse gives $h = f^{-1}$, proving that f^{-1} is multiplicative.

Exercise 2.4. Prove that $\Lambda(n) = -\sum_{d \mid n} \mu(d) \log d$ and explain why this is equivalent to $\Lambda = \mu * \log$.

Solution. From $\Lambda = \mu * \log$,

$$\Lambda(n) = \sum_{d \mid n} \mu(d) \log \frac{n}{d}.$$

For $n > 1$, expand the right-hand side:

$$\Lambda(n) = \log n \sum_{d \mid n} \mu(d) - \sum_{d \mid n} \mu(d) \log d = -\sum_{d \mid n} \mu(d) \log d,$$

because $\sum_{d|n}\mu(d)=0$. Both sides vanish when $n=1$. Conversely, substitute $d \leftrightarrow n/d$ in the last identity and use the same divisor-sum relation to recover $\Lambda(n)=\sum_{d|n}\mu(d)\log(n/d)$. The two forms are therefore equivalent.

Exercise 2.5. For $k,n\in\mathbb{N}$, let $J_k(n)$ be the number of k-tuples modulo n that are jointly coprime to n, meaning that $\gcd(a_1,\dots,a_k,n)=1$. Prove that $J_k(n)=n^k\prod_{p|n}(1-p^{-k})$.

Solution. Use the Möbius indicator formula for each k-tuple modulo n:

$$\mathbf{1}_{(a_1,\dots,a_k,n)=1}=\sum_{d|(a_1,\dots,a_k,n)}\mu(d).$$

Sum over $1\le a_j\le n$ and interchange the finite sums:

$$\begin{aligned}J_k(n)&=\sum_{d|n}\mu(d)\#\{(a_1,\dots,a_k):d\mid a_1,\dots,d\mid a_k\}\\&=\sum_{d|n}\mu(d)\left(\frac{n}{d}\right)^k=n^k\sum_{d|n}\frac{\mu(d)}{d^k}.\end{aligned}$$

The last divisor sum factors prime by prime as $\prod_{p|n}(1-p^{-k})$. Therefore

$$J_k(n)=n^k\prod_{p|n}(1-p^{-k}).$$

Exercise 2.6. Use Proposition 2.11 to compute the convolution inverse of $f(n)=n$ at $1,p,p^2,pq$, where $p\neq q$ are prime, and compare the answers with $\mu(n)n$.

Solution. Guess $g(n)=\mu(n)n$. Since id is completely multiplicative,

$$(\mathrm{id}*g)(n)=\sum_{d|n}d\,\mu(n/d)\frac{n}{d}=n\sum_{e|n}\mu(e)=\varepsilon(n).$$

Thus $g=\mathrm{id}^{-1}$, and

$$g(1)=1,\qquad g(p)=-p,\qquad g(p^2)=0,\qquad g(pq)=pq\quad(p\neq q).$$

The same values follow term by term from the recursion in Proposition 2.11.

Exercise 2.7. Let f,g be multiplicative and put $F_p(T)=\sum_{k\ge0}f(p^k)T^k$ and $G_p(T)=\sum_{k\ge0}g(p^k)T^k$. Prove that convolution at a prime p corresponds to multiplication of formal power series, and use this perspective to rederive $\mathbf{1}^{-1}=\mu$. Hint: compare the coefficients of T^k.

Solution. If $h=f*g$, the local convolution formula is

$$h(p^k)=\sum_{j=0}^{k}f(p^j)g(p^{k-j}).$$

Consequently

$$\begin{aligned}F_p(T)G_p(T)&=\left(\sum_{j\ge0}f(p^j)T^j\right)\left(\sum_{\ell\ge0}g(p^\ell)T^\ell\right)\\&=\sum_{k\ge0}\left(\sum_{j=0}^{k}f(p^j)g(p^{k-j})\right)T^k,\end{aligned}$$

whose coefficient of T^k is $h(p^k)$. This is an identity of formal power series and involves no analytic convergence. For $\mathbf{1}$, the local series is

$$1 + T + T^2 + \cdots = (1 - T)^{-1},$$

whose multiplicative inverse is $1 - T$. Thus the inverse function takes the values $1, -1, 0$ at p^0, p^1, p^k for $k \geq 2$, respectively; that is, $\mathbf{1}^{-1} = \mu$.

Chapter 3

Exercise 3.1. Use Abel summation to prove that, for fixed $\alpha > -1$, $\sum_{n \leq x} n^\alpha \sim x^{\alpha+1}/(\alpha+1)$.
Solution. In Abel summation, take $a_n = 1$, $A(t) = \lfloor t \rfloor$, and $f(t) = t^\alpha$. If $\alpha \neq 0$, then

$$\sum_{n \leq x} n^\alpha = \lfloor x \rfloor x^\alpha - \alpha \int_1^x \lfloor t \rfloor t^{\alpha-1} \, \mathrm{d}t.$$

Substitute $\lfloor t \rfloor = t + O(1)$. The part without rounding errors is

$$x^{\alpha+1} - \frac{\alpha}{\alpha+1}(x^{\alpha+1} - 1) = \frac{x^{\alpha+1}}{\alpha+1} + \frac{\alpha}{\alpha+1}.$$

The final constant can be absorbed into $O_\alpha(1)$. The endpoint rounding error is $O(x^\alpha)$, or $O(1)$ when $\alpha < 0$, and the integral error is

$$O\left(\int_1^x t^{\alpha-1} \, \mathrm{d}t\right) = \begin{cases} O(x^\alpha), & \alpha > 0, \\ O(1), & -1 < \alpha < 0. \end{cases}$$

All are $o(x^{\alpha+1})$. When $\alpha = 0$, the left-hand side is $\lfloor x \rfloor = x + O(1)$. Hence, for every fixed $\alpha > -1$,

$$\sum_{n \leq x} n^\alpha \sim \frac{x^{\alpha+1}}{\alpha+1}.$$

Exercise 3.2. Deduce $D(x) = \sum_{n \leq x} \lfloor x/n \rfloor$ from (3.2), and explain its relation to counting divisor pairs (a, b).
Solution. Since $\tau(n) = \sum_{ab=n} 1$, the finite sum can be rearranged directly:

$$D(x) = \sum_{n \leq x} \tau(n) = \sum_{ab \leq x} 1.$$

For fixed a, the possible values of b satisfy $1 \leq b \leq x/a$, and there are $\lfloor x/a \rfloor$ of them. Therefore

$$D(x) = \sum_{a \leq x} \left\lfloor \frac{x}{a} \right\rfloor.$$

This is (3.2) with $f = g = \mathbf{1}$; both expressions count the positive integer lattice points in the first quadrant beneath the hyperbola $ab = x$.

Exercise 3.3. Check the proof of Theorem 3.11 line by line, and explain why merely knowing $\sum_{d\le y}\mu(d)=O(y)$ is still insufficient to improve the $O(\sqrt{x})$ error.

Solution. The proof of Theorem 3.11 begins with

$$\mu^2(n)=\sum_{d^2\mid n}\mu(d).$$

After a finite rearrangement,

$$\sum_{n\le x}\mu^2(n)=\sum_{d\le\sqrt{x}}\mu(d)\left\lfloor\frac{x}{d^2}\right\rfloor.$$

Replacing the floor by $x/d^2+O(1)$ gives a total pointwise error $O(\sqrt{x})$. Completing the finite main sum to an infinite sum gives, by absolute estimation,

$$x\sum_{d>\sqrt{x}}\frac{1}{d^2}=O(\sqrt{x}).$$

Thus both error sources are $O(\sqrt{x})$.

Let $M(y)=\sum_{d\le y}\mu(d)$. The bound $M(y)=O(y)$ is only the trivial bound already implied by $|\mu(d)|\le 1$. Abel summation still gives only

$$\sum_{d>y}\frac{\mu(d)}{d^2}=O(1/y),$$

which contributes $O(\sqrt{x})$ when $y=\sqrt{x}$. More importantly, it does not control the variable floor error $\sum_{d\le\sqrt{x}}\mu(d)\{x/d^2\}$. Improving the error along this route requires additional control of these sums with fractional-part weights; the trivial bound $M(y)=O(y)$ alone supplies no such information.

Exercise 3.4. Suppose $a_n\ge 0$ and $A(x)=\sum_{n\le x}a_n=O(x)$. Prove that $\sum_{n\le x}a_n/n=O(\log x)$ for $x\ge 2$. Where is nonnegativity used?

Solution. Apply Abel summation with $f(t)=1/t$:

$$\sum_{n\le x}\frac{a_n}{n}=\frac{A(x)}{x}+\int_1^x\frac{A(t)}{t^2}\,\mathrm{d}t.$$

Since $A(t)=O(t)$,

$$\frac{A(x)}{x}=O(1),\qquad\int_1^x\frac{A(t)}{t^2}\,\mathrm{d}t=O\left(\int_1^x\frac{\mathrm{d}t}{t}\right)=O(\log x).$$

The required sum is therefore $O(\log x)$. Once $A(x)=O(x)$ has been included in the hypotheses, the proof does not use $a_n\ge 0$; nonnegativity usually serves only to make the partial-sum bound easy to establish in applications.

Exercise 3.5. In Theorem 3.9, take $u=x^\alpha$ and $v=x^{1-\alpha}$ with $0\le\alpha\le 1$. Do the bounds $F(t)\ll t^\rho$ and $G(t)\ll t^\eta$ alone determine an optimal α? If two error terms additionally satisfy $E_1\ll x^A u^B$ and $E_2\ll x^C u^{-D}$ with $B,D>0$, find the α that minimizes the larger exponent in these two bounds, and explain how to check the contribution of $F(u)G(v)$.

Solution. The bounds $F(t)\ll t^\rho$ and $G(t)\ll t^\eta$ alone do not determine an optimal α. They directly give only the bound for the endpoint correction,

$$F(u)G(v)\ll u^\rho v^\eta=x^{\alpha\rho+(1-\alpha)\eta}.$$

The errors in the two truncated sums also depend on the sizes of f, g and on the remainders in their partial-sum asymptotics.

Under the additional error assumptions in the question, put

$$H(\alpha) = \max\{A + B\alpha,\ C - D\alpha\}.$$

The first line is strictly increasing and the second strictly decreasing. Their intersection $\alpha_0 = (C - A)/(B + D)$ minimizes H if it lies in $[0, 1]$; otherwise the nearer endpoint does. Thus

$$\alpha_* = \min\left\{1, \max\left\{0, \frac{C - A}{B + D}\right\}\right\}.$$

When $0 \le \alpha_0 \le 1$, the common exponent is $(AD + BC)/(B + D)$.

Finally, check how $F(u)G(v)$ enters the calculation. If it has been combined with the main terms of the two truncated sums, estimate the resulting remainder. If its absolute value is instead included in the error, also compare the exponent $\alpha\rho + (1-\alpha)\eta$. In the latter case one must minimize the maximum of all three exponents, and the optimum may change. Compare $0, 1$ and all intersections of pairs of lines that lie in the interval.

Exercise 3.6. For fixed $\alpha > 0$, use $\sigma_\alpha = \mathrm{id}_\alpha * \mathbf{1}$ to predict and prove the main term of $\sum_{n \le x} \sigma_\alpha(n)$. Hint: treat separately the accumulation of the power-sum error when $\alpha < 1$, $\alpha = 1$, and $\alpha > 1$.

Solution. Since $\sigma_\alpha = \mathrm{id}_\alpha * \mathbf{1}$, a finite rearrangement gives

$$\sum_{n \le x} \sigma_\alpha(n) = \sum_{m \le x} \sum_{d \le x/m} d^\alpha.$$

The quantitative form of Exercise 3.1 is

$$\sum_{d \le y} d^\alpha = \frac{y^{\alpha+1}}{\alpha + 1} + O(y^\alpha + 1) \qquad (\alpha > 0).$$

Therefore

$$\sum_{n \le x} \sigma_\alpha(n) = \frac{x^{\alpha+1}}{\alpha + 1} \sum_{m \le x} \frac{1}{m^{\alpha+1}} + O\left(x^\alpha \sum_{m \le x} \frac{1}{m^\alpha} + x\right).$$

Absolute convergence gives

$$\sum_{m \le x} m^{-\alpha-1} = \zeta(\alpha + 1) + O(x^{-\alpha}),$$

whose tail contributes only $O(x)$ to the displayed formula. Also

$$\sum_{m \le x} m^{-\alpha} = \begin{cases} O(x^{1-\alpha}), & 0 < \alpha < 1, \\ O(\log x), & \alpha = 1, \\ O(1), & \alpha > 1. \end{cases}$$

Consequently

$$\sum_{n \le x} \sigma_\alpha(n) = \frac{\zeta(\alpha + 1)}{\alpha + 1} x^{\alpha+1} + \begin{cases} O(x), & 0 < \alpha < 1, \\ O(x \log x), & \alpha = 1, \\ O(x^\alpha), & \alpha > 1. \end{cases}$$

The implied constant may depend on the fixed parameter α.

Chapter 4

Exercise 4.1. Verify (4.2), and compute both sides when $x = 10$.

Solution. In $\operatorname{lcm}(1, \dots, \lfloor x \rfloor)$, the exponent of a prime p is the largest k such that $p^k \le x$. Thus

$$\begin{aligned} \log \operatorname{lcm}(1, \dots, \lfloor x \rfloor) &= \sum_{p \le x} \left\lfloor \frac{\log x}{\log p} \right\rfloor \log p \\ &= \sum_{p^k \le x} \log p = \psi(x). \end{aligned}$$

When $x = 10$,

$$\operatorname{lcm}(1, \dots, 10) = 2^3 \cdot 3^2 \cdot 5 \cdot 7 = 2520,$$

so the left-hand side is $\log 2520$. The right-hand side consists of the von Mangoldt weights at $2, 3, 4, 5, 7, 8, 9$ and is likewise $3\log 2 + 2\log 3 + \log 5 + \log 7 = \log 2520$.

Exercise 4.2. Prove that $\vartheta(x) \le \pi(x) \log x$ and that $\vartheta(x) \ge (\pi(x) - \pi(\sqrt{x}))\frac{1}{2} \log x$.

Solution. For each $p \le x$, $\log p \le \log x$, so

$$\vartheta(x) = \sum_{p \le x} \log p \le \pi(x) \log x.$$

For the lower bound retain only $\sqrt{x} < p \le x$. Each such prime satisfies $\log p > \frac{1}{2} \log x$, and there are $\pi(x) - \pi(\sqrt{x})$ of them. Therefore

$$\vartheta(x) \ge \sum_{\sqrt{x} < p \le x} \log p \ge \big(\pi(x) - \pi(\sqrt{x})\big) \frac{1}{2} \log x.$$

Exercise 4.3. Improve the crude argument in Proposition 4.6 to prove $\psi(x) - \vartheta(x) = O(\sqrt{x} \log x)$. Hint: first treat the square terms separately, then estimate the number and size of the terms with $k \ge 3$.

Solution. By (4.1),

$$\psi(x) - \vartheta(x) = \sum_{k \ge 2} \vartheta(x^{1/k}).$$

The square term satisfies

$$\vartheta(\sqrt{x}) \le \pi(\sqrt{x}) \log x = O(\sqrt{x} \log x).$$

For $k \ge 3$, $x^{1/k} \le x^{1/3}$, at most $\lfloor \log_2 x \rfloor$ indices give nonzero terms, and

$$\vartheta(x^{1/k}) \le \pi(x^{1/k}) \log x \le x^{1/3} \log x.$$

These terms total $O(x^{1/3} (\log x)^2) = O(\sqrt{x} \log x)$. Hence

$$\psi(x) - \vartheta(x) = O(\sqrt{x} \log x).$$

Exercise 4.4. Prove that if $\pi(x) \sim x/\log x$, then the nth prime satisfies $p_n \sim n \log n$.

Solution. Since $\pi(p_n) = n$, the hypothesis gives

$$n \sim \frac{p_n}{\log p_n}.$$

Taking logarithms,

$$\log n = \log p_n - \log\log p_n + o(1).$$

As $\log\log p_n = o(\log p_n)$, we have $\log n \sim \log p_n$. Substituting back gives

$$p_n \sim n \log p_n \sim n \log n.$$

Exercise 4.5. Extract explicit constants from the proof of Theorem 4.14. Which steps cause the largest losses?

Solution. The proof gives explicit, though nonoptimal, constants. Proposition 4.12 yields

$$\vartheta(x) \le 4(\log 2)x.$$

The central binomial coefficient gives

$$\psi(2n) \ge 2n \log 2 - \log(2n+1).$$

Take $n = \lfloor x/2 \rfloor$ and absorb a bounded range. For sufficiently large x this gives $\psi(x) \ge (\log 2/2)x$. From $\psi(x) - \vartheta(x) = O(\sqrt{x}\log x)$, after enlarging the threshold we may take

$$\vartheta(x) \ge \frac{\log 2}{4}x.$$

Thus

$$\pi(x) \ge \frac{\vartheta(x)}{\log x} \ge \frac{\log 2}{4}\frac{x}{\log x},$$

whereas

$$\pi(x) \le \sqrt{x} + \frac{2\vartheta(x)}{\log x} \le \sqrt{x} + 8(\log 2)\frac{x}{\log x} \le (8\log 2 + 1)\frac{x}{\log x}$$

for sufficiently large x. One may therefore take, for example, $c = \log 2/4$ and $C = 8\log 2 + 1$; both pairs of bounds hold for all sufficiently large x.

The main losses occur in three places: covering a general x by a dyadic interval costs about a factor 2; the upper and lower bounds for the binomial coefficient are crude; and in returning from ϑ to π, primes below $\sqrt{x}$ are discarded while all remaining weights are replaced by $\frac{1}{2}\log x$. These steps preserve only the order of magnitude and cannot recover the constant 1.

Exercise 4.6. Use Legendre's formula to compute $\nu_2(100!)$ and $\nu_5(100!)$, and find the number of trailing zeros of 100!.

Solution. Legendre's formula gives

$$\nu_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97,$$

and

$$\nu_5(100!) = 20 + 4 = 24.$$

Each trailing decimal zero requires one pair $2\cdot 5$. Since there are more factors of 2, the number of trailing zeros is

$$\min\{97, 24\} = 24.$$

Exercise 4.7. Prove that $\int_2^x \mathrm{d}t/\log t = O(x/\log x)$. Hint: split at $\sqrt{x}$; bound the integrand by $1/\log 2$ on the first part and by $2/\log x$ on the second. Use the estimate to complete the reverse implication in Proposition 4.8.

Solution. First estimate the integral for $x \geq 4$. The contribution of $[2, \sqrt{x}]$ is at most $\sqrt{x}/\log 2$. On $[\sqrt{x}, x]$, $\log t \geq \frac{1}{2}\log x$, and therefore

$$\int_{\sqrt{x}}^{x} \frac{\mathrm{d}t}{\log t} \leq \frac{2x}{\log x}.$$

Thus

$$\int_2^x \frac{\mathrm{d}t}{\log t} = O(x/\log x).$$

The bounded range $2 \leq x < 4$ is absorbed into the implicit constant.

Now assume $\pi(x) \sim x/\log x$. Abel summation applied to the indicator of the primes gives

$$\vartheta(x) = \pi(x)\log x - \int_2^x \frac{\pi(t)}{t}\,\mathrm{d}t.$$

The first term is $x + o(x)$. The hypothesis also gives $\pi(t) = O(t/\log t)$, so the integral estimate above yields

$$\int_2^x \frac{\pi(t)}{t}\,\mathrm{d}t = O\left(\int_2^x \frac{\mathrm{d}t}{\log t}\right) = O(x/\log x) = o(x).$$

Hence $\vartheta(x) \sim x$, and $\psi(x) - \vartheta(x) = O(\sqrt{x}\log x) = o(x)$ then gives $\psi(x) \sim x$, completing the reverse implication.

Chapter 5

Exercise 5.1. Prove that $\zeta(\sigma) > 1$ and $-\zeta'(\sigma) = \sum_{n\geq 2}(\log n)n^{-\sigma} > 0$ for $\sigma > 1$.

Solution. For $\sigma > 1$,

$$\zeta(\sigma) = 1 + \sum_{n\geq 2} n^{-\sigma} > 1.$$

On every closed half-plane $\sigma \geq 1 + \delta$, both $\sum n^{-1-\delta}$ and $\sum (\log n)n^{-1-\delta}$ converge. The series therefore converges normally and can be differentiated term by term. Hence

$$\zeta'(\sigma) = -\sum_{n\geq 2} \frac{\log n}{n^{\sigma}}, \qquad -\zeta'(\sigma) > 0.$$

Exercise 5.2. Use (5.3) to prove that $\lim_{s\to 1}(s-1)\zeta(s) = 1$.

Solution. By (5.3),

$$\zeta(s) = \frac{s}{s-1} - s\int_1^{\infty} \{x\}x^{-s-1}\,\mathrm{d}x.$$

The integral converges absolutely and locally uniformly in a neighborhood of $s = 1$, and is therefore bounded there. Multiplying by $s - 1$ gives

$$(s-1)\zeta(s) = s - s(s-1)\int_1^{\infty} \{x\}x^{-s-1}\,\mathrm{d}x \longrightarrow 1.$$

Thus the residue of the pole at $s = 1$ is 1.

Exercise 5.3. Expand (5.4) in detail, verify that the coefficient of every n is nonnegative, and explain why the coefficients in $3+4\cos\theta+\cos 2\theta$ are exactly strong enough to overwhelm one boundary zero.

Solution. For $\sigma > 1$, substitute (5.2) into (5.4). The coefficient attached to a fixed n is

$$\frac{\Lambda(n)}{n^{\sigma}}\bigl(3 + 4\cos(t\log n) + \cos(2t\log n)\bigr).$$

With $\theta = t\log n$,

$$3 + 4\cos\theta + \cos 2\theta = 2(1+\cos\theta)^2 \geq 0.$$

Thus every term is nonnegative; all the series converge absolutely for $\sigma > 1$.

If $t_0 \neq 0$ and $\zeta(1+it_0)$ has a zero of order m, then the simple pole at $s=1$ makes $-3\zeta'/\zeta(\sigma)$ contribute $3/(\sigma-1)$, while $-4\Re(\zeta'/\zeta)(\sigma+it_0)$ contributes $-4m/(\sigma-1)$. Since $t_0 \neq 0$, the point $1+2it_0$ in the third term is not a pole. Let its order as a zero be $m_2 \geq 0$, with $m_2 = 0$ if it is not a zero. The total principal part is therefore

$$\frac{3-4m-m_2}{\sigma-1},$$

which is negative for $m \geq 1$, contradicting termwise nonnegativity. The coefficients $3, 4, 1$ both produce the square identity and make the fourfold negative part from a zero overwhelm the threefold positive part from the pole.

Exercise 5.4. Assume $\psi(x) \sim x$. Without citing the conclusion in the text, give a complete proof that $\pi(x) \sim x/\log x$.

Solution. The estimate $\psi(x) - \vartheta(x) = o(x)$ from Chapter 4 first gives $\vartheta(x) \sim x$, so $\vartheta(t) \leq Ct$ for all $t \geq 2$ and some fixed constant C. Abel summation gives

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t\log^2 t}\,\mathrm{d}t.$$

The boundary term is $x/\log x + o(x/\log x)$. For $x \geq 4$, splitting the integral at $\sqrt{x}$ gives

$$0 \leq \int_2^x \frac{\vartheta(t)}{t\log^2 t}\,\mathrm{d}t \leq C\left(\frac{\sqrt{x}}{\log^2 2} + \frac{4x}{\log^2 x}\right) = O(x/\log^2 x) = o(x/\log x).$$

Therefore $\pi(x) \sim x/\log x$.

Exercise 5.5. Compare the Wiener–Ikehara route with the classical route using Perron's formula and a truncated integral. Which boundary properties of ζ must be controlled in each method?

Solution. Both routes use the same analytic object $-\zeta'/\zeta = \sum \Lambda(n)n^{-s}$, but turn boundary information into a statement about partial sums in different ways.

The Wiener–Ikehara route requires nonnegative coefficients $\Lambda(n) \geq 0$, analyticity in $\Re s > 1$, and continuous extension of the function after subtracting $1/(s-1)$ to the entire line $\Re s = 1$. The Euler product gives nonvanishing in the open half-plane, while analytic continuation and the boundary nonvanishing theorem give continuity on the boundary. The Tauberian theorem uses nonnegativity to pass from smoothed boundary information to a sharp asymptotic for partial sums; it does not require explicit estimates for high horizontal segments.

The classical Perron route first truncates at a finite height T. Besides the same pole at $s=1$ and boundary nonvanishing, it requires quantitative control of the vertical growth of ζ'/ζ, a choice of heights avoiding zeros, estimates for both horizontal sides and the new vertical side of the shifted rectangle, and simultaneous optimization of the Perron truncation error. It can yield an explicit error term, but demands stronger boundary estimates than mere continuous extension.

Exercise 5.6. Let $m \geq 1$ be an integer, let g be analytic near s_0 with $g(s_0) \neq 0$, and put $f(s) = (s - s_0)^m g(s)$. Compute $f'/f - m/(s - s_0)$ directly and explain why it is analytic at s_0.

Solution. Direct logarithmic differentiation gives

$$\frac{f'}{f}(s) = \frac{m}{s - s_0} + \frac{g'}{g}(s),$$

and hence

$$\frac{f'}{f}(s) - \frac{m}{s - s_0} = \frac{g'}{g}(s).$$

Since $g(s_0) \neq 0$, continuity makes g nonzero in a smaller neighborhood. Both g' and $1/g$ are analytic there, so g'/g is analytic at s_0.

Exercise 5.7. In the proof of Theorem 5.13, justify each of the five checks, citing numbered theorems, propositions, or equations where possible. Explain the logical dependencies among these checks, and determine whether $A = 1$ depends on the order or on the residue of the pole of ζ at 1.

Solution. The five checks for Theorem 5.11 are:

1. The definition of the von Mangoldt function gives $a_n = \Lambda(n) \geq 0$, making the partial sums monotone.
2. The representation $-\zeta'/\zeta = \sum \Lambda(n) n^{-s}$ and its absolute convergence in $\Re s > 1$ follow from Theorem 5.4.
3. Analyticity follows from the nonvanishing Euler product in Theorem 5.3; alternatively, it follows from the absolute convergence in the second check and Lemma 5.1.
4. Proposition 5.5 removes the singularity at $s = 1$, and Theorem 5.10 excludes zeros at every other boundary point. Together these give continuity after subtraction of the principal part.
5. Proposition 5.5 gives a pole of order one for ζ at 1. Proposition 5.9 then gives the principal part $1/(s - 1)$ for $-\zeta'/\zeta$, so $A = 1$.

These checks are not independent: the second already implies the third, and the fifth contributes to the fourth at $s = 1$. Replacing the residue of ζ by any nonzero constant while keeping a simple pole would leave the principal part of its negative logarithmic derivative equal to $1/(s-1)$. Thus A depends on the order of the pole, not on the residue of the original function.

Chapter 6

Exercise 6.1. Use Theorem 6.8 and $\mathbf{1} * \Lambda = \log$ to derive $-\zeta'(s) = \zeta(s) \sum \Lambda(n) n^{-s}$ for $\sigma > 1$.

Solution. For $\sigma > 1$, both $\sum n^{-s}$ and $\sum \Lambda(n) n^{-s}$ converge absolutely; the latter is dominated using $0 \leq \Lambda(n) \leq \log n$. Theorem 6.8 and $\mathbf{1} * \Lambda = \log$ give

$$\zeta(s) \sum_{n \geq 1} \frac{\Lambda(n)}{n^s} = \sum_{n \geq 1} \frac{\log n}{n^s}.$$

In the same region, Theorem 6.3 permits termwise differentiation of $\zeta(s) = \sum n^{-s}$. Hence

$$-\zeta'(s) = \sum_{n \geq 1} \frac{\log n}{n^s} = \zeta(s) \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}.$$

Every multiplication, rearrangement, and differentiation has taken place in the half-plane of absolute convergence $\Re s > 1$.

Exercise 6.2. Without citing Proposition 6.13, prove again that $\Gamma(1/2) = \sqrt{\pi}$, explaining step by step why Tonelli's theorem and the change to polar coordinates apply.

Solution. Let

$$I = \int_0^\infty \mathrm{e}^{-x^2}\,\mathrm{d}x.$$

The integrand is nonnegative, so Tonelli's theorem gives

$$I^2 = \int_0^\infty \int_0^\infty \mathrm{e}^{-(x^2+y^2)}\,\mathrm{d}x\,\mathrm{d}y,$$

even before the value of the integral is known. In the first quadrant, the change of variables $x = r\cos\theta$, $y = r\sin\theta$ has Jacobian r and transforms the region into $r > 0$, $0 < \theta < \pi/2$. Therefore

$$I^2 = \int_0^{\pi/2} \int_0^\infty \mathrm{e}^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta = \frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi}{4}.$$

Since $I > 0$, $I = \sqrt{\pi}/2$. The substitution $x = u^2$ in the Gamma integral is valid for this nonnegative improper integral and gives

$$\Gamma(1/2) = \int_0^\infty x^{-1/2}\mathrm{e}^{-x}\,\mathrm{d}x = 2\int_0^\infty \mathrm{e}^{-u^2}\,\mathrm{d}u = 2I = \sqrt{\pi}.$$

Exercise 6.3. Let $a_n = (-1)^{n-1}$. Find the abscissae of convergence and absolute convergence of the corresponding Dirichlet series, and use Theorem 6.2 to describe its region of analyticity.

Solution. The partial sums

$$A(N) = \sum_{n\le N} (-1)^{n-1}$$

are always 0 or 1. Taking $\alpha = 0$ in Theorem 6.2 shows that

$$\sum_{n\ge 1} \frac{(-1)^{n-1}}{n^s}$$

converges and is analytic in $\Re s > 0$. At $s = 0$, its terms are $(-1)^{n-1}$ and do not tend to zero, so

$$\sigma_{\mathrm{c}} = 0.$$

The absolute-value series is $\sum n^{-\sigma}$, which converges exactly for $\sigma > 1$. Thus

$$\sigma_{\mathrm{a}} = 1.$$

Exercise 6.4. Split the integral in (6.3) into $(0,1)$ and $(1,\infty)$, and explain how subtracting $1/x$ continues the right-hand side to a larger half-plane.

Solution. Write the right-hand side of (6.3) as

$$\int_0^1 \frac{x^{s-1}}{\mathrm{e}^x - 1}\,\mathrm{d}x + \int_1^\infty \frac{x^{s-1}}{\mathrm{e}^x - 1}\,\mathrm{d}x.$$

Exponential decay makes the second integral converge locally uniformly for every s, so it defines an entire function. Put

$$h(x) = \frac{1}{\mathrm{e}^x - 1} - \frac{1}{x}.$$

Taylor expansion gives $h(x) = O(1)$ as $x \downarrow 0$, and hence

$$\int_0^1 \frac{x^{s-1}}{\mathrm{e}^x - 1}\,\mathrm{d}x = \int_0^1 x^{s-2}\,\mathrm{d}x + \int_0^1 h(x)x^{s-1}\,\mathrm{d}x$$
$$= \frac{1}{s-1} + H(s).$$

The first identity initially holds for $\Re s > 1$, while $H(s)$ converges absolutely and locally uniformly in $\Re s > 0$. The right-hand side therefore continues $\Gamma(s)\zeta(s)$ meromorphically to $\Re s > 0$, with the displayed pole at $s = 1$. This agrees with the continuation of ζ from (5.3); the integral argument itself does not require a global nonvanishing theorem for the Gamma function. Subtracting $1/x$ removes the worst x^{s-2} singularity at 0 and moves the convergence boundary from $\Re s > 1$ to $\Re s > 0$.

Exercise 6.5. For $F(s) = \zeta(s)$, formally move the Perron integral from $c > 1$ to $0 < c' < 1$. Identify the crossed pole and its main term, and list the three types of boundary integrals that must be controlled to turn the formal argument into a rigorous estimate.

Solution. For nonintegral $x > 0$ and $c > 1$, Perron's formula formally reads

$$\sum_{n \le x} 1 = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \zeta(s)\frac{x^s}{s}\,\mathrm{d}s.$$

First truncate at height T, then move the line to $0 < c' < 1$. The only pole crossed by the rectangle is the simple pole of ζ at $s = 1$; since the new line remains to the right of zero, the pole of $1/s$ is not crossed. The residue is

$$\operatorname*{Res}_{s=1} \zeta(s)\frac{x^s}{s} = x,$$

which supplies the main term of the counting function.

A rigorous argument must separately control the new vertical side $[c'-iT, c'+iT]$, the upper horizontal side $[c'+iT, c+iT]$, and the lower horizontal side $[c-iT, c'-iT]$. Growth estimates for $\zeta(s)$ in the vertical strip are needed to make these small for a suitable T. In addition, the truncation error in Theorem 6.22 must be optimized together with these boundary terms. Writing only the residue without estimating all four sides of the finite rectangle is not a proof.

Exercise 6.6. Find the Mellin transforms and fundamental strips of $f(x) = \mathbf{1}_{(0,1)}(x)$ and $g(x) = \mathbf{1}_{(1,\infty)}(x)x^{-2}$. Why are the two fundamental strips determined by different endpoints?

Solution. For $f(x) = \mathbf{1}_{(0,1)}(x)$,

$$\mathcal{M}f(s) = \int_0^1 x^{s-1}\,\mathrm{d}x = \frac{1}{s}, \qquad \Re s > 0.$$

The endpoint 0 imposes $\Re s > 0$, while infinity contributes nothing. Thus the fundamental strip is $0 < \Re s < +\infty$.

For $g(x) = \mathbf{1}_{(1,\infty)}(x)x^{-2}$,

$$\mathcal{M}g(s) = \int_1^\infty x^{s-3}\,\mathrm{d}x = \frac{1}{2-s}, \qquad \Re s < 2.$$

There is no singularity at the lower endpoint 1; the condition comes entirely from infinity. The fundamental strip is therefore $-\infty < \Re s < 2$. The first strip is determined by the endpoint 0, the second by infinity.

Exercise 6.7. Prove the smoothed-kernel formula used in (6.6) and compute the residues at $s = 0$ and $s = -1$. Hint: treat separately the contour-closing directions for $y > 1$ and $0 < y < 1$.

Solution. Let $c > 0$ and consider

$$K(y) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{y^s}{s(s+1)}\, \mathrm{d}s.$$

The kernel is $O_y(|t|^{-2})$ on the vertical line, so the integral converges absolutely. If $y > 1$, first close to the left at finite height T and send the far vertical side to $-\infty$. Its integral vanishes by exponential decay, while the sum of the absolute horizontal integrals is at most $2y^c/(T^2|\log y|)$, which tends to zero as $T \to \infty$. Thus only the residues at $s = 0, -1$ remain:

$$\operatorname*{Res}_{s=0} \frac{y^s}{s(s+1)} = 1, \qquad \operatorname*{Res}_{s=-1} \frac{y^s}{s(s+1)} = -y^{-1},$$

and therefore $K(y) = 1 - y^{-1}$. If $0 < y < 1$, close to the right. The factor $y^{\Re s}$ decays and the right half-plane contains no pole; the same horizontal estimate gives $K(y) = 0$. At $y = 1$, an integrable bound uniform for $y \in [1/2, 2]$ permits passage to the limit, giving $K(1) = 0$. Set $y = x/n$. On $\Re s = c > 1$,

$$\sum_{n \geq 1} (x/n)^c \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{|(c+it)(c+1+it)|} < \infty,$$

so summation and integration over the entire vertical line may be interchanged, giving

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \zeta(s) \frac{x^s}{s(s+1)}\, \mathrm{d}s = \sum_{n \leq x} \left(1 - \frac{n}{x}\right),$$

where the weight at an integer endpoint is zero. This is (6.6).

Chapter 7

Exercise 7.1. List all characters modulo 3, verify both orthogonality relations, and write the first six nonzero terms of the L-series for the nonprincipal character.

Solution. The group $(\mathbb{Z}/3\mathbb{Z})^\times = \{1, 2\}$ is cyclic of order two. Its characters are the principal character χ_0 and the nonprincipal real character χ:

$n \bmod 3$	0	1	2
$\chi_0(n)$	0	1	1
$\chi(n)$	0	1	-1

The first orthogonality relation follows directly from

$$\sum_{a \bmod 3} |\chi_0(a)|^2 = \sum_{a \bmod 3} |\chi(a)|^2 = 2, \qquad \sum_{a \bmod 3} \chi_0(a)\overline{\chi(a)} = 1 - 1 = 0.$$

For the second relation, at $a = 1$,

$$\frac{1}{2}\big(\chi_0(n) + \chi(n)\big) = \mathbf{1}_{n \equiv 1 \ (3)},$$

and at $a = 2$,

$$\frac{1}{2}\big(\chi_0(n) - \chi(n)\big) = \mathbf{1}_{n \equiv 2 \ (3)}.$$

When $3 \mid n$, both sides vanish. The first six nonzero terms of the nonprincipal L-series are

$$L(s, \chi) = 1 - \frac{1}{2^s} + \frac{1}{4^s} - \frac{1}{5^s} + \frac{1}{7^s} - \frac{1}{8^s} + \cdots.$$

Exercise 7.2. Check Proposition 7.11 line by line, and use Abel summation on an infinite interval to write an integral representation of $L(1,\chi)$.

Solution. If χ is nonprincipal, a complete period sums to $\sum_{a=1}^{q}\chi(a)=0$. Split $\{1,\dots,\lfloor x\rfloor\}$ into complete periods and a tail of fewer than q terms. Then

$$A(x):=\sum_{n\le x}\chi(n),\qquad |A(x)|\le q.$$

Apply Abel summation with finite upper endpoint and $f(t)=1/t$:

$$\sum_{n\le X}\frac{\chi(n)}{n}=\frac{A(X)}{X}+\int_1^X\frac{A(t)}{t^2}\,\mathrm{d}t.$$

Since $A(X)=O(q)$, the first term tends to zero and the integral converges absolutely. Thus

$$L(1,\chi)=\sum_{n\ge1}\frac{\chi(n)}{n}=\int_1^\infty\frac{A(t)}{t^2}\,\mathrm{d}t.$$

The series is conditionally convergent; the justification comes from bounded partial sums and Abel summation on an infinite interval, not from absolute convergence of $\sum|\chi(n)|/n$.

Exercise 7.3. Prove (7.3) in detail. Hint: for every p^k, use $3+4\Re(\chi(p)^k)+\Re(\chi(p)^{2k})\ge0$.

Solution. For real $\sigma>1$, the logarithm of the Euler product converges absolutely, so

$$\log L(\sigma,\chi)=\sum_p\sum_{k\ge1}\frac{\chi(p)^k}{kp^{k\sigma}}.$$

Consequently

$$\begin{aligned}&3\log\zeta(\sigma)+4\Re\log L(\sigma,\chi)+\Re\log L(\sigma,\chi^2)\\&\quad=\sum_p\sum_{k\ge1}\frac{3+4\Re(\chi(p)^k)+\Re(\chi(p)^{2k})}{kp^{k\sigma}}.\end{aligned}$$

If $p\nmid q$, write $\chi(p)^k=\mathrm{e}^{i\theta}$. The numerator is

$$3+4\cos\theta+\cos2\theta=2(1+\cos\theta)^2\ge0.$$

If $p\mid q$, then $\chi(p)=0$ and the numerator is 3. The entire real sum is nonnegative. Exponentiating and using $\exp(\Re\log z)=|z|$ gives

$$\zeta(\sigma)^3|L(\sigma,\chi)|^4|L(\sigma,\chi^2)|\ge1,$$

which is (7.3).

Exercise 7.4. For $q=4$ and $a=3$, write out (7.4) and explain how $L(1,\chi_4)=\pi/4$ forces infinitely many primes $p\equiv3\pmod4$.

Solution. Modulo 4 there are only χ_0 and χ_4. For $a=3$, $\overline{\chi_0(3)}=1$ and $\overline{\chi_4(3)}=-1$, so (7.4) becomes

$$\log L(\sigma,\chi_0)-\log L(\sigma,\chi_4)=2\sum_{p\equiv3\ (4)}\frac{1}{p^\sigma}+O(1).$$

The principal character satisfies

$$L(\sigma,\chi_0)=\zeta(\sigma)(1-2^{-\sigma}),$$

so $\log L(\sigma,\chi_0) = \log(1/(\sigma-1)) + O(1)$. On the other hand, $L(1,\chi_4) = \pi/4 > 0$, and hence $\log L(\sigma,\chi_4) = O(1)$ as $\sigma \downarrow 1$. Therefore

$$\sum_{p\equiv 3 \pmod 4} \frac{1}{p^\sigma} = \frac{1}{2}\log\frac{1}{\sigma-1} + O(1) \longrightarrow \infty.$$

If there were only finitely many primes $p \equiv 3 \pmod 4$, the left-hand side would remain bounded as $\sigma \downarrow 1$, a contradiction.

Exercise 7.5. Use Proposition 7.13 to relate the L-function of the character modulo 8 induced by χ_4 to $L(s,\chi_4)$, and verify that the finite product is empty, hence nonzero at $s = 1$.

Solution. Let $\widetilde{\chi}$ be the character modulo 8 induced by χ_4 modulo 4. Its conductor is $q^* = 4$, while $q = 8$. Proposition 7.13 gives

$$L(s,\widetilde{\chi}) = L(s,\chi_4)\prod_{\substack{p\mid 8\\ p\nmid 4}}\left(1-\frac{\chi_4(p)}{p^s}\right).$$

The only prime dividing 8 is 2, but $2 \mid 4$, so the index set is empty and the finite product equals 1. Thus

$$L(s,\widetilde{\chi}) = L(s,\chi_4).$$

The empty product has value 1 at $s = 1$ and does not affect nonvanishing of $L(1,\chi_4)$.

Exercise 7.6. Find the conjugate of each character in the character table modulo 5, and verify that $\overline{\chi(n)} = \chi(n)^{-1}$ when $(n,5) = 1$. Which characters are real?

Solution. Taking conjugates row by row in the character table gives

$$\overline{\chi_0} = \chi_0, \qquad \overline{\chi_1} = \chi_1^3, \qquad \overline{\chi_1^2} = \chi_1^2, \qquad \overline{\chi_1^3} = \chi_1.$$

When $(n,5) = 1$, $\chi(n)$ is a root of unity, so

$$\chi(n)\overline{\chi(n)} = |\chi(n)|^2 = 1, \qquad \overline{\chi(n)} = \chi(n)^{-1}.$$

The real characters are χ_0 and χ_1^2. The characters χ_1 and χ_1^3 are nonreal conjugates of one another.

Exercise 7.7. Give a complete justification for interchanging the Taylor series and the double nonnegative sum in Landau's lemma, and explain why $a(m^2) \geq 1$ forces the abscissa of convergence of $\zeta(s)L(s,\chi)$ to be at least $1/2$.

Solution. Let $F(s) = \sum a_n n^{-s}$ with $a_n \geq 0$ and abscissa of convergence σ_c. Choose a real $s_0 > \sigma_c$. Termwise differentiation in the half-plane of absolute convergence gives

$$(-1)^k F^{(k)}(s_0) = \sum_{n\geq 1} a_n (\log n)^k n^{-s_0} \geq 0.$$

The positive distance from s_0 to the abscissa justifies differentiation: choose $0 < \eta < (s_0 - \sigma_c)/2$. For $|s - s_0| \leq \eta$,

$$a_n(\log n)^k |n^{-s}| \ll_{k,\eta} a_n n^{-(s_0-2\eta)}.$$

The right side is summable because $s_0 - 2\eta > \sigma_c$, giving uniform control throughout this neighborhood. If F were analytic in $|s-\sigma_c| < r$, take $s_0 = \sigma_c + r/4$. The Taylor disk of radius $r/2$ about s_0 lies in that neighborhood and crosses the boundary line. For $h > 0$ in this disk,

$$\begin{aligned} F(s_0-h) &= \sum_{k\geq 0}\frac{(-h)^k}{k!}F^{(k)}(s_0) \\ &= \sum_{k\geq 0}\sum_{n\geq 1}\frac{h^k}{k!}a_n(\log n)^k n^{-s_0}. \end{aligned}$$

Every term of the double series is nonnegative, so Tonelli's theorem permits the interchange of sums without a prior assumption of absolute convergence. Hence

$$F(s_0 - h) = \sum_{n \ge 1} a_n n^{-s_0} \sum_{k \ge 0} \frac{(h \log n)^k}{k!} = \sum_{n \ge 1} a_n n^{-(s_0 - h)}.$$

The left-hand side is a finite analytic value within the disk, so the series on the right converges. Choosing $r/4 < h < r/2$ makes the original Dirichlet series converge to the left of its abscissa, a contradiction. This completes the summation step in Landau's lemma.

For a real nonprincipal character, the text writes

$$\zeta(s) L(s, \chi) = \sum_{n \ge 1} \frac{a(n)}{n^s}, \qquad a(n) \ge 0, \qquad a(m^2) \ge 1.$$

Therefore, for real σ,

$$\sum_{n \ge 1} \frac{a(n)}{n^\sigma} \ge \sum_{m \ge 1} \frac{a(m^2)}{m^{2\sigma}} \ge \sum_{m \ge 1} \frac{1}{m^{2\sigma}}.$$

At $\sigma = 1/2$, the right-hand side is the harmonic series, so the left-hand side diverges. The abscissa of convergence of this nonnegative-coefficient Dirichlet series therefore satisfies $\sigma_{\mathrm{c}} \ge 1/2$, exactly the lower bound needed for the application of Landau's lemma.

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